



Sec: Sr. IIT

Date: 14-02-24

Race -1

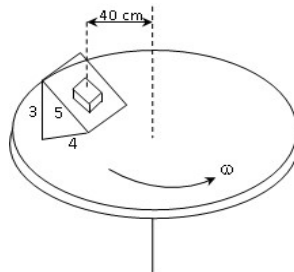
PHYSICS SECTION - I

(SINGLE CORRECT ANSWER TYPE)

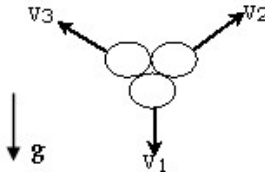
This section contains 10 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +2 for correct answer, 0 if not attempted and 0 in all other cases.

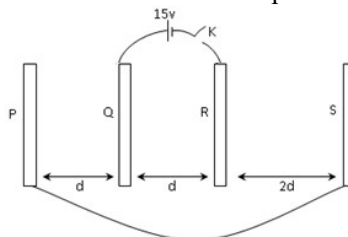
1. A 3 – 4 – 5 inclined plane is fixed to a rotating turn table as shown in figure. A block rests on the inclined plane and the co-efficient of static friction between the inclined plane and the block is $\mu_s = 1/4$. The block is to remain at a position 40 cm from the axes of the turn table. Then the minimum angular speed required to keep the block from sliding down the plane ($g = 9.8 \text{ m/s}^2$)



- A) $3.8 \frac{\text{rad}}{\text{sec}}$ B) $4 \frac{\text{rad}}{\text{sec}}$ C) $3.2 \frac{\text{rad}}{\text{sec}}$ D) $4.2 \frac{\text{rad}}{\text{sec}}$
2. A rocket is projected straight up and explodes into three equally massive fragments just as it reaches the top of its flight. One of its fragment is observed to come straight down in time t_1 . While other two land simultaneously at a time t_2 , after the burst. The height at which fragmentation occurs is: (Assume g is constant)

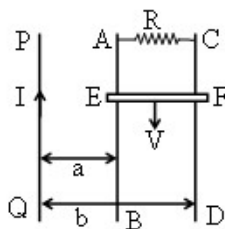


- A) $\frac{gt_1t_2}{2} \left(\frac{t_1 + 2t_2}{2t_1 + 2t_2} \right)$ B) $\frac{gt_1t_2}{2} \left(\frac{2t_1 + t_2}{t_1 + 2t_2} \right)$ C) $\frac{gt_1t_2}{2} \left(\frac{t_1 + t_2}{2t_1 + t_2} \right)$ D) $\frac{gt_1t_2}{2} \left(\frac{t_1 + 2t_2}{2t_1 + t_2} \right)$
3. Consider a rotating spherical planet. The velocity of a point on its equator is V . The effect of rotation of the planet is to make g at the equator $\frac{1}{2}$ of g at poles. The escape velocity for a polar particle on the planet is
A) V B) $2V$ C) $3V$ D) $V/2$
4. Consider a system of four identical uncharged infinite metal plates marked P, Q, R and S arranged as shown. The key is now closed what is potential difference between plates Q and S in steady state



- A) 10 V B) 5 V C) 8 V D) 12 V

5. PQ is an infinite current carrying conductor. AB and CD are smooth conducting rods on which a conductor EF moves with constant velocity V as shown. The force along the direction of velocity needed to maintain constant speed of EF is (No gravity is there)



- A) $\frac{1}{VR} \left[\frac{\mu_0 IV}{2\pi} \ln\left(\frac{b}{a}\right) \right]^2$ B) $\frac{1}{R} \left[\frac{\mu_0 IV}{2\pi} \ln\left(\frac{b}{a}\right) \right]^2$ C) $\frac{V}{R} \left[\frac{\mu_0 IV}{2\pi} \ln\left(\frac{b}{a}\right) \right]^2$ D) $\frac{V}{R} \left[\frac{\mu_0 IV}{2\pi} \ln\left(\frac{b}{a}\right)^2 \right]$

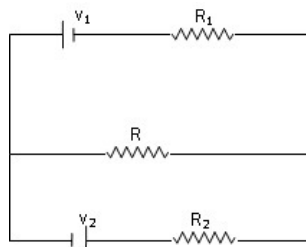
SECTION – II

(MULTIPLE CORRECT ANSWER TYPE)

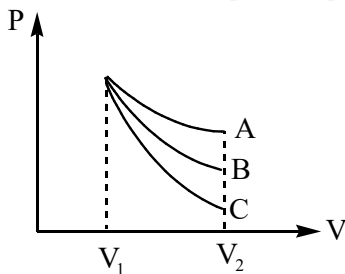
This section contains 5 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct.

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6. Consider a circuit as shown in figure; choose the correct option(s). ($V_1 \neq V_2$)



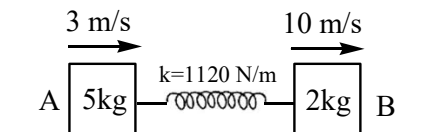
- A) For maximum power development across R , if $R_1 = R_2$ then $R = 2R_1 = 2R_2$
 B) For maximum power development across R , if $R_1 = R_2$ then $R = \frac{R_1}{2} = \frac{R_2}{2}$
 C) For no power across R , $\frac{V_1}{R_1} = \frac{V_2}{R_2}$ and R can attain any value
 D) For no power across R , $\frac{V_1}{R_1} = \frac{V_2}{R_2}$ and $R = \frac{R_1 R_2}{R_1 + R_2}$ only
7. An ideal gas undergoes an expansion from a state with temperature T_1 and volume V_1 to V_2 through three different polytropic processes A, B and C as shown in the P - V diagram. If $|\Delta E_A|$, $|\Delta E_B|$ and $|\Delta E_C|$ be the magnitude of changes in internal energy along the three paths respectively, then :



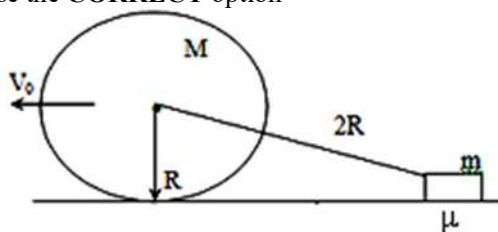
- A) $|\Delta E_A| < |\Delta E_B| < |\Delta E_C|$ if temperature in every process decreases
 B) $|\Delta E_A| > |\Delta E_B| > |\Delta E_C|$ if temperature in every process decreases
 C) $|\Delta E_A| > |\Delta E_B| > |\Delta E_C|$ if temperature in every process increases

D) $|\Delta E_B| < |\Delta E_A| < |\Delta E_C|$ if temperature in every process increases

8. Two blocks A(5kg) and B(2kg) attached to the ends of an ideal massless spring of spring constant 1120N/m are placed on a smooth horizontal plane with the spring undeformed. Simultaneously velocities of 3m/s and 10m/s along the line of the spring in the same direction are imparted to A and B respectively as shown in the fig, then



- A) when the extension of the spring is maximum the velocities of A and B are zero
 B) the maximum extension of the spring is 25cm.
 C) maximum extension and maximum compression occur alternately.
 D) the maximum compression occur for the first time after $\frac{\pi}{56}$ sec
9. An equilateral prism deviates a ray through 45° for two angles of incidence differing by 20° . The angles of incidence are :
 A) 60° B) $42^\circ 30'$ C) $62^\circ 30'$ D) $40^\circ 30'$
10. A uniform solid cylinder of mass M and radius R, rolling without sliding on a rough horizontal plane, is pulled at its axis with a horizontal velocity of V_0 . By means of a string of length 2R attached to its axis, the cylinder is dragging a thin plate of mass $m = 2M$ lying on the plane. If the pulling force is removed and the coefficient of friction between the cylinder and ground is sufficient enough for rolling always. ($\mu = 0.4$, $V_0 = 2\text{m/sec}$, $R = 0.5\text{ m}$, $g = 10\text{ m/sec}^2$). Choose the **CORRECT** option



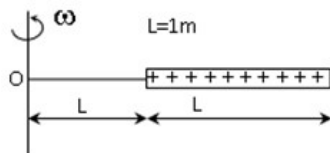
- A) Distance covered before stopping is 0.96 m B) Distance covered before stopping is 0.49 m
 C) Time taken to stop is 0.96 sec D) Average speed during the motion is 1 m/sec

SECTION – III INTEGER TYPE

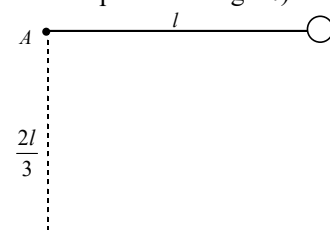
(This section contains **5 questions**. The answer to each question is a single digit integer ranging from 0 to 9. The correct digit below the question number in the ORS is to be bubbled).

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11. A circuit draws a power of 550 watt from a source of 220 volt, 50 Hz. The power factor of the circuit is 0.8 and the current lags in phase behind the potential difference. To make the power factor of the circuit as 1.0, the capacitance will have to be connected with it is 15k then find the value of k (in μF) to the nearest integer.
12. In a slow reaction, heat is being evolved at a rate about 10mW in a liquid. If the heat were being generated by the decay of ^{32}P , a radioactive isotope of phosphorus that has half-life of 14 days and emits only beta-particles with a mean energy of 700KeV, estimate the number of ^{32}P atoms in the liquid. Express your answer in form of $A \times 26 \times 10^{15}$ and fill A in OMR sheet. Round off A to nearest integer.
 [Take : $\ln 2 = 0.7$]
13. A thin rod of length $L = 1\text{ m}$ is being rotated about point O in horizontal plane as shown in figure. The rod carries uniformly distributed charge over it having linear charge density $\frac{2}{\pi}$ coulomb/metre. The rod is rotating with a constant angular velocity 2π rad/sec. If magnetic field strength at O is $\frac{\mu_0 X}{4\pi}$ ln 2 Tesla the value of x



14. A is a fixed point at a height $\frac{2l}{3}$ above a perfectly inelastic smooth horizontal plane. A light inextensible string of length l has one end attached to A and other to a heavy ball. The ball is held at the level of A with string just taut and released from rest. The speed of ball just after striking the plane is n times $\sqrt{\frac{gl}{27}}$. Find n . (Neglect the size of the heavy ball when compared to length l)



15. A flexible steel cable of total length 'L' and mass per unit length μ hangs vertically (under its own weight) from a support at upper end. If a small transverse pulse starts to move down in the wire from its support. The ratio of acceleration of the pulse at distance $\frac{L}{4}$ from the support end to the acceleration of the pulse at distance $\frac{L}{2}$ from the support end is K : 1 then the value of K is

CHEMISTRY

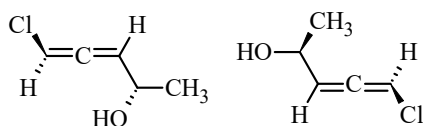
SECTION - I

(SINGLE CORRECT ANSWER TYPE)

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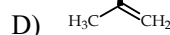
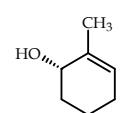
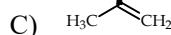
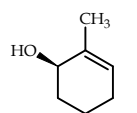
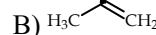
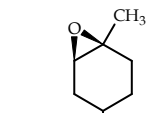
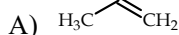
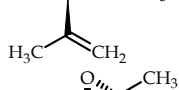
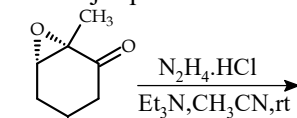
Marking scheme: +2 for correct answer, 0 if not attempted and 0 in all other cases.

16. The correct relation between the following compounds is:

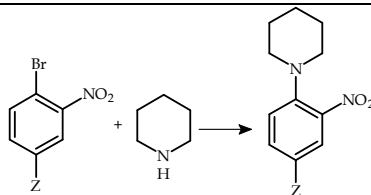


- A) Enantiomers B) Diastereomers C) Homomers (Identical) D) Constitutional isomers

17. The major product in the following reaction is:



18. The correct order of the rate constants for the following series of reactions ($Z = \text{CF}_3 / \text{CH}_3 / \text{OCH}_3$) is



A)	$\text{CF}_3 > \text{CH}_3 > \text{OCH}_3$	B)	$\text{CF}_3 > \text{OCH}_3 > \text{CH}_3$
C)	$\text{OCH}_3 > \text{CF}_3 > \text{CH}_3$	D)	$\text{CH}_3 > \text{OCH}_3 > \text{CF}_3$

19. The left half-cell of the electrochemical cell consists of an iron electrode (in excess) that is oxidized during the action and an aqueous solution of iron (III) nitrate. The right half-cell consists of graphite electrode and a mixture of iron (II) and (III) nitrates. Using the information on ion mobility (μ°), select the most suitable ionic compound(s) for the salt bridge

Cation	$\mu^\circ, m^2 V^{-1} s^{-1} \cdot 10^9, 25^\circ C$	Anion	$\mu^\circ, m^2 V^{-1} s^{-1} \cdot 10^9, 25^\circ C$
$\text{Ca}^{2+} (\text{aq})$	61.6	$\text{F}^- (\text{aq})$	51.5
$\text{K}^+ (\text{aq})$	73.5	$\text{CO}_3^{2-} (\text{aq})$	70.3
$\text{NH}_4^+ (\text{aq})$	76.1	$\text{NO}_3^- (\text{aq})$	74.1
$\text{H}^+ (\text{aq})$	205	$\text{Br}^- (\text{aq})$	81.8

- A) Calcium carbonate B) Ammonium nitrate
C) Potassium carbonate D) Ammonium bromide
20. 50.0 ml of 0.100 M HX has been mixed with 100.0 ml of 0.0500 M NaOH then concentration of conjugate base of HX in the resultant solution can't be
- A) 0.032M B) 0.028M C) 0.036M D) both A and B

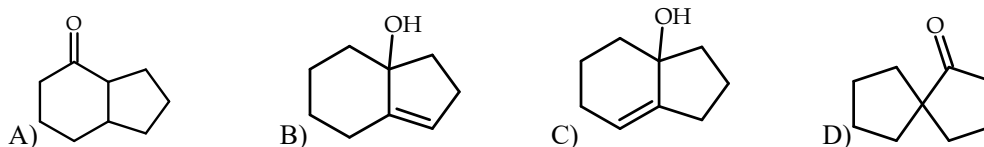
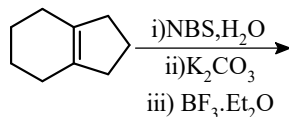
SECTION – II

(MULTIPLE CORRECT ANSWER TYPE)

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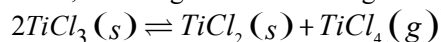
21. Which of the following reaction involve carbanion as intermediate?
A) Perkin condensation B) Sandmeyer reaction
C) Cannizzaro reaction D) Hofmann bromamide reaction
22. The major product in the following reaction is



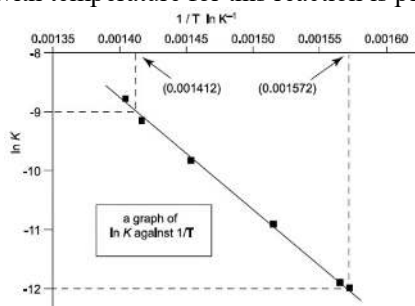
23. On adding potassium iodide to an aqueous solution containing complex ion $[\text{Ag}(\text{NH}_3)_2]^+$ yellow coloured precipitate of AgI is formed. Similarly when hydroiodic acid was added to the complex ion $[\text{Ag}(\text{NH}_3)_2]^+$ again precipitates AgI. Then correct statements for these observations
- A) Complex ion $[\text{Ag}(\text{NH}_3)_2]^+$ is less stable than AgI
B) Complex ion $[\text{Ag}(\text{NH}_3)_2]^+$ ionizes less than AgI
C) HI neutralizes NH_3 ligands and precipitates AgI
D) IUPAC name of the complex is diammine silver (I) ion
24. Which of the following statement(s) is/are correct?
A) Cu^{2+} salts form soluble complex with excess KCN
B) Cu^{2+} salts form soluble complex with aqueous ammonia
C) Cu^{2+} salts form soluble complex with KI

D) A piece of iron or zinc when placed in Cu^{2+} salt solution, precipitates copper.

25. When titanium(III) chloride is heated, it undergoes the following reaction:



A graph showing how K varies with temperature for this reaction is plotted below.



Select the correct options about the reversible reaction.

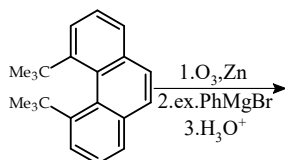
- A) Enthalpy of the reaction under standard conditions is $+155.9 \text{ kJ mol}^{-1}$
- B) Standard free energy change of reaction increases with increase in temperature if Standard enthalpy change of reaction and standard entropy change of reaction are considered to be constant for given temperature interval
- C) Heating the equilibrium mixture after addition of inert gas at constant volume can't effect the equilibrium composition.
- D) Standard entropy of reaction is equal to Standard enthalpy of reaction at equilibrium

SECTION – III INTEGER TYPE

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26. The number of possible stereoisomers obtained in the following reaction is



27. In a solution of 0.010 M barium nitrate and 0.010 M lead(II) nitrate, When the more soluble fluoride (other than NaF) begins to precipitate, the concentration of the cation for the less soluble fluoride that remains in solution, when NaF is added, is $a \times 10^{-b}$. Value of b is... ($K_{sp}(\text{BaF}_2) = 1.5 \times 10^{-6}$ and $K_{sp}(\text{PbF}_2) = 4.0 \times 10^{-8}$).
28. A drug (M. Mass 100) decomposes by first order kinetics in an animal body. If animal body is supplied by 10 IU of drug at regular intervals such that the time between two successive loadings is $T_{25\%}$. Maximum mass of drug in the animal body is 'x' times 10 IU. Value of 'x' is _____.
29. Compound $\text{PdCl}_4 \cdot 6\text{H}_2\text{O}$ is a hydrated complex; 1 m aqueous solution of it has freezing point 269.28 K. Assuming 100% ionization of complex, calculate the number of ions furnished by complex in the solution.
30. How many of the following statements regarding oxides (M_2O), peroxides (M_2O_2) and super oxides (MO_2) of alkali metals are correct.

- i) Order of size of oxide, peroxide and super oxide anions is $\text{O}^{2-} < \text{O}_2^{2-} < \text{O}_2^-$
- ii) Thermal stability order is $\text{O}^{2-} < \text{O}_2^{2-} < \text{O}_2^-$
- iii) Thermal stability with a particular alkali metal is super oxide < peroxide < monoxide
- iv) Lattice energy of oxides, peroxides and super oxides decreases down the group for alkali metals
- v) In the formation of oxides, peroxides and super oxides the entropy decreases in the order oxide < peroxide < super oxide
- vi) Heat of formation is in the order oxide > peroxide > super oxide
- vii) Heat of formation decreases down the group for oxides, peroxides and super oxides
- viii) Compared to the difference in lattice energies of monoxides and peroxides is less than monoxides and super oxides

- ix) Compared to the difference in lattice energies between monoxides and peroxides is less than peroxides and super oxides.
 x) All the three 1-oxides, 2-peroxides and super oxides form basic solution when dissolved in water
 xi) Thermal stability of oxides and 3-peroxides decreases down the group while those of super oxides increases down the group.
 xii) The difference between the lattice enthalpies of monoxides and peroxides decreases down the group.

MATHEMATICS**SECTION – I****(SINGLE CORRECT ANSWER TYPE)**

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31. The least integer value of $f(x) = \frac{(x-1)^7 + 3(x-1)^6 + (x-1)^5 + 1}{(x-1)^5} \quad \forall x > 1$ is
 A) 8 B) 6 C) 12 D) 18
32. If $a, b \in N$, $a \neq b$ and the two quadratic equations $(a-1)x^2 - (a^2 + 2)x + a^2 + 2a = 0$ and $(b-1)x^2 - (b^2 + 2)x + (b^2 + 2b) = 0$ have a common root. Then the value of ab is ____
 A) 4 B) 6 C) 8 D) 10
33. If $z = e^{i2\pi/5}$ then $1 + z + z^2 + z^3 + 5z^4 + 4z^5 + 4z^6 + 4z^7 + 4z^8 + 5z^9 =$
 A) 0 B) $4z^3$ C) $5z^4$ D) $-4z^2$
34. If $\Delta_1 = \begin{vmatrix} ap^2 & 2ap & 1 \\ aq^2 & 2aq & 1 \\ ar^2 & 2ar & 1 \end{vmatrix}$ and $\Delta_2 = \begin{vmatrix} apq & a(p+q) & 1 \\ aqr & a(q+r) & 1 \\ arp & a(r+p) & 1 \end{vmatrix}$ then
 A) $\Delta_1 = \Delta_2$ B) $\Delta_2 = 2\Delta_1$ C) $\Delta_1 = 2\Delta_2$ D) $\Delta_1 + 2\Delta_2 = 0$
35. All the five digit numbers $N = abcde$ having property that $a < b < c < d < e$ are arranged in the increasing order of their magnitude. The 97th number in the list does not contain the digit
 A) 4 B) 5 C) 7 D) 8

SECTION – II**(MULTIPLE CORRECT ANSWER TYPE)**

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36. The line $\frac{x+6}{5} = \frac{y+10}{3} = \frac{z+14}{8}$ is the hypotenuse of an isosceles right angled triangle whose opposite vertex is $(7, 2, 4)$. Then which of the following is/are side(s) of the triangle
 A) $\frac{x-7}{2} = \frac{y-2}{-3} = \frac{z-4}{6}$ B) $\frac{x-7}{3} = \frac{y-2}{6} = \frac{z-4}{2}$
 C) $\frac{x-7}{3} = \frac{y-2}{5} = \frac{z-4}{-1}$ D) $\frac{x-7}{1} = \frac{y-2}{3} = \frac{z-4}{2}$
37. If $f(x)$ is a real function satisfies $f(x)f^1(-x) = f(-x)f^1(x) \quad \forall x$ and $f(0) = 3$ then which of the following is/are true
 A) $f(4)f(-4) = f(16)f(-16)$ B) The function $h(x) = f(x)f(-x)$ is differentiable $\forall x$
 C) $\int_{-51}^{51} \frac{dx}{3+f(x)} = 17$ D) $f(x) = 0$ has no solution in $[-2, 2]$

38. If $f(x) = \begin{cases} -e^{-x} + k & x \leq 0 \\ e^x + 1 & 0 < x < 1 \\ ex^2 + \lambda & x \geq 1 \end{cases}$ is monotonically increasing $\forall x \in R$, then
- A) Maximum value of k is 1
B) Maximum value of k is 3
C) Minimum value of λ is 0
D) Minimum value of λ is 1
39. Let the end points M and N of the diameter $4x - y = 15$ of the circle $x^2 + y^2 - 6x + 6y - 16 = 0$ are also on the tangents at the end points of the major axis of the ellipse respectively, such that MN is also tangent to the same ellipse at the point P . The major axis is along the line $y = x$ then
- A) Eccentricity of the ellipse is $\frac{4}{5}$
B) Length of latusrectum is $\frac{18}{5}$
C) Centre of the ellipse is $(0, 0)$
D) Equation of directrix of the ellipse $x + y = \frac{25\sqrt{2}}{4}$
40. $A = \begin{bmatrix} a & b & c \\ b & c & a \\ c & a & b \end{bmatrix}$ and $\text{trace}(A) = 9$ and $a, b, c \in N$ such that $ab + bc + ca = 26$. Let A_1 denote $\text{adj}A$ and A_2 denote $\text{adj}A_1$ and so on..... and if $|A_4| = K$ then
- A) $K = 3^{48}$
B) K is divisible by 3^{24}
C) Last two digits of K are 6 and 1
D) Sum of last three digits of K is 15

SECTION – III

INTEGER TYPE

(This section contains **5 questions**. The answer to each question is a single digit integer ranging from 0 to 9.

The correct digit below the question number in the ORS is to be bubbled).

Marking scheme +4 for correct answer , 0 if not attempted and -1 in all other cases.

41. The shortest distance between the parabola $2y^2 = 2x - 1$ and $2x^2 = 2y - 1$ is $\frac{1}{d}$ then d^2 ____
42. If $f(x) = \begin{cases} 1 - \cos\left(1 - \cos\frac{x}{2}\right) & x \neq 0 \\ \frac{2^m x^n}{1} & x = 0 \end{cases}$ is continuous at $x = 0$ then the absolute value of $m + n$ is ____
43. The graph of $xy = 1$ reflected in $y = 2x$ and obtained as $12x^2 + rxy + sy^2 + t = 0$ then $|r - s| =$
44. If $\tan 9\theta = \frac{3}{4}; \left(0 < \theta < \frac{\pi}{18}\right)$ then the value of $\frac{3\cos 3\theta - 4\sec 3\theta}{2}$ is equal to
45. The area of the region enclosed by curves $x = y^2 - 1$ and $x = |y|\sqrt{1 - y^2}$ is ____ units

1	C	2	D	3	B	4	B	5	A
6	BC	7	AC	8	BC	9	BC	10	ACD
11	5	12	6	13	4	14	4	15	1
16	C	17	D	18	A	19	B	20	C
21	D	22	D	23	A	24	B	25	C
26	6	27	4	28	4	29	2	30	8
31	B	32	C	33	C	34	D	35	B
36	AB	37	ABCD	38	BD	39	ABC	40	ABC
41	8	42	3	43	5	44	5	45	2



Sec: Sr. IIT

Date: 18-02-24

Race -2
PHYSICS
SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 06 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +2 for correct answer, 0 if not attempted and 0 in all other cases.

- The density of a sphere is measured by measuring its mass and diameter. If, it is known that the maximum percentage errors in the measurement are 2% and 3% respectively, then find the maximum percentage error in the measurement of density?
 A) 15% B) 18% C) 9% D) 11%
- A shot is fired with velocity $\sqrt{2gh}$ horizontally h m above the top of a mountain which is in the form of hemisphere of radius r . The point on the mountain which can be reached by the shot is at a distance (measured in a straight line) from top of the mountain
 A) $(r - \sqrt{r^2 - 4rh})$ B) $(r + \sqrt{r^2 - 4rh})$ C) $(r - \sqrt{r^2 + 4rh})$ D) $(r - \sqrt{r^2 - 2rh})$
- A cart initially moving on a frictionless track filled with sand has a hole at its bottom from where sand leaks out
 Student A says : The velocity of cart remains constant although mass of sand in cart decrease.
 Student B says: Since mass is decreasing, the velocity of cart must increase.
 A) Student –A is correct, Student –B is wrong
 B) Student A is wrong, Student –B is correct
 C) Both are correct D) Both are wrong
- A block of mass 2kg was moving along a straight line on a smooth surface with a speed of 5 m/s. At $t=0$, a force given by $F=(3+2t)$ N directed in the direction of motion of the body starts acting on the block. The kinetic energy of the block after 2sec is
 A) 20J B) 200J C) 100J D) 120J
- A ring of radius R rolls without sliding with a constant velocity with respect to ground. The radius of curvature of the path followed by any particle of the ring at the highest point of its path will be
 A) R B) $2R$ C) $4R$ D) $9R$

6. Six stars of equal mass each m are moving about the centre of mass of the system under mutual gravitational force of attraction such that they are always on the vertices of a regular hexagon of side length a . Their common time period will be

A) $4\pi\sqrt{\frac{a^3}{Gm}}$ B) $2\pi\sqrt{\frac{4\sqrt{3}a^3}{Gm(5\sqrt{3}+4)}}$ C) $4\pi\sqrt{\frac{\sqrt{3}a^3}{Gm}}$ D) $2\pi\sqrt{\frac{5\sqrt{3}a^3}{Gm(5\sqrt{3}+4)}}$

SECTION - II

(COMPREHENSION TYPE)

This section contains **3 multiple choice questions** relating to three paragraphs with two questions on each paragraph. Each question has 4 choices A), B), C) and D) for its answer, out of which **ONLY ONE** is correct.

Marking scheme +3 for correct answer, 0 if not attempted and -1 in all other cases.

Paragraph for Questions 7

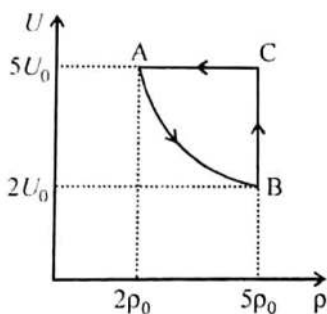
Waves $y_1 = A\cos(0.5\pi x - 100\pi t)$ and $y_2 = A\cos(0.46\pi x - 92\pi t)$ are travelling along x-axis. (Here x is in m and t is in seconds)

7. Find the number of times intensity is maximum in time interval of 1s.

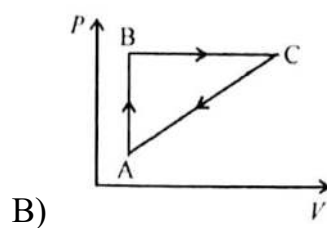
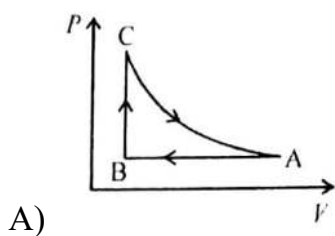
A) 4 B) 6 C) 8 D) 10

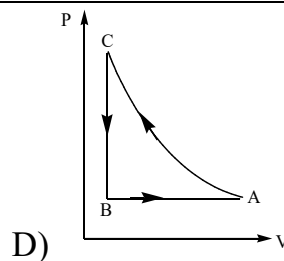
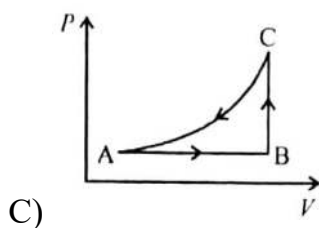
Paragraph for Questions 8

Figure shows the variation of the internal energy U with the density ρ of one mole of ideal mono atomic gas for a thermodynamic cycle ABCA. Here process AB is a part of rectangular hyperbola.



8. The P-V diagram of above process





Paragraph for Questions 9

Three concentric conducting shells A, B and C are shown in figure. Radii of the shells are R , $2R$ and $3R$ respectively. Initial charges given to the shells are Q , $-Q$ and $3Q$ respectively.

9. If V_A and V_B are potentials of shells A and B respectively, then

A) $V_A = V_B$ B) $2V_A = 3V_B$ C) $3V_A = 2V_B$ D) $V_A = 2V_B$

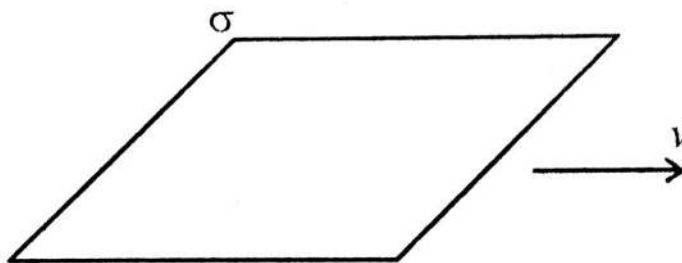
SECTION – III

(MULTIPLE CORRECT CHOICE TYPE)

This section contains **6 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE is/ are correct**.

Marking scheme +4 for correct answer , 0 if not attempted and 0 in all other cases.

10. A large thin non conducting plate with uniform surface charge density σ is moving with constant speed v as shown in the figure. The magnetic field at a small distance above or below the plate in the middle region is



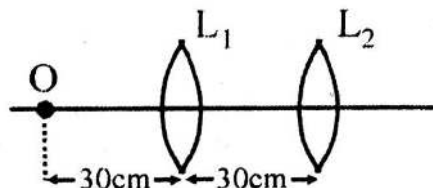
- A) $\mu_0 \sigma v$ in magnitude B) $\frac{\mu_0 \sigma v}{2}$ in magnitude
 C) perpendicular to the plate D) parallel to plate
11. A non conducting ring of mass m and radius R has a charge Q uniformly distributed over its circumference. The ring is placed on a rough horizontal surface such that the plane of the ring is parallel to the surface. A vertical magnetic field $B = B_0 t^2$ tesla is switched on. After 2 seconds from switching on the magnetic field, the ring is just about to rotate about vertical axis through its centre. Then
- A) the induced electric field is quadratic in time t

B) the torque on the ring is $B_0 QR^2 t$

C) until 2 seconds, the friction force does not come into play.

D) the friction coefficient between the ring and the surface is $\frac{2B_0 QR}{mg}$

12. Two thin convex lenses are separated as shown. The focal length of lens L_1 is 15cm and that of lens L_2 is 30cm. An object is placed at distance 30cm from lens L_1 . The location of image formed finally will be at



- A) infinity
B) 15 cm behind the lens L_1
C) optical centre of lens L_2
D) 60cm away from object
13. While light is used to illuminate the two slits in a Young's double slit experiment. The separation between the slits is b and the screen is at a distance $d(>b)$ from the slits. At a point on the screen directly in front of one of the slits, certain wavelengths are missing. Some of these missing wavelengths are
- A) $\lambda = \frac{b^2}{d}$ B) $\lambda = \frac{2b^2}{d}$ C) $\lambda = \frac{b^2}{3d}$ D) $\lambda = \frac{2b^2}{3d}$
14. The initial nucleus of uranium series is ${}_{92}\text{U}^{238}$ and the final nucleus is ${}_{82}\text{Pb}^{206}$. When uranium decays to lead by emitting α and β particles then number of
- A) α -particles produced is 8 B) β -particles produced is 6
C) α -particles produced is 6 D) β -particles produced is 8
15. In an X-ray tube, if the accelerating potential difference is increased, then
- A) the frequency of characteristic X-rays of a material will get changed.
B) number of electrons reaching the anode per second will change.
C) number of characteristics X-ray lines may increase.
D) the difference between λ_0 (minimum wavelength) and $\lambda_{K\alpha}$ (wavelength of K_α X-ray) will get changed.

CHEMISTRY**SECTION – I****(SINGLE CORRECT ANSWER TYPE)**

This section contains 06 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +2 for correct answer, 0 if not attempted and 0 in all other cases.

16. Consider the following statements.

I) Second law of thermodynamics helps in supporting the 3rd law of thermodynamics.

II) $2H_2(g) + O_2(g) \rightarrow 2H_2O(l); \Delta_{Rxn}S = -ve$ so we can conclude that reaction is impossible.

III) Melting point of substances always increases with increase in pressure.

IV) $C_p - C_v = R$ is valid only for ideal gases.

The correct code is:

A) I, III & IV only B) I & IV only C) II, IV only D) II, III & IV

17. The only incorrect statement about enzyme catalysis is:

A) Enzymes do not alter the amount (or) nature of the product.

B) Enzymes retain identify at the end of the reaction, as in the beginning.

C) Most enzymes can't work in either directions.

D) Enzymes are specific in their action.

18. Consider the following statements:

I) The measurement of cataphoresis of a hydrophobic colloid is more difficult than hydrophilic colloid.

II) Brownian motion is responsible for the stability of colloidal solution.

III) Gelatin has a isoelectric point at PH =4.7. So, at PH = 4.7 the zeta potential is zero.

IV) The conductivity of an o/w emulsion is lesser than that of w/o emulsion.

The code for the correct statement is: (T – True, F – False)

A) F T T F B) T F F T C) F T F T D) T F T F

19. In the specific conductance measurement, D.C (direct current) is not used because:

A) Ohm's law can't be applied to D.C

B) D.C heats up the solution.

C) D.C causes polarization at the electrode.

D) D. C is expensive.

20. The only incorrect statement is:

A) Rise in the compression factor with increasing pressure is due to the factors, 'b'.

B) A gas can be liquefied at $T = T_C$, $P < P_C$.

C) Average speed of molecules of a gas in a container moving only in one dimension will be zero.

D) For two gases, if pressures are equal, $T_1 = 2T_2$ and $\sigma_2 = 2\sigma_1$, then $\lambda_1 > \lambda_2$.

21. The correct set (T – True, F – False) about the following statement is:

I) All tetrahedral complexes are high – spin.

II) Tetrahedral complexes are favoured where the ligands are large and bulky.

III) Whenever dz^2 and $dx^2 - y^2$ orbitals are unequally occupied, Jahn – teller distortion occurs in a non – linear molecule.

IV) An aqueous solution of $[Co(NH_3)_5Br]SO_4$ gives a white Precipitate of $BaSO_4$ with $BaCl_2$ solution.

A) T T T T

B) T F T F

C) F F T T

D) T T F F

SECTION - II

(COMPREHENSION TYPE)

This section contains **3 multiple choice questions** relating to two paragraphs with three questions on each paragraph. Each question has 4 choices A), B), C) and D) for its answer, out of which **ONLY ONE** is correct.

Marking scheme +3 for correct answer , 0 if not attempted and -1 in all other cases.

Paragraph for Questions 22

Element (A) burns in nitrogen to give an ionic compound (B). Compound (B) reacts with water to give (C) and (D). A solution of (C) becomes milky on bubbling CO_2 . The chloride of elements (A) imparts brick red colour to Bunsen flame.

Now answer the following questions:

22. The element (A) is:

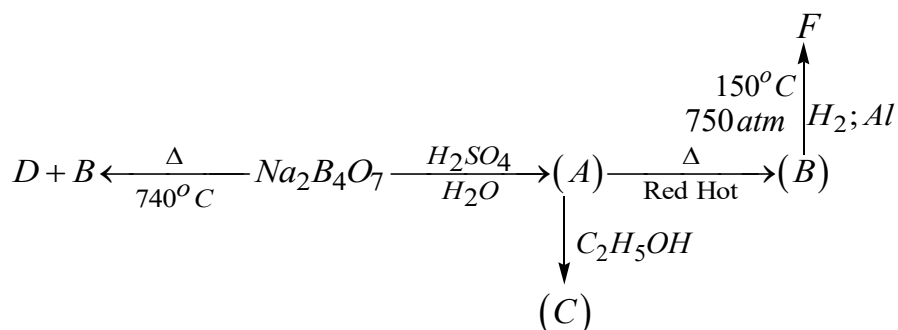
A) Ca

B) Mg

C) Al

D) K

Paragraph for Questions 23



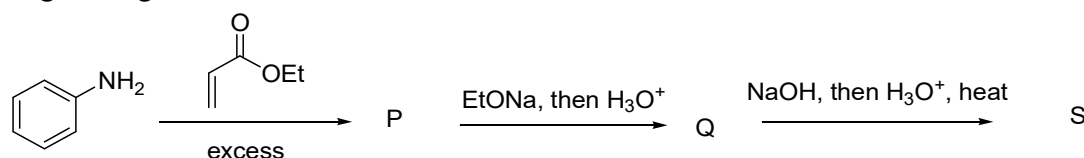
Now answer the following questions:

23. The incorrect set is:

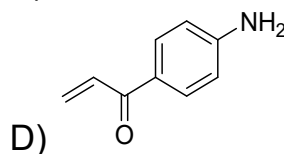
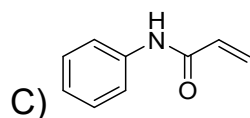
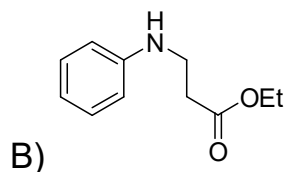
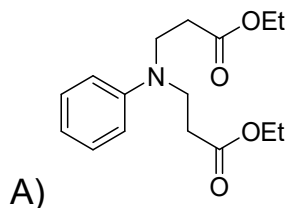
- A) $A-H_3BO_3$ B) $B-B_2O_3$ C) $C-B(OC_2H_5)_3$ D) $D-Na_2O$

Paragraph for Questions 24

Observe the following reaction scheme and answer the following question regarding it.



24. Identify the product 'P' in the above reaction scheme.



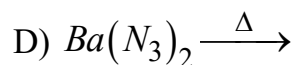
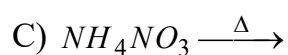
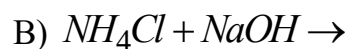
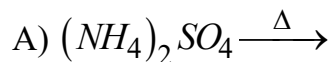
SECTION – III

(MULTIPLE CORRECT CHOICE TYPE)

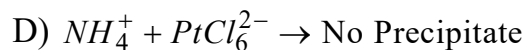
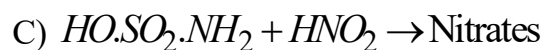
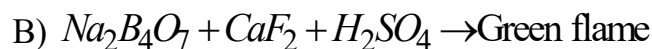
This section contains **6 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/ are correct.

Marking scheme +4 for correct answer , 0 if not attempted and 0 in all other cases.

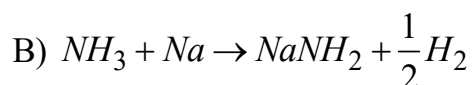
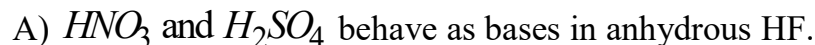
25. Which of the following reactions will lead to the formation of N_2 ?



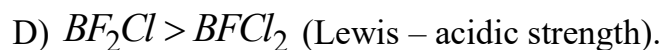
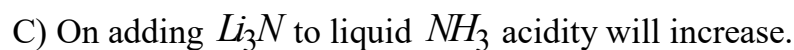
26. Pick up the correct observations?



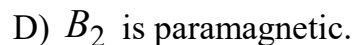
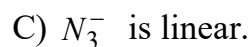
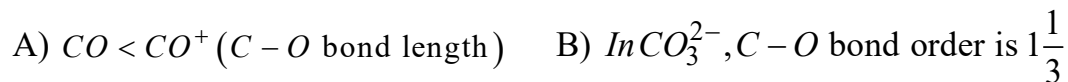
27. The correct statement(s) is/are



In the above reaction, NH_3 acts an acid.



28. The correct statement(s) is / are:



29. The **CORRECT** statement(s) about carbene is (are).

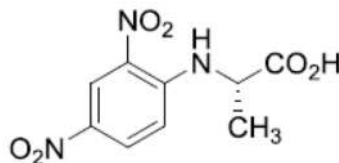
A) Carbene is a neutral species.

B) Carbene is an intermediate in the Curtius rearrangement.

C) Carbene can insert into both σ and π - bonds.

D) Carbene is generated from amines on reaction with nitrous acid.

30. Tetrapeptide(s) that gives (give) the following product on reaction with Sanger's reagent followed by hydrolysis is (are)



A) Ala-Gly-Leu-Phe

B) Asp-Phe-Leu-Pro

C) Asp-Gly-Tyr-Phe

D) Ala-Phe-Tyr-Pro

MATHEMATICS

SECTION – I

(SINGLE CORRECT ANSWER TYPE)

This section contains 06 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +2 for correct answer, 0 if not attempted and 0 in all other cases.

31. If $\tan \frac{\pi}{15} + \tan \left(\frac{\pi}{15} + \frac{\pi}{5} \right) + \tan \left(\frac{\pi}{15} + \frac{2\pi}{5} \right) + \tan \left(\frac{\pi}{15} + \frac{3\pi}{5} \right) + \tan \left(\frac{\pi}{15} + \frac{4\pi}{5} \right) = k\sqrt{3}$ then the value of k is
- A) 2 B) 3 C) 4 D) 5
32. $\lim_{a \rightarrow \infty} \frac{1}{a} \int_0^a \frac{x^2 + ax + 1}{1 + x^4} \tan^{-1} \left(\frac{1}{x} \right) dx$ is equal to $\frac{\pi^2}{k}$ where $k \in \mathbb{N}$ equals
- A) 4 B) 8 C) 16 D) 32
33. The number of three digit numbers between the two A.P's 3, 11, 19, 27..... and 7, 30, 53..... that are common is
- A) 5 B) 4 C) 6 D) 8
34. The abscissa of A and B on the circle $x^2 + y^2 = 25$ are respectively 5 and -5. Let MN be a variable chord perpendicular to AB and if the locus of point of intersection of the lines BM and AN is a conic then its eccentricity is
- A) 1 B) $\sqrt{2}$ C) $\frac{1}{5}$ D) $\sqrt{8}$
35. Let $f(x) = 8x^3 + 3x$ and $f^{-1}(x)$ be the inverse function of $f(x)$, then $\lim_{x \rightarrow \infty} \frac{x^{\frac{1}{3}}}{f^{-1}(8x) - f^{-1}(x)} =$
- A) 0 B) 1 C) 2 D) 4
36. Number of ways in which 4A's, 1B, 1C and 1D can be distributed among 4 persons such that everyone gets atleast one letter and no one gets all 4A's, is
- A) 832 B) 2432 C) 2077 D) 608

SECTION - II

(COMPREHENSION TYPE)

This section contains 3 multiple choice questions relating to three paragraphs with two questions on each paragraph. Each question has 4 choices A), B), C) and D) for its answer, out of which ONLY ONE is correct.

Marking scheme +3 for correct answer , 0 if not attempted and -1 in all other cases.

Paragraph for Questions 37

$$\int_a^b f(x) dx = \int_a^b f(a+b-x) dx,$$

$$\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx \text{ when } f(x) \text{ is even}$$

$$= 0 \text{ when } f(x) \text{ is odd}$$

37. $\int_{-\pi}^{\pi} \frac{\sin^2 x}{x + 5 + \sqrt{x^2 + 25}} dx$ is equal to

- A) 0 B) $\frac{\pi}{5}$ C) $\frac{\pi}{10}$ D) $\frac{2\pi}{5}$

Paragraph for Questions 38

Let $x^2 + y^2 \pm 4x + 3 = 0$ be two fixed circles, A triangle ABC has vertices A(0, 2), B $(2, 2\sqrt{3} + 2)$, C $(-2, 2\sqrt{3} + 2)$. It starts moving downwards (i.e, in the negative direction of y-axis) and stops when its sides touch the circles (AB at P_1 and AC at P_2)

38. The ratio in which P_1 divides new positions of A and B is

- A) $\frac{3-\sqrt{3}}{\sqrt{3}}$ B) $\frac{4-\sqrt{3}}{\sqrt{3}}$ C) $\frac{1+\sqrt{3}}{\sqrt{3}}$ D) $\frac{2+\sqrt{3}}{\sqrt{3}}$

Paragraph for Questions 39

Suppose A,B,C are three-non collinear points corresponding to complex numbers $z_0 = ai$, $z_1 = \frac{1}{2} + bi$, $z_2 = 1 + ci$ ($a, b, c \in R$) respectively. Now consider the curve $z = z_0 \cos^4 t + 2z_1 \cos^2 t \sin^2 t + z_2 \sin^4 t$ ($t \in R$)

39. The locus of z is a

- A) straight line B) straight line or parabola
C) parabola D) segment of a parabola

SECTION – III**(MULTIPLE CORRECT CHOICE TYPE)**

This section contains **6 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/ are correct.

Marking scheme +4 for correct answer , 0 if not attempted and 0 in all other cases.

40. Equation of a plane which is at a distance of $\frac{1}{5}$ units from origin and at a distance of $\frac{4}{5}$ units from $(1, 2, 1)$ is $ax + by + cz + d = 0$. The plane also passes through the intersection of $x + y + z - 3 = 0$, $2x + 2y + z - 5 = 0$ and $4x - y + 3z - 6 = 0$, $a \neq 0$. Then

A) $c + d = 0$

B) $a + b = 0$

C) $c = 0$

D) $a + b + c + d = 0$

41. If $x \sin A + y \sin B + z \sin C = 0$ and $x^2 \sin 2A + y^2 \sin 2B + z^2 \sin 2C = 0$, then the value of $x^3 \sin 3A + y^3 \sin 3B + z^3 \sin 3C$ (where $x, y, z, A, B, C \in \mathbb{R}$ and $A + B + C = \pi$)
- A) is equal to 0
B) depends on A, B, C
C) depends on x, y, z
D) None of these
42. Let A be a 3×3 invertible matrix such that $A^3 - pA^2 + p^2A = 0$ where p is a prime natural number then
- A) Number of factors of $(-\det(A))$ are 4
B) Adjoint of A is square of A
C) Trace of AA^T is positive
D) A^4 is a scalar multiple of A
43. $f(x) = e^x, 0 \leq x \leq 1$
 $= 2 - e^{x-1}, 1 < x \leq 2$
 $= x - e, 2 < x \leq 3$ and $g(x) = \int_0^x f(t) dt, x \in [1, 3]$ then $g(x)$ has
- A) local maximum at $x = 1 + \ln 2$ and local minima at $x = e$
B) local minima at $x = 1$ and local maxima at $x = 3$
C) no local maxima
D) no local minima
44. The value(s) of k for which $f(x) = (k + |x|)^2 e^{(5-|x|)^2}$ is differentiable at all $x \in \mathbb{R}$ is
- A) 0
B) $\frac{1}{5}$
C) 1
D) 5
45. A circle touches the parabola $y^2 = 4ax (a > 0)$ at P . It also passes through the focus S of the parabola and intersects its axis at Q . If angle SPQ is 90° . Then which of the following is/are correct?
- A) abscissa of centre of the circle = $10a$
B) ordinate of centre of the circle = 0
C) radius of the circle = $4a$
D) circle is passing through origin

KEY SHEET

1	D	2	A	3	A	4	C	5	C
6	B	7	A	8	A	9	B	10	BD
11	BD	12	CD	13	AC	14	AB	15	CD
16	B	17	C	18	A	19	C	20	B
21	A	22	A	23	D	24	A	25	D
26	AB	27	AB	28	BCD	29	AC	30	AD
31	D	32	C	33	B	34	B	35	C
36	A	37	C	38	B	39	D	40	CD
41	A	42	ABCD	43	A	44	AB	45	BC



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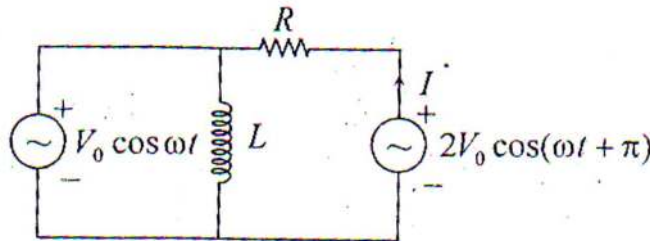
Race -3

PHYSICS
SECTION - I
(SINGLE CORRECT ANSWER TYPE)

This section contains 05 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +2 for correct answer, 0 if not attempted and 0 in all other cases.

1. The diagram shows an AC circuit with two voltage sources of same frequency. Find out the value of current I shown in the fig.



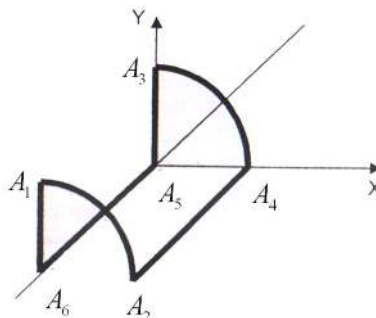
A) $I = \frac{V_0 \cos(\omega t)}{R}$

B) $I = \frac{V_0 \cos(\omega t + \pi/2)}{R}$

C) $I = \frac{V_0 \cos \omega t}{R}$

D) $I = \frac{-3V_0 \cos \omega t}{R}$

2. A time varying magnetic field $B = B_0 t \hat{k}$ is confined in a cylindrical region, cutting the XY plane on a circle of radius $x^2 + y^2 = 4$. We have placed a wire frame as shown. Segment A_1A_2 and A_3A_4 are identical quarter circles. The net emf induced in the wire frame is equal to



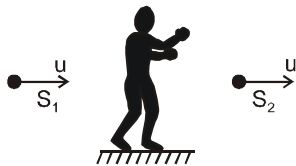
A) zero

B) $2B_0$

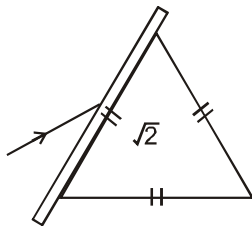
C) $4B_0$

D) B_0

3. Two sources S_1 and S_2 of same frequency f emits sound. The sources are moving as shown with speed u each. A stationary observer hears that sound. The beat frequency is (v = velocity of sound)



- A) $\frac{2 u^2 f}{v^2 - u^2}$ B) $\frac{2 v^2 f}{v^2 - u^2}$ C) $\frac{2 u v f}{v^2 - u^2}$ D) $\frac{2u}{v} f$
4. A parallel glass slab of refractive index $\sqrt{3}$ is placed in contact with an equilateral prism of refractive index $\sqrt{2}$. A ray is incident on left surface of slab as shown. The slab and prism combination is surrounded by air. The magnitude of minimum possible deviation of this ray by slab-prism combination is



- A) 30° B) 45° C) 60° D) $60^\circ - \sin^{-1} \sqrt{\frac{2}{3}}$
5. A light weight particle of charge Q is fixed at one end of an electrically insulated uniform elastic rod of natural length L , cross-sectional area A and Young's modulus Y . The rod is placed in space having uniform electric field of magnitude E and directed parallel to length of the rod as shown. Neglecting gravity, the magnitude of extension in this rod is



- A) $\frac{QEL}{YA}$ B) $\frac{QEL}{2YA}$ C) $\frac{QEL}{4YA}$ D) None of these

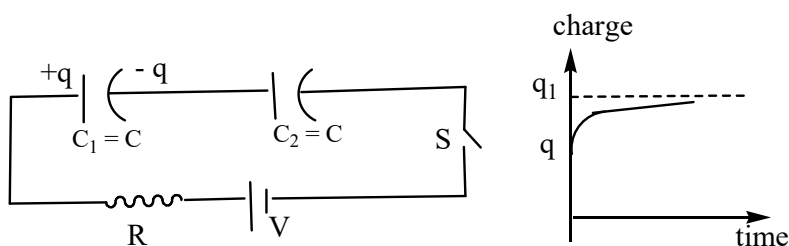
SECTION – II

(MULTIPLE CORRECT CHOICE TYPE)

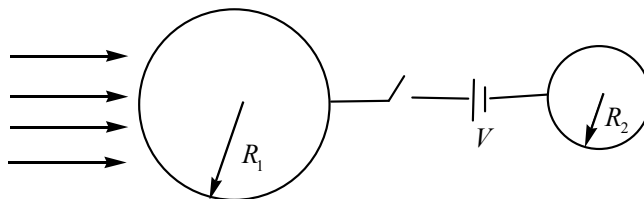
This section contains **5 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/ are correct.

Marking scheme +4 for correct answer , 0 if not attempted and 0 in all other cases.

6. In the diagram shown, both the capacitors have the same capacitance C . One capacitor has charge $+q$ and other is uncharged. The switch is closed at time $t = 0$. The graph shows the variation of the charge on C_1 as a function of time. Then



- A) Initial value of current in the circuit is $\frac{|V - \frac{q}{C}|}{R}$
- B) Charge present in second capacitor in steady state is $\frac{CV - q}{2}$
- C) Value of q_1 in steady state is $\frac{CV + q}{2}$
- D) Heat energy will be generated in the circuit on closing the switch
7. Two conducting uncharged spheres of radius R_1 and R_2 ($R_1 > R_2$) are connected to a battery with a switch as shown. Light rays of frequency f are incident on the bigger sphere and simultaneously the switch is closed work function of bigger sphere is ϕ . After some time the charge on bigger sphere becomes q_1 , and on smaller sphere becomes $-q_2$ remains constant there after. Then



A) $\frac{q_1}{4\pi\epsilon_0 R_1} + \frac{q_2}{4\pi\epsilon_0 R_2} = V$

B) The number of electrons emitted by larger sphere is $\frac{q_1 - q_2}{e}$ when e is the charge of an electron

C) $\frac{eq_1}{4\pi\epsilon_0 R_1} = hf - \phi$

D) The number of electrons emitted by larger sphere is independent of the potential of the battery

8. A proton of charge ' e ' and mass ' m ' enters a uniform and constant magnetic field $\vec{B} = B \hat{i}$ with an initial velocity $\vec{V} = V_{ox}\hat{i} + V_{oy}\hat{j}$. Which of the following will be correct at any later time ' t ' :

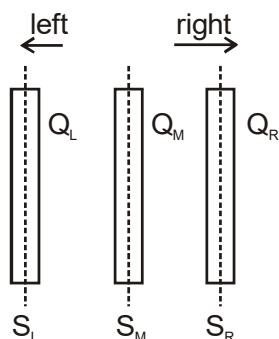
A) x - component of velocity of the proton will be $V_{ox} \cos\left(\frac{qBt}{m}\right)$

B) y - component of velocity will be $V_{oy} \cos\left(\frac{qBt}{m}\right)$

C) z-component of velocity will be $V_{oy} \sin\left(\frac{qBt}{m}\right)$ along $+\hat{k}$ direction .

D) x-component of velocity will remains unchanged.

9. Three large identical conducting plates of area A are closely placed parallel to each other as shown (the area A is perpendicular to plane of diagram). The net charge on left, middle and right plates are Q_L , Q_M and Q_R respectively. Three infinitely large parallel surfaces S_L , S_M and S_R are drawn passing through middle of each plate such that surfaces are perpendicular to plane of diagram as shown. Then pick up the correct option(s).



- A) The net charge on left side of surface S_L is equal to net charge on right side of surface S_R .
- B) The net charge on left side of surface S_L is equal to net charge on right side of surface S_M .
- C) The net charge on left side of surface S_L is equal to net charge on right side of surface S_L .
- D) The net charge on right side of surface S_L is equal to net charge on left side of surface S_R .
10. If the tension in a string is changed by 21 per cent, the fundamental frequency of the string changes by 15 Hz. Which of the following statements be correct?
- A) The original fundamental frequency is nearly 143 Hz
- B) The velocity of propagation changes nearly by 4.5%
- C) The velocity of propagation changes by 10.5%
- D) The fundamental wavelength changes nearly by 10%

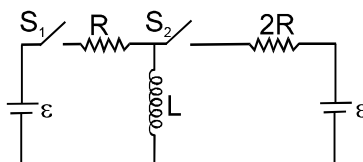
SECTION III

(Integer Answer Type)

This section contains **5 questions**. The answer to each question is single digit integer, ranging from 0 to 9 (both inclusive). The correct digit below the question number in the ORS is to be bubbled).

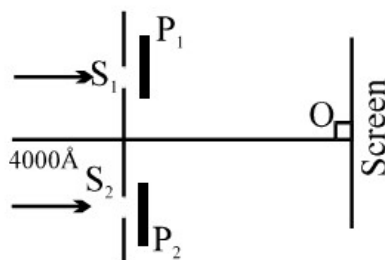
Marking scheme +4 for correct answer , 0 if not attempted and 0 in all other cases.

11. A sample has two isotopes A and B having masses 50g and 30g respectively. A is radioactive and B is stable. A decays to A' by emitting α particles. The half life of A is 2hrs. Mass numbers of A and A' are 150 and 146 respectively. In 4 hour, if the number of α particles emitted is $\frac{N_A}{n}$ where N_A = Avogadro's number then $n = \underline{\hspace{1cm}}$.
12. In the circuit shown S_1 and S_2 are switches. S_2 remains closed for a long time and S_1 open. Now S_1 is also closed. It is given that $R = 10 \Omega$, $L = 1 \text{ H}$ and $\varepsilon = 3\text{V}$. Just after S_1 is closed, find the magnitude of rate of change of current (in ampere/sec.) that is $\frac{di}{dt}$, in the inductor L .

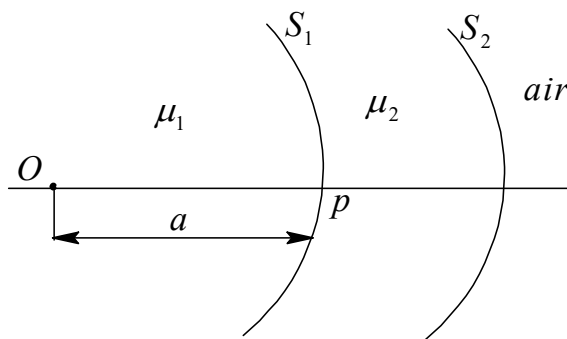


13. P_1, P_2 are transparent plates having equal thickness $20 \mu\text{m}$ and refractive indices $\mu_1 = 1.55$, $\mu_2 = 1.5$. S_1, S_2 are identical slits. The plate P_1 transmits 100% whereas

P_2 transmits 25% of energy incident. Without P_1 and P_2 , intensity at O is I. The intensity at O after placing P_1 and P_2 is $\frac{I}{4N}$. Then 'N' is



14. A point object 'O' is placed in a medium of refractive index $\mu_1 = 1.4$. S_1 and S_2 are two concentric spherical surfaces of radii 1m and 2m. To the right of 'O' contains a medium of refractive index $\mu_2 = 1.5$ between the interfaces S_1 and S_2 . Find the object distance of O from S_1 (in meter) so that an image of 'O' as seen by observer from air, coincides with O.



15. A particle of positive charge 1 mC and mass 1gm is projected from origin with velocity 20 m/s towards +ve x – axis in a non – uniform magnetic field $\vec{B} = -b_0 x \hat{k}$, where $b_0 = \frac{1}{10} T/m$ and x is meter. If the maximum positive x co-ordinate of the particle during its motion is $4x_0$ (in m) then x_0 is (Neglect gravity and air drag)

CHEMISTRY SECTION – I

(SINGLE CORRECT ANSWER TYPE)

This section contains 05 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +2 for correct answer, 0 if not attempted and 0 in all other cases.

16. Which of the following is incorrect statement?

- A) If a solution is prepared by dissolving excess of AgNO_3 solution in KI solution, the charge likely to develop on colloidal particle is positive.
- B) Increase in molecular weight of sodium alkyl sulphate surfactant in water, its CMC decreases.
- C) The molar conductivity of a solution of surfactant decreases sharply at the CMC.
- D) During micelle formation ΔH is positive and ΔS is negative.
17. The following are some statements related to VA group hydrides. Incorrect statement is:
- A) Reducing property increases from NH_3 to BiH_3 .
- B) Tendency to donate lone pair increases from NH_3 to BiH_3 .
- C) Thermal stability of hydrides decreases from NH_3 to BiH_3 .
- D) Bond angle of hydrides decreases from NH_3 to BiH_3 .
18. Which statement is correct for the following complexes?
1. $[\text{Co}(\text{en})_2 \text{Cl}_2]^+$, 2. $[\text{Co}(\text{NH}_3)_3 \text{Cl}_3]$, 3. $[\text{Co}(\text{H}_2\text{O})_4 \text{Cl}_2] \text{Cl}$.
- A) 1, 2 complexes can show both geometrical and optical isomerism
- B) 1, 2 complexes can show geometrical and optical isomerism
- C) 1, 2, 3 complexes can show geometrical isomerism and optical isomerism is shown by cis forms of 2 and 3.
- D) 1, 2, 3 complexes can show geometrical isomerism, cis form of complex 1 shows optical isomerism.
19. Which of the following statement is correct when mixture of chloride salt of a metal and $\text{K}_2\text{Cr}_2\text{O}_7$ is gently warmed with conc. H_2SO_4 ?
- A) A deep red vapour is evolved in which the oxidation state of central atom is +4
- B) The reaction is not a redox reaction
- C) chloride of only Hg, Pb, Ag, Sb, Sn can give this test
- D) when yellow solution is treated with lead acetate, a red precipitate is formed.
20. How many unit cells are present in a cube-shaped ideal crystal of NaCl of mass 1.00 g [Atomic masses: Na = 23. Cl = 35.5]
- A) 2.57×10^{21} unit cells
- B) 5.14×10^{21} unit cells
- C) 1.28×10^{21} unit cells
- D) 1.71×10^{21} unit cells

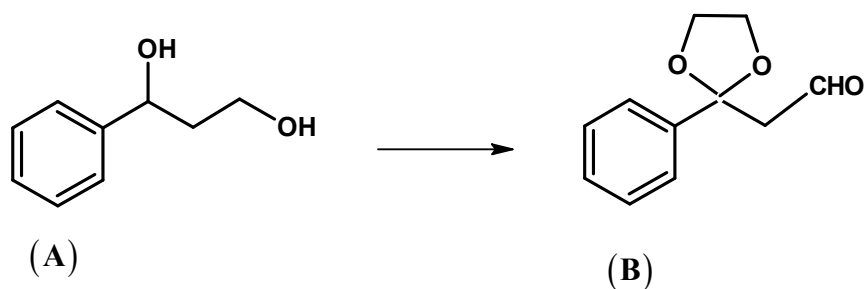
SECTION – II

(MULTIPLE CORRECT CHOICE TYPE)

This section contains 5 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/ are correct.

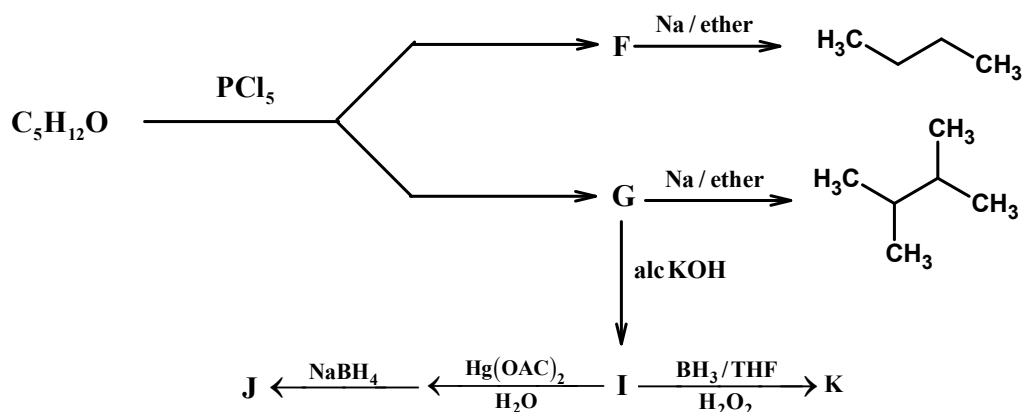
Marking scheme +4 for correct answer , 0 if not attempted and 0 in all other cases.

21. For the following three step conversion of A to B, the appropriate sequence of reaction is _____

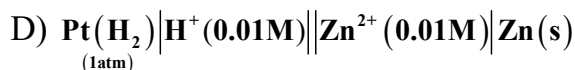
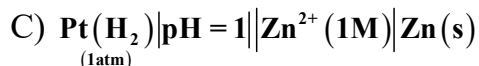
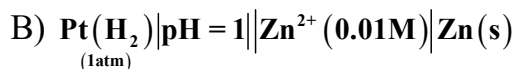


- A) MnO_2 ; $(\text{CH}_2\text{OH})_2$ / HCl ; PCC B) PCC ; MnO_2 ; $(\text{CH}_2\text{OH})_2$ / HCl
- C) PCC ; $(\text{CH}_2\text{OH})_2$ / HCl ; Jones's reagent D) Jones's reagent; $(\text{CH}_2\text{OH})_2$ / HCl ; MnO_2
22. Which of the following chemical method for purification is/are correctly matched?
- A) Cupellation — silver metal extraction
- B) Van arkel method — Ti metal extraction
- C) McArthur Forest method — lead metal extraction
- D) Mond process — Nickel metal extraction.

23.



- A) I is $\text{CH}_3-\text{CH}=\text{CH}_2$ B) J is $\text{CH}_3-\text{CH}(\text{OH})-\text{CH}_3$
- C) F is $\text{CH}_3\text{CH}_2-\text{Cl}$ D) K is $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{OH}$
24. In which of the following cells: $E_{\text{cell}} = E_{\text{cell}}^0$?



25. The correct statements are

A) Both anomers of D-glucopyranose can be crystallized and purified.

B) When pure α -D-glucose dissolved in aqueous solution, Its optical rotation Value changes to $+52.7^\circ$ approximately.

C) PHBV is biodegradable polymer.

D) Novolac is a linear polymer

SECTION III (Integer Answer Type)

This section contains **5 questions**. The answer to each question is single digit integer, ranging from 0 to 9 (both inclusive).

The correct digit below the question number in the ORS is to be bubbled).

Marking scheme +4 for correct answer , 0 if not attempted and 0 in all other cases.

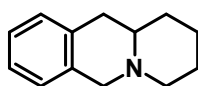
26. Phosphorescence emission of acetone (0.05M) in acetonitrile at 20°C was measured at 450 nm. The average life time of triplet state of acetone is $x \times 10$ microseconds. (on plotting the graph $\log I$ vs time the slope is $-0.72 \times 10^4 \text{ s}^{-1}$, I = intensity) what is the value of 'x'. This emission follows first order kinetics.

27. The conductivity of 0.1 N NaOH solution is 0.022 scm^{-1} when equal volume of 0.1N HCl solution is added, the conductivity of resultant solution is decreased to 0.005 scm^{-1} . The equivalent conductivity in $\text{scm}^2 \text{ equivalent}^{-1}$ of NaCl solution is $x \times 10^2$. The x value

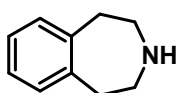
28. At 25°C vapour pressure of a volatile liquid A is 60 torr. To 4 mol of A, x mol of another volatile liquid B is added at 25°C . The vapour pressure of the resulting solution was found to be 72 torr and mole fraction of A in the vapour phase was $\frac{1}{3}$.

The value of x is

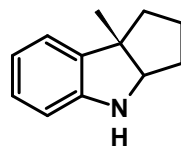
29. For some of the following amines (A through H), repetitive exhaustive methylation, followed by Hofmann elimination removes the nitrogen atom, for others this is not possible. In how many of the following nitrogen can't be removed?



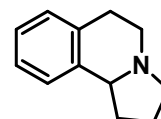
(A)



(B)



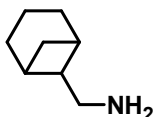
(C)



(D)



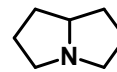
(E)



(F)

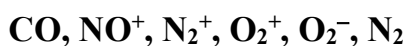


(G)



(H)

30. Copper sulphate when dissolved in excess of KCN forms a pseudohalogen dimer. The pseudo halide ion of this pseudo halogen is isoelectronic with how many of following chemical species.



MATHEMATICS

SECTION - I

(SINGLE CORRECT ANSWER TYPE)

This section contains 05 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +2 for correct answer, 0 if not attempted and 0 in all other cases.

31. If $\frac{2}{1!13!} + \frac{2}{3!11!} + \frac{2}{5!9!} + \frac{1}{7!7!} = \frac{2^m}{n!}$ then find the value of $m + n$.
- A) 25 B) 27 C) 29 D) 26
32. $[x + 0.19] + [x + 0.20] + \dots + [x + 0.91] = 546$, (where $[.]$ denotes the greatest integer function). Then $[100x]$ is
- A) 740 B) 741 C) 742 D) 743
33. Inside a semi-circle of unit radius, two circle of radii r_1 and r_2 are drawn, touching the diameter and the semi-circle and also touching each other externally $\text{Max } (r_1 + r_2) =$
- A) $\sqrt{2} + 1$ B) $\sqrt{2} - 1$ C) $2(\sqrt{2} - 1)$ D) $2(\sqrt{2} + 1)$
34. $\int_0^{100} [e^{\sin \pi x}] dx$, where $[.]$ denotes greatest integer function.
- A) $200 \left(1 - \frac{\sin^{-1} \ln 2}{\pi} \right)$ B) $100 \left(2 - \frac{\sin^{-1} \ln 2}{\pi} \right)$ C) $200 \left(1 + \frac{\sin^{-1} \ln 2}{\pi} \right)$ D) $100 \left(1 + \frac{\sin^{-1} \ln 2}{\pi} \right)$

35. Let $f(x) = \frac{x}{1 + (\ln x)(\ln x) \dots \dots \infty} \forall x \in [1, \infty)$. Find $\int_1^{2e} f(x) dx$.

A) $\frac{e-1}{2}$

B) $\frac{e+1}{2}$

C) $\frac{e^2+1}{2}$

D) $\frac{e^2-1}{2}$

SECTION – II

(MULTIPLE CORRECT CHOICE TYPE)

This section contains **5 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/ are correct.

Marking scheme +4 for correct answer , 0 if not attempted and 0 in all other cases.

36. One coin and a die are rolled for five times. Probability for 3 heads and 2 tails turn up along with the sum 26 on the dice is $N/12^5$ then factors of N are

A) 2

B) 3

C) 5

D) 7

37. Solution of D.E $y(2xy+1)dx + x(1+2xy+x^2y^2)dy = 0$ is $a \ln|y| = c + \frac{b}{xy} + \frac{1}{2(xy)^2}$.

Then which of the following is/are correct? (c is an arbitrary constant)

A) $a=1$

B) $b=2$

C) $a^2 + b^2 = 5$

D) $a^2 - b^2 = 5$

38. A circle touches the parabola $y^2 = 4ax$ ($a > 0$) at P. It also passes through the focus S of the parabola and intersects its axis at Q. if angle SPQ is 90° . Then which of the following is/are correct?

A) abscissa of centre of the circle $= 10a$ B) ordinate of centre of the circle $= 0$

C) radius of the circle $= 4a$

D) circle is passing through origin

39. Number of words can be formed using all the letters of the word 'HIPHIPHURRAY' in which all H's lie somewhere between R's is $(abc)(d!)$ where abc is a 3-digit number and d is a maximum possible prime number then which of the following is/are correct?

A) $a + b = 10$

B) $b + c = 17$

C) $b + d - a = 15$

D) $a + d = 8$

40. A quadratic equation with roots m and n satisfies the equation $\frac{m}{m-1} + \frac{n}{n-1} = \frac{a^2-7}{a^2-4}$. Given

that $m, n \in (1, 4)$, $mn = 4$ ($a \in \mathbb{R}$), which of the following is/are incorrect?

A) The number of integral values of a is 2

B) The least positive value of a equals $\sqrt{3}$

C) The maximum integral value of a is 4

D) The minimum integral value a is -3

SECTION III

(Integer Answer Type)

This section contains **5 questions**. The answer to each question is single digit integer, ranging from 0 to 9 (both inclusive). The correct digit below the question number in the ORS is to be bubbled).

Marking scheme +4 for correct answer , 0 if not attempted and 0 in all other cases.

41. Let $f(n), g(n)$ and $h(n)$ be polynomial in n and let

$$\Delta_r = \begin{vmatrix} r^2 \cdot {}^nC_r & n(n+1)2^{n-2} & f(n) \\ 6r(n-r+1) & n(n+1)(n+2) & g(n) \\ 4(5^{r-1}) & 5^n - 1 & h(n) \end{vmatrix}, \quad f(n) = \sum_{r=1}^n \Delta_r \quad \text{then } f\left(\frac{100}{100}\right) =$$

42. Let z_1, z_2, z_3 be the complex numbers such that $|z_1 - 1| = |z_2 - 1| = |z_3 - 1|$ and $\arg\left(\frac{z_3 - z_1}{z_2 - z_1}\right) = \frac{\pi}{6}$, then $z_2(z_2 - 1) - z_3(z_2 + 1) + (z_3 + 1)(z_3 - 1) = s$ find $|s|$
43. Area of triangle formed by common tangents to parabola $y^2 = 8x$ and hyperbola $\frac{x^2}{1} - \frac{y^2}{3} = 1$ and directrix of the given parabola is Δ , then (2Δ) is
44. Two different packs of card are shuffled together. Cards are distributed equally among 4 players, each getting 13 card. Number of ways a player can get his cards if no two cards are from the same suit and with the same denomination is N. Max value of k for which 3^k that can divide N
45. Two queens, placed at random on a chess board, the probability that they are not in clashing position is p then $\left[\frac{3}{p}\right] =$ (queens can move in row, column and diagonally and $[.]$ is GIF)

KEY SHEET

1	D	2	A	3	C	4	A	5	B
6	ABCD	7	ABCD	8	BD	9	ABCD	10	AC
11	4	12	1	13	4	14	6	15	5
16	D	17	B	18	D	19	B	20	A
21	ABC	22	ABD	23	ABCD	24	AB	25	ABCD
26	6	27	1	28	6	29	2	30	3
31	B	32	D	33	C	34	B	35	D
36	ACD	37	ABC	38	BC	39	ABCD	40	ACD
41	0	42	2	43	9	44	2	45	4



Sec: Sr. IIT

Date: 20-02-24

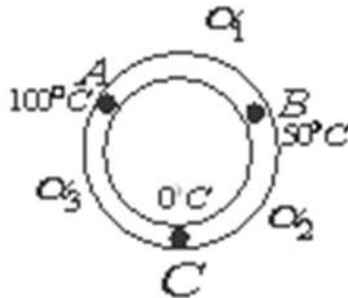
Race -4

PHYSICS**SECTION-1****(ONE OR MORE OPTIONS CORRECT TYPE)**

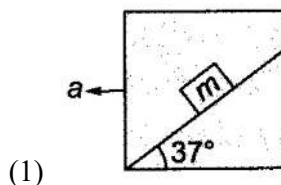
This section contains 10 multiple choice questions. Each question has four choices (A) (B), (C) and (D) out of which **ONE** or **MORE THAN ONE** are correct.

Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases.

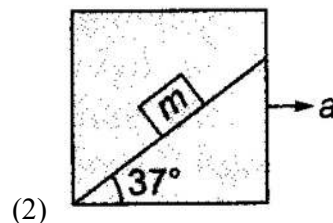
1. A circular ring of uniform cross section is made up of three different metallic rods AB, BC and CA (joined together at the points A, B and C) of thermal conductivities α_1, α_2 and α_3 , respectively. The junctions A, B and C are maintained at the temperatures 100°C , 50°C and 0°C , respectively. All the rods are of equal lengths and cross sections. Under steady state conditions, assume that no heat is lost from the sides of the rods. Let Q_1, Q_2 and Q_3 be the rates of transmission of heat along the three rods AB, BC and CA. Then



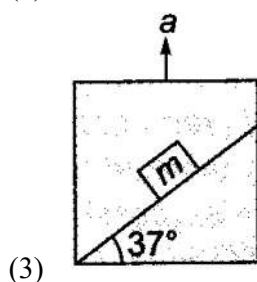
- A) $Q_1 = Q_2 = Q_3$ and all are transmitted in the clockwise sense
 B) Q_1 and Q_2 flow in clockwise sense and Q_3 in the anticlockwise sense.
 C) $Q_1 : Q_2 : Q_3 :: \alpha_1 : \alpha_2 : 2\alpha_3$
 D) $\frac{Q_1}{\alpha_1} = \frac{Q_2}{\alpha_2} + \frac{Q_3}{\alpha_3}$
2. If N_1, N_2, N_3 and N_4 are the normal reaction forces and b_1, b_2, b_3 and b_4 are the accelerations with which the block slides on the smooth fixed wedge in the above situations respectively, then



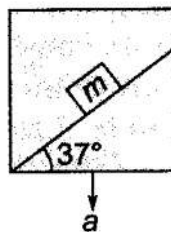
(1)



(2)



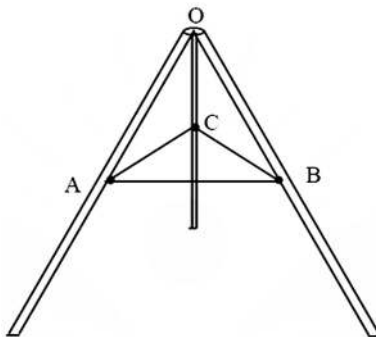
(3)



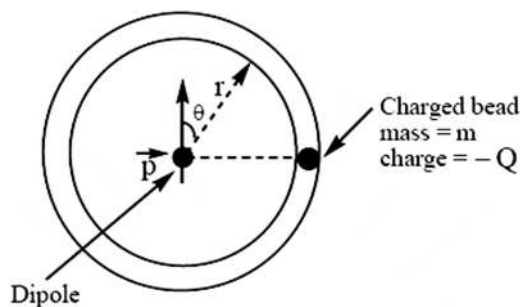
(4)

- A) $N_3 > N_1 > N_2 > N_4$ B) $N_4 > N_3 > N_1 > N_2$
 C) $b_2 > b_3 > b_4 > b_1$ D) $b_2 > b_3 > b_1 > b_4$

3. A ladder is constructed with three identical uniform rods each of mass m and length l , with their ends hinged to a light, smooth rigid ring of negligible size. The ring lies in horizontal plane. Three identical light strings, each of length $l/\sqrt{3}$ is connected to the rods as shown, between AB, BC and CA, at a distance $2l/3$ from the top most point O ($OA = OC = OB = 2l/3$). If the ladder lies on smooth horizontal surface



- A) The tension in each string is $\frac{mg}{4}$
 B) The tension in each string is $\frac{mg}{12}$
 C) The normal force exerted by ground on any rod is mg
 D) The normal force exerted by ground on any rod is $\sqrt{3} mg$
4. A small charged bead can slide on a circular frictionless, insulating fixed wire frame. A point like dipole is fixed at the centre of circle, dipole moment is $\vec{P}$. Initially the bead is on the plane of symmetry of the dipole. Bead is released from rest. Ignore the effect of gravity. Select the correct options:



- A) Magnitude of velocity of bead as function of its angular position is $\sqrt{\frac{QP \cos \theta}{2\pi\epsilon_0 m r^2}}$
 B) Normal force exerted by the wire on bead is zero
 C) Bead would move along a circular path until it reaches the position opposite to its starting position and then executes periodic motion
 D) If the wire frame were not present bead executes circular motion and returns to initial point after tracing a complete circle
5. A simple circuit that can be used for multi-meter (with A and B as terminals) is shown in figure-1. It consists of a galvanometer coil G of resistance 48Ω and two resistors $R = 952 \Omega$ and $r = 2 \Omega$ and two switches k_1 and k_2 as shown in figure-1. It can be used as ammeter or voltmeter. Figure-2 shows another circuit with battery of emf $10V$ and internal resistance $r_0 = 2 \Omega$.

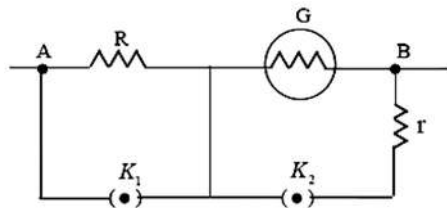


Figure-1

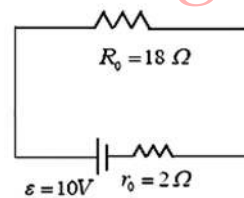


Figure-2

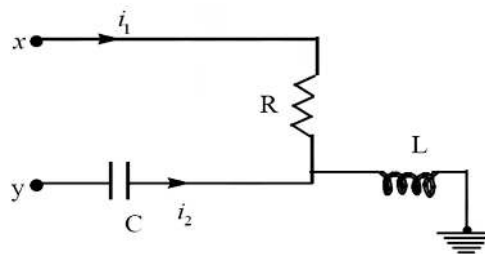
A student uses this multi-meter to measure current in the circuit shown in figure-2 and potential difference across R_0 .

- A) Current measured by the student is $\frac{50}{105}$ A
 B) Voltage measured by the student is 9V
 C) Current measured by the student is $\frac{500}{1096}$ A
 D) Voltage measured by the student is $\frac{18 \times 10^4}{20036}$ V

6. In a resonance column experiment initially tube is completely filled with water. Reading for 1st resonance and 2nd resonance is obtained as 15.2 cm and 46.0 cm respectively.

- A) Diameter of tube used is $\frac{2}{3}$ cm
 B) Diameter of tube used is $\frac{4}{3}$ cm
 C) Reading for 3rd resonance will be 77 cm
 D) Reading for 3rd resonance will be 75.6 cm

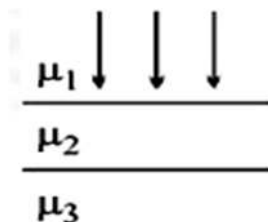
7. Figure shows a part of A.C. network. Current in upper and lower branch $i_1 = 5 \sin\left(100\pi t + \frac{\pi}{4}\right)$ and $i_2 = 5 \sin\left(100\pi t - \frac{\pi}{4}\right)$. Resistance of resistor $R = 10\Omega$, Capacitance $C = \frac{1000}{\pi} \mu F$ and inductance of inductor $L = \frac{50}{\pi} mH$.



- A) The potential difference $V_{xy} = 0V$
 B) The potential difference $V_{xy} = 100 \sin\left(100\pi t + \frac{\pi}{4}\right)$
 C) Net rate of heat loss across given part of network is 250W
 D) Net rate of heat loss across given part of network is 125W
8. For L – R circuit, the time constant is equal to
- A) Twice the ratio of the energy stored in the magnetic field to the rate of the dissipation of energy in the resistance
 B) The ratio of the energy stored in the magnetic field to the rate of dissipation of energy in the resistance
 C) Half of the ratio of the energy stored in the magnetic field to the rate of dissipation of energy in the resistance

D) Square of the ratio of the energy stored in the magnetic field to the rate of dissipation energy in the resistance.

9. Consider a case of thin film interference as shown. Thickness of film is equal to wavelength of light in μ_2 .



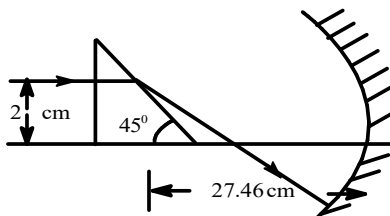
- A) Reflected light will be maxima if $\mu_1 < \mu_2 < \mu_3$
 B) Reflected light will be maxima if $\mu_1 < \mu_2 > \mu_3$
 C) Transmitted light will be maxima if $\mu_1 > \mu_2 > \mu_3$
 D) Transmitted light will be maxima if $\mu_1 > \mu_2 < \mu_3$
10. A solid spherical ball of radius r made of a material of a given density ρ is dropped in a viscous liquid of coefficient viscosity η and density ' σ '. Then choose the correct option(s):
 A) Linear momentum of the ball after achieving terminal velocity is proportional to r^2
 B) Instantaneous power developed by the drag force is proportional to the r^5
 C) After steady state, angular momentum of the ball about the centre of the ball is proportional to r^3
 D) Power developed by the force of buoyancy is proportional to the r^5

SECTION-II INTEGER TYPE

This section contains 10 questions. The answer to each question is a **single digit integer, ranging from 0 to 9** (both inclusive).

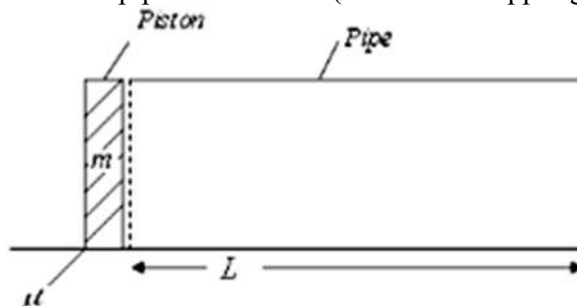
Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases.

11. Force acting on a particle executing oscillations along x-axis about origin is given as $F = -kx^n$, where k is a positive constant and x is the position. The energy (E) of the particle depends upon mass (m) of the particle, amplitude of oscillation (A) and k . If $E \propto A^4$, then the value of n
12. A police Cruiser, approaching a right-angled intersection from the north, is chasing a speeding car that has turned around corner and is now moving straight east. When the Cruiser is 0.6 mile north of the intersection and the car is 0.8 mile to the east, the police with radar determine the distance between them and the car is increasing at 20 mph. If the cruiser is moving at 60 mph at the instant of measurement, what is the speed (as multiple of 10mph) of the car?
13. A ray of light falls normally on a right angled prism whose $\mu = \sqrt{1.5}$ and prism angle 45° . After passing through prism, it falls on a concave mirror which is at a distance of 27.46 cm from the point of emergence of prism. The deviation of the ray produced at the mirror is $n \times 30^\circ$, (if focal length of the mirror is 10 cm is) then value of n is
 (Take $\sqrt{3} = 1.73$)

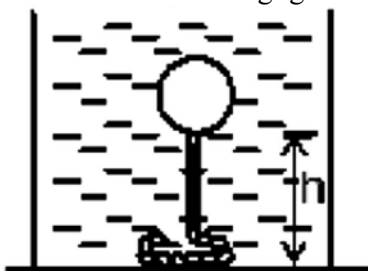


14. A piston of mass $m = 0.005$ kg and area of cross of section $S = 0.1$ m² is placed very close to one end of an open organ pipe of length $L = 5\pi$ m and area of cross section S . At the other end, a tuning fork of frequency f produces a travelling sound wave of displacement amplitude A , which excites the pipe into second overtone. The coefficient of friction between piston and ground is $\mu = 0.1$. Find the

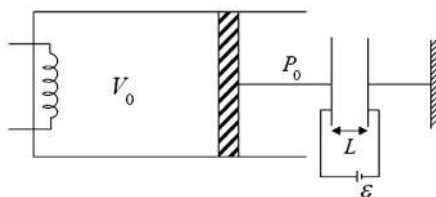
value of A (in micrometer) at which the piston is about to shift towards left from the pipe. Given bulk modulus of medium in the pipe is 10^4 N/m^2 . (Assume no toppling)



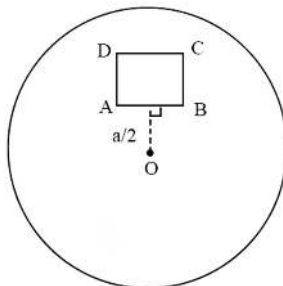
15. One end of a uniform long thick iron chain of linear mass density λ is fixed to a sphere of mass m and specific density $1/3$ while the other end is free. The value of $\frac{m}{\lambda} = \frac{15}{7}$ metres. The sphere along with the chain is immersed in a deep lake. The specific density of iron is 7. Find the height h (in metres) above the bed of the lake at which the sphere will float in equilibrium (Assume that the part of the chain lying on the bottom of the lake exerts negligible force on the upper part of the chain).



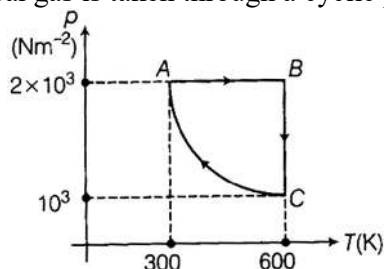
16. An ideal gas is enclosed in a cylinder fitted with a frictionless piston. The piston is connected with a light rod to one plate of capacitor whose other plate is fixed as shown. Initially volume of the gas inside the cylinder is V_0 . Atmospheric pressure is P_0 , separation between the plates is L , area of the piston as well as of the capacitor plates is A and emf of battery is $\mathcal{E}$. A heater supplies heat to the gas so that final pressure of the gas is given as $P = P_0 - \frac{n\epsilon_0\mathcal{E}^2}{L^2}$, when piston is displaced by a distance $\frac{L}{2}$. Find the value of n .



17. Figure shows cross-sectional view of a solenoid. There exists uniform time dependent magnetic field $B_t = at^2 + bt$ along the axis of solenoid. A uniform square ($a \times a$) conducting frame is kept in solenoid keeping its plane perpendicular to axis of solenoid. The side of square closest to axis of solenoid is at a distance $a/2$ from axis. Find the ratio of magnitude of the potential difference between C and D (V_{CD}) to the potential difference between A and B (V_{AB})? (Resistance per unit length of square frame is uniform).



18. Two moles of a monatomic ideal gas is taken through a cyclic process as shown.

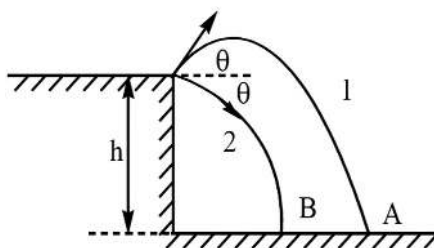


Process CA follows $PT = \text{constant}$. If cycle efficiency is $1 - \frac{21}{12\ln 2 + 3x}$, then find the value of x ?

19. A vernier calliper has 1 mm marks on the main scale. It has 20 equal divisions on the vernier scale which match with 16 main scale divisions. For this vernier callipers, the least count is $\frac{1}{x}(\text{mm})$.

Find $x = ?$

20. Two balls are thrown with the same speed $v_0 = 10 \text{ m/s}$ from the top of a cliff. The angle of their initial velocities is $\theta = 15^\circ$ above and below the horizontal, as shown in figure. The two balls hit the ground at the points 'A' and 'B'. The distance AB (in m) is? Given height of cliff = 50m, $g = 10 \text{ m/s}^2$.



CHEMISTRY

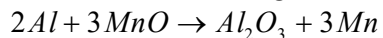
SECTION-1

(ONE OR MORE OPTIONS CORRECT TYPE)

This section contains 10 multiple choice equations. Each question has four choices (A) (B),(C) and (D) out of which **ONE** or **MORE THAN ONE** are correct.

Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases.

21. Which of the following statement is/are correct? The following reaction occurs:



108g of Al and 213g of MnO was heated to initiate the reaction.

(Mwt of MnO = 71, at wt of Al = 27)

A) Al is present in excess

B) MnO is present is excess

C) 54g of Al is required

D) 159 g of MnO is in excess

22. Select correct statements

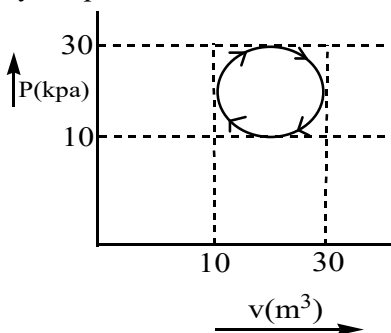
A) NaClO₄ is water soluble but KClO₄ is not

B) Na₃ [Co(NO₂)₆] is water soluble but K₃[Co(NO₂)₆] is not

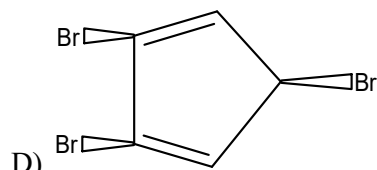
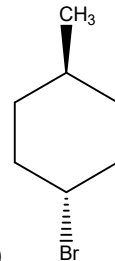
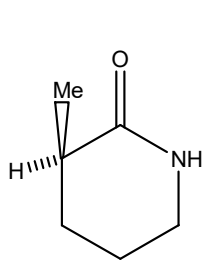
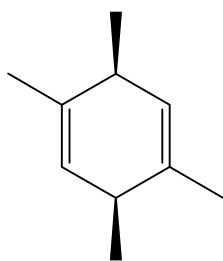
C) Na₂CO₃ is water soluble but CaCO₃ is not

D) NaCl is water soluble but CaCl₂ is not

23. Which of the following statements is/are correct?
 A) The cell constant of an electrolytic cell is measured as the product; $[k.R]$ rather using (l/a)
 B) As an electrolytic solution diluted, its conductance, equivalent conductance, molar conductance increases
 C) Kohlrausch's law may be applied at any concentration of the electrolyte
 D) Kohlrausch's law may be applied to calculated molar conductance at infinite dilution for both weak and strong electrolytes.
24. The carbon based reduction method is not used for the extraction of
 A) tin from SnO_2
 B) Fe from Fe_2O_3
 C) Al from Al_2O_3
 D) Mg from $\text{MgCO}_3 \cdot \text{CaCO}_3$
25. Select the incorrect order for given properties
 A) $\text{LiH} > \text{NaH} > \text{KH} > \text{RbH}$ (Thermal stability)
 B) $\text{CdCO}_3 > \text{CaCO}_3 > \text{K}_2\text{CO}_3$ (Covalent character)
 C) $\text{BeCO}_3 > \text{MgCO}_3 > \text{CaCO}_3$ (ease of oxide formation on heating)
 D) $\text{BeF}_2 > \text{MgF}_2 > \text{CaF}_2 > \text{SrF}_2$ (Solubility in water)
26. Which one is not correct for a cyclic process as shown in the figure?



- A) $du=0$ B) $q = -W$ C) $w = 314 \text{ J}$ D) 31.4 J
27. $\text{B}_3\text{N}_3\text{H}_6 + \text{solution of Hydrochloric acid} \rightarrow ?$
 Select incorrect about above reaction
 A) No reaction
 B) $\text{B}_3\text{N}_3\text{H}_6$ show substitution reaction and produce $\text{B}_3\text{N}_3\text{Cl}_6$
 C) $\text{B}_3\text{N}_3\text{H}_6$ show addition reaction and produce $\text{B}_3\text{N}_3\text{H}_9\text{Cl}_3$ in which Cl is bonded to boron
 D) $\text{B}_3\text{N}_3\text{H}_6$ show addition reaction and produce $\text{B}_3\text{N}_3\text{H}_9\text{Cl}_3$ in which Cl is bonded to nitrogen
28. Which of the following compounds is chiral?



29. Acetamide is treated with the following reagents separately. Which one of these yield methyl amine.
 A) PCl_5 B) $\text{NaOH} + \text{Br}_2$ C) Sodalime D) Con. H_2SO_4
30. Which of the following statements are correct?
 A) 4-chlorophenol is soluble in aq. NaOH but 4-chlorobenzaldehyde is not soluble in aq. NaOH .
 B) 4-Methyl benzoic acid is soluble in NaHCO_3 but 4-methyl phenol is not soluble in NaHCO_3
 C) 2,4,6- trinitrophenol is stronger acid than 4-methyl phenol
 D) Difference between O-cresol and anisole can be identified by aq. NaOH

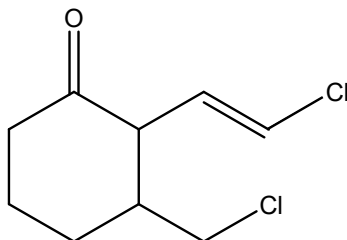
SECTION-II

INTEGER TYPE

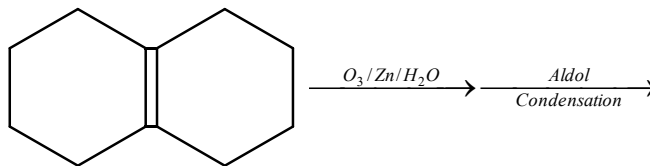
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Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases.

31. The oxidation number of Mn in the product of alkaline oxidative Fusion of MnO_2 is.....
32. The Co-ordination number of Al in the crystalline state of AlCl_3 is.....
33. What is the degeneracy of the level of the hydrogen atom that has the energy $-\frac{R_H}{9}$?
34. Calculate the total number of stereoisomers when alkene having trans configuration



35. Stachyose + water $\xrightarrow[\text{enzyme}]{H^+}$ 'X' (monosaccharide molecules), X =
36. 'P' gram of urea is dissolve in 500g of water. If the difference in boiling point and freezing point of the solution is 100.46k. What is the 'P' value?
 $\left(\text{for water } K_b = 0.5 \frac{K \text{ kg}}{\text{Mole}}, K_f = 1.8 \frac{K \text{ kg}}{\text{Mole}} \right)$
37. A person takes 6.1 g of an anti-acid tablet comprising bicarbonate ion at 20.8%. The volume of CO_2 evolved at (1 atm, 25°C) in the stomach (on neutralization) multiplied by a factor of '10' will be x L. Calculate the approximate value of 'x'?
38. (i) $\text{Ag}_2\text{S} + \text{NaCN} \rightarrow [\text{Ag}(\text{CN})_x]^{\pm n}$
 (ii) $\text{Au} + \text{NaCN} + \text{O}_2 \xrightarrow{\text{H}_2\text{O}} [\text{Au}(\text{CN})_y]^{\pm n}$
 (iii) $[\text{Ag}(\text{CN})_x]^{\pm n} / [\text{Au}(\text{CN})_y]^{\pm n} \xrightarrow{\text{Zn}} \text{Ag} / \text{Au} + [\text{Zn}(\text{CN})_z]^{\pm m}$
 What is the value of $x + y + z$
39. How many ions are obtained by ionization of iron (III) hexacyanoferrate (II)
40. In the scheme given below, the total number of intramolecular aldol condensation product formed from 'y' is.....



MATHS

SECTION-1

(ONE OR MORE OPTIONS CORRECT TYPE)

This section contains 10 multiple choice equations. Each question has four choices (A) (B),(C) and (D) out of which **ONE** or **MORE THAN ONE** are correct.

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41. $f: \mathbb{R} \rightarrow \mathbb{R}$, $g: \mathbb{R} \rightarrow [0, \infty)$ be two functions such that $f''(x)$ is continuous and $f(x) + f''(x) + x g(x) f'(x) = 0$
 A) there exists M such that $f(x) \leq M \forall x \in \mathbb{R}$
 B) there exists m such that $f(x) \geq m \forall x \in \mathbb{R}$

- C) Let $f(0) = 2, f'(0) = 4$ then, $|f(x)| \leq 5 \quad \forall x \in \mathbb{R}$
 D) Let $f(0) = 3, f'(0) = 4$ then, $|f(x)| \leq 6 \quad \forall x \in \mathbb{R}$
42. Let $f''(x)$ be a continuous function such that $f(2x) = (e^x + 1)f(x) \quad \forall x \in \mathbb{R}$.
 Then the value of $f(0) + f'(0) + f''(0)$ must be
 A) 2 B) $2f'(0)$ C) $2f(0)$ D) $2f''(0)$
43. α, β are roots of $3x^2 + ax + b = 0, (\alpha \neq \beta)$
 Let $f(t) = \lim_{x \rightarrow t} (1 + \tan(3x^2 + ax + b))^{\frac{1}{x-t}}$ then which of the following is/are INCORRECT
 A) $f(\alpha) = f(\beta)$
 B) $f(\alpha) \cdot f(\beta) = 1$
 C) f has only two elements in domain
 D) f has only two elements in Range
44. A hyperbola with focus at $S(6,8)$ and directrices $3x + 4y + 10 = 0$ and $3x + 4y - 10 = 0$ has centre C , then which of the following is/are CORRECT?
 A) $CS = 15$ B) eccentricity = $\sqrt{2}$
 C) eccentricity = $\sqrt{5}$ D) $CS = 8$
45. Circle S_1, S_2, S_3 with radii $r_1 > r_2 > r_3$ are such that $S_i - S_j = 0$ represents same line $\forall i \neq j$. S_1 lies on one side of that line & S_2, S_3 on another side. Line passing through centers $O_1 O_2 O_3$ of circles S_1, S_2, S_3 intersects the circles S_1, S_2, S_3 at $A_1, B_1; A_2, B_2; A_3, B_3$ respectively and the line $S_i - S_j$ at C , then which of the following is/are CORRECT?
 A) $\max(\min(CA_1, CB_1), \min(CA_2, CB_2), \min(CA_3, CB_3))$ is CA_1 or CB_1
 B) $\max(\min(CA_1, CB_1), \min(CA_2, CB_2), \min(CA_3, CB_3))$ is CA_3 or CB_3
 C) $\max(\max(CA_1, CB_1), \max(CA_2, CB_2), \max(CA_3, CB_3))$ is CA_3 or CB_3
 D) $\max(\max(CA_1, CB_1), \max(CA_2, CB_2), \max(CA_3, CB_3))$ is CA_1 or CB_1
46. Let A be set of distinct points B_1, B_2, B_3, B_4 in space, $A = A_0$ and $A_n = \{P: P \text{ lies in segment } QR \text{ (including } Q \text{ and } R), Q, R \in A_{n-1}\}$. Then which of the following is/are true.
 A) If B_1, B_2, B_3, B_4 are co-planar, then $A_2 = A_3$
 B) for all A , $A_3 = A_4$
 C) for all A , $A_2 = A_3$
 D) If $B_1 B_2 B_3 B_4$ is a tetrahedron $A_1 = A_2$
47. n badminton players participate in a tournament, each player plays with every other player once. If there are no ties or draws and a_k, b_k are number of games won, lost by k^{th} player respectively, then
 A) $\sum a_k = \sum b_k$ B) $\sum a_k = \frac{n(n-1)}{2}$ C) $\sum a_k^2 = \sum b_k^2$ D) $\sum \frac{1}{a_k} = \sum \frac{1}{b_k}$
48. Which of the following functions defined in $[0, 1] \rightarrow \mathbb{R}$ satisfy $\left(\frac{1}{x} \int_0^x f(x) dx\right)^2 = f(0)f(x)$
 $\forall x \in (0, 1]$
 A) $\frac{3}{(x-1)^2}$ B) $\frac{3}{(x-2)^2}$ C) $-\frac{5}{(x-3)^2}$ D) $\frac{6}{(2x-1)^2}$

49. Distinct points P, Q, R, S in xy plane are such that every circle passing through P, Q has at least one point in common with every circle passing through R, S. Then
 A) P, Q, R, S are collinear
 B) P, Q, R, S satisfy equation $ax^2 + ay^2 + bx + cy + d = 0$ for some $a, b, c, d \in \mathbb{R}$ (not all zero)
 C) If P, Q, R are given, S can be determined uniquely.
 D) P, Q, R, S must lie on an ellipse.
50. Which of the following statements are true ($\text{Arg } z \in (-\pi, \pi] \forall z \in \text{complex numbers}$)
 A) Only one complex number $z = a + ib$ where $a, b \in \text{Integer}$ satisfy $\text{Arg}\left(\frac{z-1}{z-i}\right) = \frac{\pi}{4}$
 B) Only one complex numbers $z = a + ib$ where $a, b \in \text{Integer}$ satisfy $\text{Arg}(z-1) - \text{Arg}(z-i) = \frac{7\pi}{4}$
 C) Two complex numbers $z = a + ib$ where $a, b \in \text{Integer}$ satisfy $\text{Arg}\left(\frac{z-1}{z-i}\right) = \frac{\pi}{4}$
 D) Two complex number $z = a + ib$ where $a, b \in \text{Integer}$ satisfy $\text{Arg}(z-1) - \text{Arg}(z-i) = \frac{\pi}{4}$

SECTION-II INTEGER TYPE

This section contains 10 questions. The answer to each question is a **single digit integer, ranging** from 0 to 9 (both inclusive).

Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases.

51. Given $T_0 = 0, T_{n+1} = 1 + \sin(T_n - 1)$, $\lim_{n \rightarrow \infty} \frac{\sum_{r=0}^n T_r}{n} = b$ then $\left[\left(\frac{dy}{dx} \right)_{x=b} \right]$ for curve $y = \ln \tan(x+y) - x$ is.....
52. Let $f(x)$ be a polynomial function such that $f \circ g \equiv g \circ f \forall$ polynomial functions $g, h(n)$ be defined as highest exponent of 2 in n , $P(n) = h(1) + h(2) + h(3) + \dots + h(n)$ and $2b = \sum_{n=2}^{\infty} e^{-(P \circ f)(n)}$ then $\int_0^b [(e-1)x] dx = \dots$, (where $[.]$ is GIF)
53. $a_1, a_2, a_3, \dots, a_n, \dots$ are positive real numbers such that $\frac{1}{a_1} + \frac{1}{a_2} + \frac{1}{a_3} + \dots + \frac{1}{a_n} = k$ the minimum value of $\lim_{n \rightarrow \infty} \frac{a_1 + a_2 + a_3 + \dots + a_n}{n} = h$ and $\lim_{n \rightarrow \infty} \frac{a_1}{a_2} + \frac{a_2}{a_3} + \dots + \frac{a_{n-1}}{a_n} = k$ the minimum value of hk is
54. $2-d, 2, 2+d$ are roots of equation, $x^3 - ax^2 + bx - c = 0$ the maximum value of $b+c-9$ is _____ ($d \in \mathbb{N}$)
55. Let $A = \{1, 2, 3, \dots, 11\}$, P is power set of A , Q is largest subset of P such that any two members of Q have atleast one element in common. The sum of last two digits (i.e., unit digit and 10s place digit) in $n(Q)$ is
56. Three integral sides of a right angled triangle are a, b, c such that $a < b < c$ and $ab = 4(a+b+c)$. If $c = d(m^2 + n^2)$; a, b are $2dmn, d(m^2 - n^2)$ [$d, m, n \in \mathbb{Z}$], then number of such triangle(s) is.....

57. A bag contains 6 fair six face dice $D_1, D_2, \dots, D_6$. D_i has i white faces and $(6-i)$ black faces. A dice is selected at random and rolled, (assuming only the top face is visible) shows white. The probability that all faces adjacent to the white face are black is $\frac{p}{q}$ (p, q are coprimes), the unit's digit of q is
58. Let L be locus of midpoint of focal chords of the ellipse $\frac{x^2}{9} + \frac{y^2}{25} = 1$. The integer nearest to the area enclosed by curve L is
59. A 2×2 matrix B commutes with $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. ($AB = BA$). If trace of the matrix B is 6 then $|B| = \underline{\hspace{2cm}}$
60. A is the region of the complex plane, $A = \{z: z/4 \text{ and } 4/\bar{z} \text{ have real and imaginary part in } (0,1)\}$, then $[p]$ (where p is the area of the region A and $[.]$ denotes the greatest integer function) is

1	BC	2	AC	3	AC	4	ABC	5	CD
6	AC	7	BD	8	A	9	AD	10	BD
11	3	12	7	13	6	14	5	15	5
16	2	17	1	18	5	19	5	20	5
21	AC	22	ABC	23	ABD	24	CD	25	D
26	ABC	27	ABD	28	AB	29	B	30	ABCD
31	6	32	6	33	9	34	4	35	4
36	6	37	7	38	8	39	5	40	1
41	ABCD	42	BD	43	ACD	44	C	45	BD
46	ABC	47	ABC	48	BC	49	B	50	BCD
51	1	52	0	53	1	54	8	55	6
56	3	57	5	58	8	59	9	60	5



Sec: Sr. IIT

Race -5

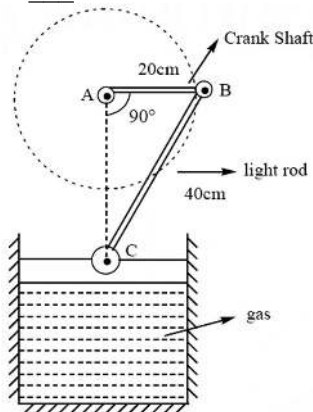
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PHYSICS
SECTION - I
(SINGLE CORRECT ANSWER TYPE)

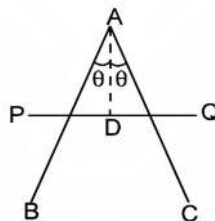
This section contains 10 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

- An aeroplane is flying up in geographic meridian at an angle of 30° with the horizontal (north) and wind is blowing from west. A package is dropped from the aeroplane. Find the velocity of the wind if package hits a kite flying in the space with a position vector $\vec{R} = (400\sqrt{3}\hat{i} + 80\hat{j} + 200\hat{k})$ m with respect to the point of dropping. (Here $\hat{i}$ and $\hat{j}$ are the unit vectors along north and vertically up respectively and $\hat{k}$ be the unit vector due east. Assume the bag is light enough to get carried away by the wind):
 A) 50 m/sec B) 25 m/sec C) 20 m/sec D) 10 m/sec
- A light rod of length 40cm is hinged between (BC) light smooth piston and crank shaft. Crank shaft is free to rotate about A as shown. At the instant shown pressure of gas inside the cylinder is $5P_0$ where $P_0 (=10^5 \text{ Pa})$ is the atmospheric pressure. The area of piston is 20 cm^2 . At the given moment, the torque acting on shaft about A is ____

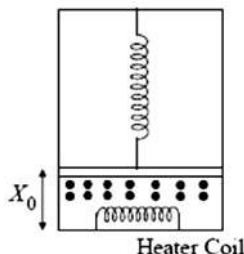


- A) 120 Nm B) 160 Nm C) 200 Nm D) 240 Nm
- In the figure shown AB, AC and PQ are thin, smooth, rigid wires. AB and AC are joined at A and fixed in a vertical plane. $\angle BAC = 2\theta = 90^\circ$ and line AD is angular bisector of angle BAC. A liquid of surface tension $T = 0.025 \text{ N/m}$ forms a thin film in the triangle formed by intersection of the wires AB, AC and PQ. In the figure shown the uniform wire PQ of mass 1 gm is horizontal and in equilibrium under the action of surface tension and gravitational force. Find the time period of SHM of PQ in vertical plane for small displacement from its mean position.



- A) $\frac{\pi}{10}s$ B) $\frac{\pi}{5}s$ C) $\frac{2\pi}{5}s$ D) $\frac{\pi}{2}s$

4. Certain amount of oxygen is contained in a cylindrical vessel shown. The system is initially (as shown) in equilibrium with gas pressure equal to p_0 and initial compression in spring equal to x_0 . Now, heat is slowly supplied to the gas so that gas pressure becomes $8p_0$. The spring force F is related to compression x in the spring as $F = -kx^3$, where k is a constant. What fraction of heat supplied goes into increasing in spring energy?



(Vessel with insulated walls with Frictionless mass less insulated piston)

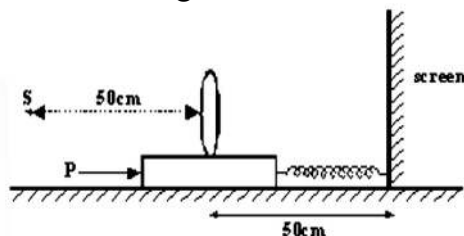
- A) $\frac{1}{11}$ B) $\frac{10}{11}$ C) $\frac{8}{9}$ D) $\frac{1}{9}$
5. A particle of charge $+q$ and mass m moving under the influence of a uniform electric field $E\hat{i}$ and a magnetic field $B\hat{k}$ enters in 1st quadrant of a coordinate system at a point $(0, a)$ with initial velocity $v\hat{i}$ and leaves the quadrant at a point $(2a, 0)$ with velocity $-2v\hat{j}$. Rate of work done by both the fields at $(2a, 0)$
- A) Zero B) $\frac{3mv^3}{4a}$ C) $\frac{3mv^3}{2a}$ D) $\frac{mv^3}{4a}$

SECTION - II (PARAGRAPH TYPE)

This section contains **3 Paragraph of questions**. Each paragraph has 2 multiple choice questions based on a paragraph. Each question has 4 choices A), B), C) and D) for its answer, out of which **ONLY ONE IS** correct.
Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

Paragraph for Question Nos. 11 & 12

A point object 'S' is located at a distance 100 cm from the screen. A lens of focal length 23 cm, mounted on a movable frictionless stand is kept between the source and screen. The stand is attached to a massless spring of natural length 50 cm and spring constant 800 N/m as shown in the figure. Mass of the stand with lens is 2 kg.



11. An impulse P is imparted to the stand so that a real image of the object S is formed on the screen after a fixed time gap. (Image is formed four times in an oscillation). This time gap is:

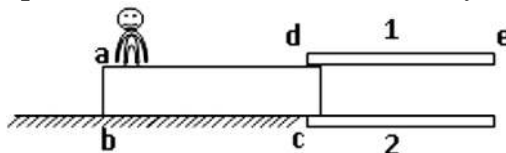
- A) $\frac{\pi}{10}$ sec B) $\frac{\pi}{20}$ sec C) $\frac{\pi}{30}$ sec D) $\frac{\pi}{40}$ sec

12. Amplitude of oscillation so that real image of the object is formed on the screen after a fixed time gap. (Image is formed 4 times in one oscillation)

- A) 25 cm B) 30 cm C) 20 cm D) 10 cm

Paragraph for Question Nos. 13 & 14

The diagram shows two capacitor plates 1 and 2 lying horizontally. The plates are squares of side L and separated by a distance d . The plates are fixed rigidly and the potential difference between them is maintained V . A dielectric slab $abcd$ of mass m and dielectric constant 2 is on a smooth horizontal surface which can just fill the gap between the plates completely without friction. In the condition shown the slab is slightly inserted into the gap between the plates. An insect of mass m is on the slab at a as shown. The insect and the horizontal surface do not affect the charge distribution. Initially the system is held at rest and at time $t=0$ the system is released, simultaneously the insect starts running towards d such that slab is at rest. Upon reaching the end d the insect jumps into air such that it lands at the far end e of the plate 1. During the jump the insect's horizontal velocity remains same.



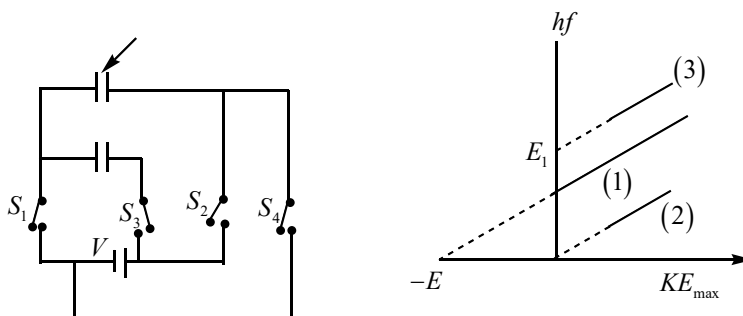
13. The time taken by the insect to move from a to d on the dielectric slab is:

- A) $\left[\frac{4dm}{\epsilon_0 V^2} \right]^{1/2}$ B) $\left[\frac{2dm}{\epsilon_0 V^2} \right]^{1/2}$ C) $\left[\frac{dm}{\epsilon_0 V^2} \right]^{1/2}$ D) $\left[\frac{dm}{2\epsilon_0 V^2} \right]^{1/2}$

14. The time for which the insect will be in air during the jump is

- A) $\left[\frac{4dm}{\epsilon_0 V^2} \right]^{1/2}$ B) $\left[\frac{2dm}{\epsilon_0 V^2} \right]^{1/2}$ C) $\left[\frac{dm}{\epsilon_0 V^2} \right]^{1/2}$ D) $\left[\frac{dm}{2\epsilon_0 V^2} \right]^{1/2}$

Paragraph for Question Nos. 15 & 16



The circuit shown is placed in vacuum. Both the capacitors are identical and they have the same capacitance C . Light is incident on the left plate of the upper capacitor. When all the switches are open then the hf versus $KE_{\max}$ is shown by the straight line (1). In all the cases, we are measuring the $KE_{\max}$ when the electron reaches the opposite plate.

When only S_1 and S_2 are closed, the graph becomes (2).

When only S_3 and S_4 are closed, the graph becomes (3).

15. What is the value of eV ?
 A) E B) E_1 C) $E + E_1$ D) $E - E_1$
16. What is the value of E_1 ?
 A) $3E$ B) $3E/2$ C) $E/2$ D) $2E$

SECTION - III**(Matching List Type)**

This section contains four questions, each having two matching lists (List-I & List-II). The options for the correct match are provided as (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

17. Three holes A, B and C are made at depths 1m, 2m and 5m as shown. Total height of liquid in the container is 8m. Let v is the speed with which liquid comes out of the hole and R is the range on ground. Match the following two columns, for the liquid coming out of holes A, B and C



Column – I		Column-II	
A	V is maximum for	P	hole A
B	V is minimum for	Q	hole B
C	R is maximum for	R	hole C
D	R is minimum for	S	For a hole made half way between A and B

- A) A-P; B-R, C-P; D-S
 B) A-S; B-Q, C-R; D-S
 C) A-R; B-Q, C-R; D-P
 D) A-R; B-P, C-R; D-P
18. A wire of length l , area of cross section A and Young's modulus of elasticity Y is stretched by a longitudinal force F . The change in length is Δl . Match the following two columns. (Only one of the parameters in column-I is changed at a time)

Column – I		Column-II	
A	F is increased	P	Δl will increase
B	l is increased	Q	Stress will increase
C	A is increased	R	Δl will decrease
D	Y is increased	S	Stress will decrease

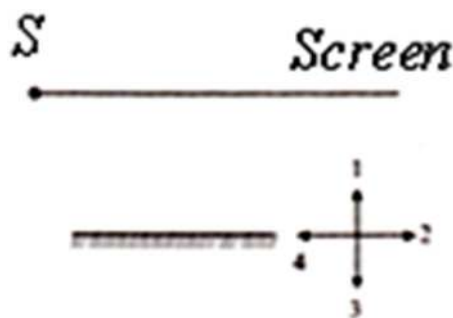
A) A-PQ; B-P, C-RS; D-R

B) A-PS; B-Q, C-RS; D-R

C) A-PQ; B-P, C-P; D-S

D) A-PS; B-S, C-RS; D-S

19. A plane mirror is arranged parallel to the screen and a point source S is kept on the screen as shown in figure. The mirror is moved towards the screen. Corresponding to the direction of motion of the mirror match the column (consider the moment the motion starts)



COLUMN-I	COLUMN-II
A) Direction of motion of mirror along 1	P) length of spot increases
B) Direction of motion of mirror along 2	Q) length of spot decreases
C) Direction of motion of mirror along 3	R) length of spot does not change
D) Direction of motion of mirror along 4	S) brightness of spot increases

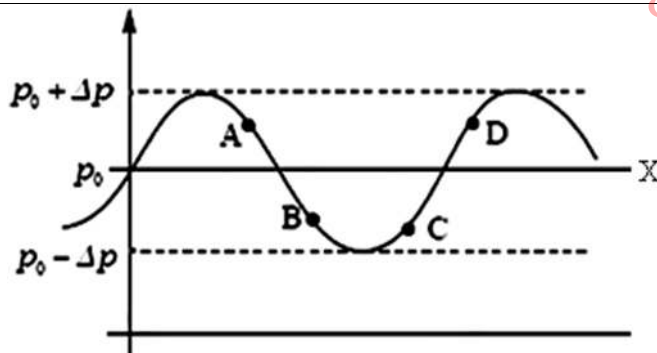
A) A-PS, B-R, C-P, D-RS

B) A-RS, B-R, C-R, D-RS

C) A-PS, B-P, C-R, D-PS

D) A-RS, B-P, C-R, D-QS

20. In a long organ pipe of small cross-section longitudinal progressive wave is propagating in positive X-direction. Figure shows the instantaneous pressure variation in the organ pipe. 'A', 'B', 'C' and 'D' are fluid elements in the pipe. Instantaneously, (X is the position of particle on X-axis)

**Column-I**

A) Element A

B) Element B

C) Element C

D) Element D

A) A-QS,B-PS,C-QR,D-PR

C) A-PR,B-QR,C-PS,D-QS

Column-II

P) Speed is increasing

Q) Speed is decreasing

R) Element is present left to its equilibrium position

S) Element is present right to its equilibrium position

B) A-PR,B-QR,C-QS,D-PS

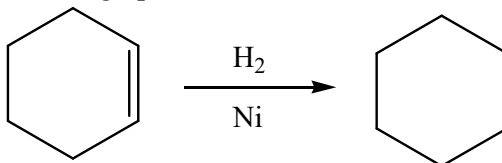
D) A-PS,B-QS,C-QR,D-PR

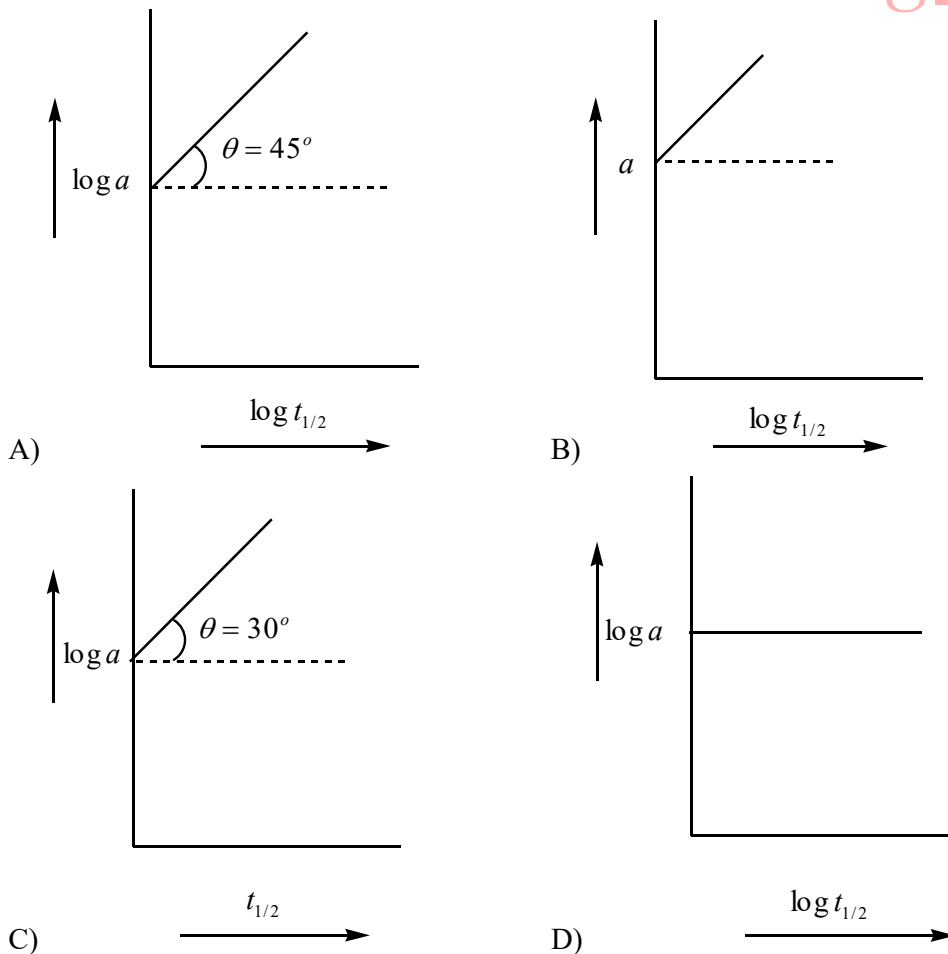
CHEMISTRY**SECTION – I****(SINGLE CORRECT ANSWER TYPE)**

This section contains 10 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

21. In a AB_xL_y (sp^3d hybridization of 'A') have "T" shape geometry. What will be the geometry of molecule. If 'A' form AB_yL_x type molecule? (B = Bond, L – Lone pair)
 A) 'T' shape B) sea - saw C) T.B.P D) Linear
22. Only gases remains after 12g of carbon is treated with 49.26 Lit of air at 300 K and 2atm pressure (assume 20% by volume O_2 , N_2 and 1% CO_2)
 ΔH_f for CO & CO_2 are – 26 Kcal, - 94 Kcal.
 The amount of heat evolved under constant pressure is.....
 A) -96.8 Kcal B) -66.8 Kcal C) -23.4 Kcal D) -33.4 Kcal
23. Which of the following is correct graph for the reaction?

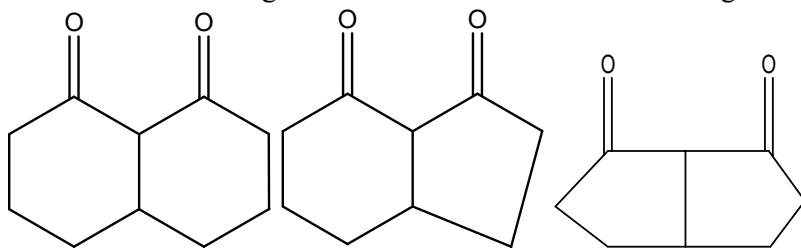




24. Fructose has a specific rotation of (-92.4°) . An aqueous solution of a fructose has an observed angle of rotation of (-27.7°) when placed in a polarimeter tube 10cm long. How much amount of fructose is dissolved in 100 ml of aqueous solution?

A) 0.299 g/ml B) 3.33 g/ml C) 2.99 g/ml D) 33.3 g/ml

25. The correct decreasing order in the enol content of following molecules is



(I) A) I > II > III (II) B) II > I > III (III) C) III > II > I D) II > III > I

26. Which of the following option is incorrect regarding XeF_6 ?

A) It acts as Lewis acid when it reacts with RbF
 B) It undergoes complete hydrolysis to give XeO_3
 C) It Fluorinates Silica (SiO_2) to give $XeOF_4$
 D) When it react with HCl gives HF, $XeCl_2$ & F_2

27. Which of the following statement is wrong?

A) Melting and Boiling points of H_2O_2 are greater than water
 B) Dipolemoment of H_2O_2 is greater than H_2O

- C) Dihedral angle of H_2O_2 in gas phase is more than solid phase
 D) H_2O_2 is strong oxidizing agent in acidic medium than in alkaline medium.
28. Combination of which parameter explain more oxidising strength of F_2 as compared to Cl_2 in water?
 A) Bond dissociation Enthalpy, electronegativity, hydration enthalpy
 B) Bond dissociation enthalpy, ionization enthalpy, hydration enthalpy
 C) Sublimation enthalpy, electronegativity, hydration enthalpy
 D) Bond dissociation enthalpy, electronegativity, hydration enthalpy
29. $(C_6H_{10})^P \xrightarrow[\text{ii) } NaBH_4 / \text{ethanol; iii) dil. acid}]{\text{i) dil. } H_2SO_4 / HgSO_4} Q \xrightarrow[\text{ii) } O_3 ; \text{ iii) } Zn / H_2O]{\text{i) con. } H_2SO_4} 2H_3C - \overset{\overset{O}{||}}{C} - CH_3$
 What is 'P'.....
 A) $CH_3CH_2CH_2CH_2 - C \equiv C - H$ B) $H_3CCH_2 - C \equiv C - CH_2CH_3$
 C) $\begin{array}{c} H_3C \diagdown \\ H - C - C \equiv C - CH_3 \\ H_3C \diagup \end{array}$ D) $\begin{array}{c} H_3C \diagdown \\ H_3C - C - C \equiv C - H \\ H_3C \diagup \end{array}$
30. The total number of alkenes possible by dehydrobromination of 3-bromo-3-cyclopentyl hexane using alcoholic. KOH is
 A) 4 B) 5 C) 6 D) 7

SECTION - II

(PARAGRAPH TYPE)

This section contains **3 Paragraph of questions**. Each paragraph has 2 multiple choice questions based on a paragraph. Each question has 4 choices A), B), C) and D) for its answer, out of which **ONLY ONE IS** correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

Paragraph for Question Nos. 31 & 32

Two students were given mixture of $Al^{+3}, Cr^{+3}, Fe^{+3}, Zn^{+2}, Co^{+2}$ and Mn^{+2} ions as their soluble salts which were not interfering radicals. They performed the experiment as follows.

STUDENT-I: In the aqueous solution of ions he added first NH_4Cl then NH_4OH and then passed H_2S gas. He got the precipitate (x).

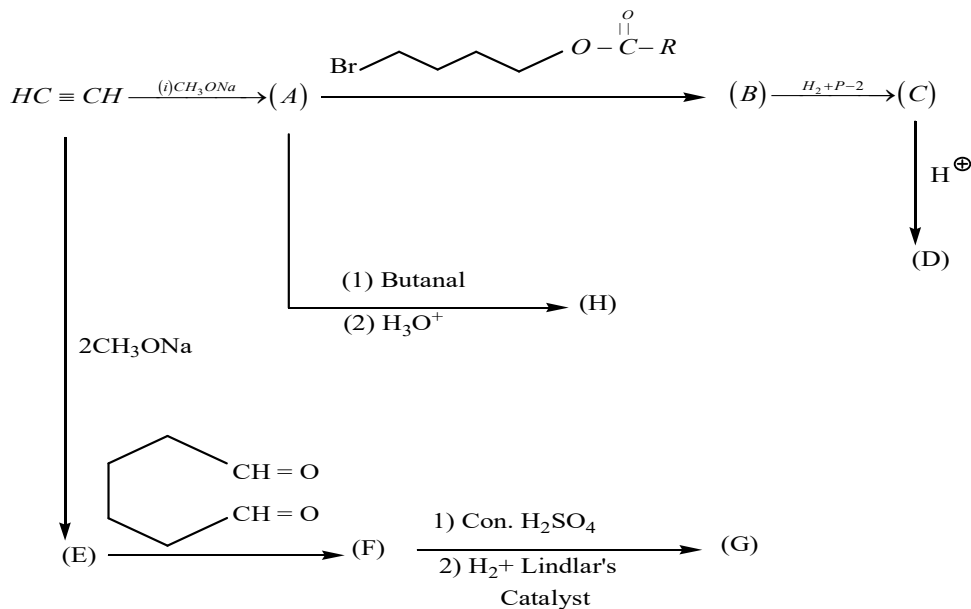
STUDENT-II: In the aqueous solution of ions he first added NH_4Cl and then NH_4OH . He got the precipitate (y) and a Filtrate. In Filtrate, he passed H_2S gas and again got a precipitate (z).

31. Which of the following is not the precipitate (x)?
 A) ZnS B) MnS C) $Cr(OH)_3$ D) Al_2S_3
32. Precipitate (y) when treated with aqueous Na_2O_2 solution, gives a yellow colour solution due to formation of which of the following?
 A) Na_2CrO_4 B) ZnO C) $Na_2Cr_2O_4$ D) $Fe(OH)_3$

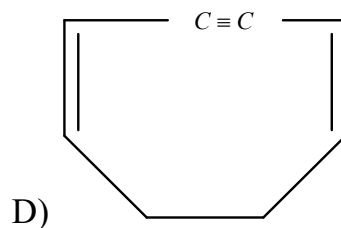
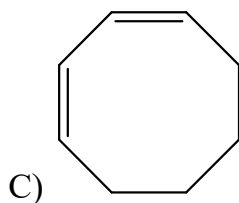
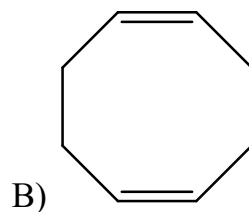
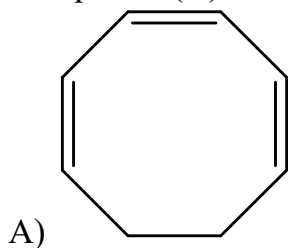
Paragraph for Question Nos. 33 & 34

A balloon of radius 3m is at a pressure of 1 atm and 25°C at sea level density of air is 1.22 kg/m^3 if air presents in balloon is replaced by hydrogen gas, the balloon lifts at sea level.

33. How many moles of Hydrogen are required to fill the balloon?
 A) $2.31 \times 10^3 \text{ mole}$ B) $4.62 \times 10^3 \text{ mole}$ C) $6.93 \times 10^3 \text{ mole}$ D) $9.24 \times 10^3 \text{ mole}$
34. What would be the payload of balloon?
 A) 128.7 kg B) 182.7 kg C) 62.4 kg D) 91.4 kg

Paragraph for Question Nos. 35 & 36

35. Compound (B) is....
 A) $\text{H}-\text{C} \equiv \text{C}-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CHO}$
 B) $\text{H}-\text{C} \equiv \text{C}-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{OH}$
 C) $\text{H}-\text{C} \equiv \text{C}-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{COOH}$
 D) $\text{H}-\text{C} \equiv \text{C}-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{OH}$
36. Compound (G) is



SECTION - III

(Matching List Type)

This section contains four questions, each having two matching lists (List-I & List-II). The options for the correct match are provided as (A), (B), (C) and (D) out of which ONLY ONE is correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

37. Match the following.

Column-I		Column II (Characteristics of complex ion compound)	
A)	$[Pt(NO_2)_2(en)_2]^{2+}$	P)	Stable according to E.A.N rule
B)	$[Cr(\pi-C_6H_6)(NO_2)]$	Q)	Ligand acts as ambidentate
C)	$[Ir(SCN)(SO_4)(NH_3)_4]$	R)	Bond order of Ligand decreases
D)	$[Cr(C_2H_4)(CO)_5]$	S)	Bond order of M – L bond > 1.0

A) A- (P), B- (PRS), C- (PQ), D-(PRS)

B) A- (R), B- (PRS), C- (QS), D-(PS)

C) A –(P), B-(RS), C-(Q), D-(PRS)

D) A-(S), B-(RS), C-(S), D-(RS)

38. Match the following:

Column-I		Column II	
A)	Sp^3	P)	Trimerisation of SO_3
B)	Sp^3d^2	Q)	Solidification of $PCl_{5(g)}$
C)	Sp	R)	Addition of $F^\ominus$ to F_3ClO
D)	No change in hybridisation	S)	Removal of $F^\ominus$ from $BeF_3^\ominus$
		T)	Solidification of N_2O_5

A) A-(PQ), B-(QR), C-(ST), D-(T)

B) A-(Q), B-(R), C-(ST), D-(T)

C) A-(ST), B-(R), C-(ST), D-(S)

D) A-(PQ), B-(R), C-(S), D- (P,Q)

39. Match the following:

Column-I		Column II	
A)	Simple cubic and FCC	P)	Have these cell parameters $a = b = c$, $\alpha = \beta = \gamma$
B)	Cubic and rhombohedral	Q)	Are two crystal systems
C)	Cubic and tetragonal	R)	Have only two crystallographic angles of 90°
D)	Hexagonal and Monoclinic	S)	Belongs to same crystal system

A) A-PS, B-PQ, C-QR, D-QR

B) A-PS, B- SQ, C-Q, D-SQ

C) A-PS, B-PQ, C-Q, D-QR

D) A-S, B-S, C-Q, D-PR

40. Match each of the compounds in Column-I with its characteristic reaction in Column-II

Column-I		Column II	
A)	$CH_3CH_2CH_2CN$	P)	Reduction with $pd-c/H_2$
B)	$CH_3CH_2O\overset{O}{\underset{ }{C}}CH_3$	Q)	Reduction with $SnCl_2/HCl$
C)	$CH_3-CH=CH-CH_2OH$	R)	Development of foul smell on treatment with chloroform and alcoholic. KOH
D)	$CH_3CH_2CH_2CH_2NH_2$	S)	Reduction with di isobutylaluminium hydride (DIBAL-H)
		T)	Alkaline hydrolysis

A) A- PQST, B-S, C-P, D- Q,R

B) A- PQST, B-ST, C-P, D-R

C) A- PQ, B-ST, C-R, D-ST

D) A-QST, B-ST, C-PR,D-QS

MATHEMATICS

SECTION – I

(SINGLE CORRECT ANSWER TYPE)

This section contains 10 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

41. Consider the binomial expansion $\left(\sqrt{x} + \frac{1}{2\sqrt[4]{x}}\right)^n$, $n \in \mathbb{N}$, $n > 1$, Arrange terms of expansion in decreasing powers of x . If the coefficients of the first three terms form an A.P., then in the binomial expansion, there are.....**number of distinct** terms of x with integral power

A) 1

B) 2

C) 3

D) 4

42. Value of $\int_0^1 (x^3 + x^5) \left(\frac{d^2}{dx^2} \ln(x + \sqrt{1+x^2}) \right)^2 dx$ is

A) $\frac{3}{4} \ln e$ B) $\ln \left(\frac{e^{\frac{3}{4}}}{2} \right)$ C) $\ln \left(\frac{e^{\frac{1}{4}}}{2} \right)$ D) $\ln 2$

43. The following integral $\int_0^{\frac{\pi}{4}} (2 \sec x)^{2019} dx$ is equal to

A) $\int_1^{\sqrt{2}+1} \left(t + \frac{1}{t} \right)^{2018} \frac{dt}{t}$ B) $\int_1^{\sqrt{2}-1} \left(t - \frac{1}{t} \right)^{2018} \frac{dt}{t}$ C) $\int_1^{\sqrt{2}+1} 2 \left(t - \frac{1}{t} \right)^{2018} \frac{dt}{t}$ D) $\int_1^{\sqrt{2}-1} 2 \left(t + \frac{1}{t} \right)^{2018} \frac{dt}{t}$

44. In triangle ABC, $\angle ABC = 120^\circ$, $AB = 3$ and $BC = 4$. If perpendiculars constructed on the side AB at A and to the side BC at C meet at D then CD is equal to
 A) 3 B) $\frac{8\sqrt{3}}{3}$ C) 5 D) $\frac{10\sqrt{3}}{3}$
45. Four boys and three girls stand in queue, then probability that number of boys ahead of every girl is atleast one more than the number of girls ahead of her, is
 A) $\frac{2}{3}$ B) $\frac{2}{5}$ C) $\frac{3}{4}$ D) $\frac{3}{7}$
46. Let f be any real valued function on real line with continuous third derivative. Let $h(x) = f(x)f'(x)f''(x)f'''(x)$. Which of the following must be true
 A) there must exist atleast one a such that $h(a) \leq 0$
 B) there must exist atleast one a such that $h(a) \geq 0$
 C) there exist no a such that $h(a) \geq 0$
 D) Nothing can be said
47. How many integers between 1000 and 9999 have exactly one pair of equal digit such as 4049 or 9902 but no 4449 or 4040?
 A) 4104 B) 3888 C) 4320 D) 3915
48. Let p be a point in the plane, let $d_1(p)$ and $d_2(p)$ be the distances of the point p from lines $3x - 4y = 0$ and $4x + 3y = 0$ respectively. Area of region R consisting of all points p lying in the first quadrant of the plane and satisfying $1 \leq d_1(p) + d_2(p) \leq 3$ is
 A) 2 B) 4 C) 6 D) 8
49. $f(x) = \begin{cases} x^2 \sin \frac{1}{x} & ; x \neq 0 \\ 0 & ; x = 0 \end{cases}$ and $g(x) = \begin{cases} \frac{|\sin x|}{\tan^{-1} x} & ; x \neq 0 \\ 1 & ; x = 0 \end{cases}$; which of the following is
CORRECT
 A) $y = f(x).g(x)$ is discontinuous B) $y = g(x)$ is discontinuous at $x = 1$
 C) $y = f(x).g(x)$ is differentiable at $x = 0$ D) $y = f(x).g(x)$ is differentiable $\forall x \in \mathbb{R}$
50. Assume that $M = \{(x, y) : x^2 + 2y^2 = 3\}$ and $N = \{(x, y) : y = mx + b\}$. If $M \cap N \neq \emptyset, \forall m \in \mathbb{R}$, then true set of values of b is?
 A) $\left[-\frac{\sqrt{6}}{2}, \frac{\sqrt{6}}{2}\right]$ B) $\left[\frac{-\sqrt{6}}{2}, \frac{\sqrt{6}}{4}\right]$ C) $\left[\frac{-2\sqrt{3}}{3}, \frac{-\sqrt{3}}{3}\right]$ D) $\left[\frac{-2\sqrt{3}}{3}, \frac{2\sqrt{3}}{3}\right]$

SECTION - II

(PARAGRAPH TYPE)

This section contains **3 Paragraph of questions**. Each paragraph has 2 multiple choice questions based on a paragraph. Each question has 4 choices A), B), C) and D) for its answer, out of which **ONLY ONE IS** correct.
Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

Paragraph for Question Nos. 51 & 52

In $\mathbb{R}^3$, Let L be straight a line passing through $(1, 1, 1)$.

Consider another line L_2 whose unsymmetrical form is $x + 2y + z = 1, x + y + 2z = 3$

51. Suppose that all points on L are at constant distance from two planes, used in L_2 , then equation of L is
- A) $\frac{x-1}{3} = \frac{y-1}{1} = \frac{z-1}{1}$ B) $\frac{x-1}{-3} = \frac{y-1}{1} = \frac{z-1}{1}$
 C) $\frac{x-1}{3} = \frac{y-1}{-1} = \frac{z-1}{1}$ D) $\frac{x-1}{3} = \frac{y-1}{-1} = \frac{z-1}{-1}$
52. Equation of L, if it is parallel to $x+y+z=1$ and intersect L_2
- A) $\frac{x-1}{4} = \frac{y-1}{-3} = \frac{z-1}{1}$ B) $\frac{x-1}{4} = \frac{y-1}{3} = \frac{z-1}{1}$
 C) $\frac{x-1}{+4} = \frac{y-1}{3} = \frac{z-1}{1}$ D) $\frac{x-5}{-4} = \frac{y+2}{3} = \frac{z}{1}$

Paragraph for Question Nos. 53 & 54

A player throwing a dice scores as many points as on the top face of the die and is to play until his score reaches or passes a total n. Let $p(n)$ the probability of making exactly the total n. By using this information answer following

53. Value of $p(3) + p(4)$
- A) $\frac{7^3}{6^4}$ B) $\frac{13 \times 7^2}{6^4}$ C) $\frac{7^2}{6^4}$ D) none
54. $\lim_{n \rightarrow \infty} p(n)$
- A) $\frac{1}{6}$ B) $\frac{2}{7}$ C) $\frac{1}{7}$ D) $\frac{1}{5}$

Paragraph for Question Nos. 55 & 56

$$\text{Let } P = \begin{bmatrix} 3 & -1 & -2 \\ 2 & \beta & \alpha \\ 3 & -5 & 0 \end{bmatrix}; \alpha, \beta \in R$$

$Q = [q_{ij}]_{3 \times 3}$ is a matrix such that $PQ = KI$

where $k \in R, k \neq 0$ and I is the identity matrix of order 3.

55. If $q_{23} = \frac{-k}{8}$ and $\det(Q) = \frac{k^2}{2}$, value of $\det(P \text{adj } Q)$ is, if $\beta = 0$
- A) 2 B) 2^5 C) 2^4 D) 2^9
56. If $\beta = 1, q_{23} = \frac{-k}{3}$, value of α is
- A) $\alpha = -\frac{1}{3}$ B) $\alpha = -\frac{14}{3}$ C) $\alpha = -\frac{7}{3}$ D) $\alpha = -\frac{5}{3}$

SECTION-3

(MATCHING LIST TYPE)

This section contains four questions, each having two matching lists (List-I & List-II). The options for the correct match are provided as (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

57. Match the following.

COLUMN - I		COLUMN - II	
P	If $x = \sin(2 \tan^{-1} 2)$; $y = \sin\left(\frac{1}{2} \tan^{-1} \frac{4}{3}\right)$ value of $x + y^2$ is	1	$\frac{1}{2}$
Q	Length of common chord of $2y^2 = 3(x+1)$, $x^2 + y^2 + 2x = 0$ is 'a', then value of $[a+1]$ is; $[.]$ denotes G.I.F	2	2
R	If $\vec{a}, \vec{b}, \vec{c}$ are non coplanar vectors such that $\vec{b} \times \vec{c} = \vec{a}$, $\vec{a} \times \vec{b} = \vec{c}$ $\vec{c} \times \vec{a} = \vec{b}$, value of $ \vec{a} + \vec{b} + (\vec{c})$ is	3	1
S	If $(1+x^2)\frac{dy}{dx} + 2xy - 1 = 0$, passes through origin, value of $y(1)$	4	3

A) P-3, Q-2, R-4, S-1

B) P-2, Q-1, R-3, S-4

C) P-3, Q-2, R-4, S-3

D) P-2, Q-1, R-4, S-3

58. Let $Z \in \mathbb{C}$ and satisfying

$$\begin{vmatrix} |Z-1|^2 & |Z+1|^2 & -4 \\ |Z+1|^2 & -4 & |Z-1|^2 \\ -4 & |Z-1|^2 & |Z-1|^2 \end{vmatrix} = 0$$

List - I		List - II	
P	Least value of $ Z-3i $ is	1	2
Q	Greatest value of $ Z+3i $ is	2	4
R	Least value of $ Z+i + Z-i $ is	3	5
S	If Z_1, Z_2 are the solutions of the given equation such that $1+2i$, Z_1 and Z_2 are collinear then $ Z_1-1-2i Z_2-1-2i $ is equal to	4	8

A) P-2, Q-1, R-3, S-4

B) P-1, Q-3, R-2, S-1

C) P-1, Q-2, R-1, S-2

D) P-3, Q-1, R-1, S-2

59. Match the columns

COLUMN-I		COLUMN-II	
P	In $\mathbb{R}^2$, Let $\sqrt{3}\hat{i} + \hat{j}$, $\hat{i} + \sqrt{3}\hat{j}$ and $\beta\hat{i} + (1-\beta)\hat{j}$ be position vectors of X, Y, Z with respect to the origin O, respectively If the distance of Z from the bisector of acute angle of $\overline{OX}$ and $\overline{OY}$ is $\frac{3}{\sqrt{2}}$, sum of possible value of $ \beta $ is	1	4
Q	Number of solution of the equation $2^x + x = 2^{\sin x} + \sin x$ in $[0, 10\pi]$ is	2	2

R	Let A and B are two non singular matrices of order 3 such that $A+B=I$ $A^{-1}+B^{-1}=2I$, then $ adj(2AB) +1$, is	3	1
S	If $\tan^{-1} \alpha = \tan^{-1}(\alpha + k) + \frac{\pi}{4}$ is true for atleast one $\alpha \in [0, 2]$ and $\alpha \in I$ then value of $ K-3 $ can be	4	3

A) P-3, Q-4, R-2, S-1

B) P-4, Q-3, R-2, S-1

C) P-3, Q-4, R-1, S-2

D) P-4, Q-3, R-1, S-2

60. Match the following

COLUMN - I		COLUMN - II	
P	Number of polynomials $f(x)$ with non(-ve) integral coefficients of degree ≤ 3 . $f(0)=0$ and $\int_0^1 f(x)dx=1$, is	1	8
Q	for $x \in (0, \pi)$, number of solutions of equation $\sin x + 2 \sin 2x - \sin 3x = 3$, is	2	0
R	The graphs $y = 2x^3 - 4x + 2$ and $y = x^3 + 2x - 1$ distinct intersect in exactly three distinct points. The slope of the line passing through two of these points	3	7
S	Let the harmonic means of two positive real numbers a and b be 4, If q is a positive real number such that a, 5, q, b is an A.P. then sum of values of $ q-a $ is	4	4

A) P-4, Q-2, R-3, S-1

B) P-2, Q-4, R-3, S-1

C) P-2, Q-4, R-1, S-3

D) P-4, Q-2, R-1, S-3

1	B	2	B	3	B	4	A	5	A
6	A	7	D	8	B	9	C	10	C
11	D	12	C	13	A	14	C	15	A
16	B	17	D	18	A	19	B	20	C
21	D	22	B	23	A	24	C	25	A
26	D	27	A	28	A	29	D	30	B
31	D	32	A	33	B	34	A	35	B
36	A	37	A	38	A	39	C	40	B
41	C	42	B	43	C	44	D	45	B
46	B	47	B	48	B	49	C	50	A
51	B	52	D	53	B	54	B	55	D
56	B	57	A	58	C	59	B	60	D



Sec: Sr. IIT

Race -5

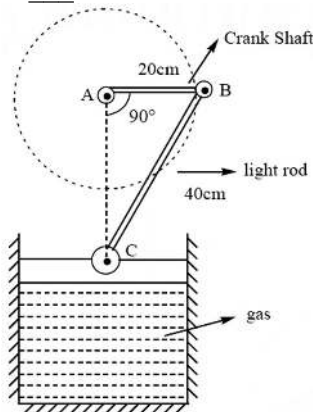
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PHYSICS
SECTION - I
(SINGLE CORRECT ANSWER TYPE)

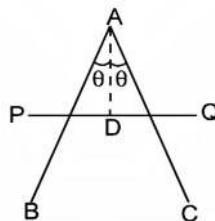
This section contains 10 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

- An aeroplane is flying up in geographic meridian at an angle of 30° with the horizontal (north) and wind is blowing from west. A package is dropped from the aeroplane. Find the velocity of the wind if package hits a kite flying in the space with a position vector $\vec{R} = (400\sqrt{3}\hat{i} + 80\hat{j} + 200\hat{k})$ m with respect to the point of dropping. (Here $\hat{i}$ and $\hat{j}$ are the unit vectors along north and vertically up respectively and $\hat{k}$ be the unit vector due east. Assume the bag is light enough to get carried away by the wind):
 A) 50 m/sec B) 25 m/sec C) 20 m/sec D) 10 m/sec
- A light rod of length 40cm is hinged between (BC) light smooth piston and crank shaft. Crank shaft is free to rotate about A as shown. At the instant shown pressure of gas inside the cylinder is $5P_0$ where $P_0 (=10^5 \text{ Pa})$ is the atmospheric pressure. The area of piston is 20 cm^2 . At the given moment, the torque acting on shaft about A is ____

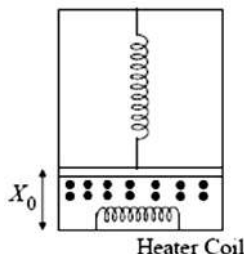


- A) 120 Nm B) 160 Nm C) 200 Nm D) 240 Nm
- In the figure shown AB, AC and PQ are thin, smooth, rigid wires. AB and AC are joined at A and fixed in a vertical plane. $\angle BAC = 2\theta = 90^\circ$ and line AD is angular bisector of angle BAC. A liquid of surface tension $T = 0.025 \text{ N/m}$ forms a thin film in the triangle formed by intersection of the wires AB, AC and PQ. In the figure shown the uniform wire PQ of mass 1 gm is horizontal and in equilibrium under the action of surface tension and gravitational force. Find the time period of SHM of PQ in vertical plane for small displacement from its mean position.



- A) $\frac{\pi}{10}s$ B) $\frac{\pi}{5}s$ C) $\frac{2\pi}{5}s$ D) $\frac{\pi}{2}s$

4. Certain amount of oxygen is contained in a cylindrical vessel shown. The system is initially (as shown) in equilibrium with gas pressure equal to p_0 and initial compression in spring equal to x_0 . Now, heat is slowly supplied to the gas so that gas pressure becomes $8p_0$. The spring force F is related to compression x in the spring as $F = -kx^3$, where k is a constant. What fraction of heat supplied goes into increasing in spring energy?



(Vessel with insulated walls with Frictionless mass less insulated piston)

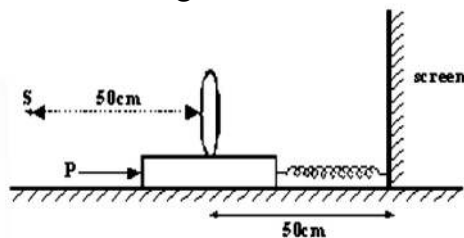
- A) $\frac{1}{11}$ B) $\frac{10}{11}$ C) $\frac{8}{9}$ D) $\frac{1}{9}$
5. A particle of charge $+q$ and mass m moving under the influence of a uniform electric field $E\hat{i}$ and a magnetic field $B\hat{k}$ enters in 1st quadrant of a coordinate system at a point $(0, a)$ with initial velocity $v\hat{i}$ and leaves the quadrant at a point $(2a, 0)$ with velocity $-2v\hat{j}$. Rate of work done by both the fields at $(2a, 0)$
- A) Zero B) $\frac{3mv^3}{4a}$ C) $\frac{3mv^3}{2a}$ D) $\frac{mv^3}{4a}$

SECTION - II (PARAGRAPH TYPE)

This section contains **3 Paragraph of questions**. Each paragraph has 2 multiple choice questions based on a paragraph. Each question has 4 choices A), B), C) and D) for its answer, out of which **ONLY ONE IS** correct.
Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

Paragraph for Question Nos. 11 & 12

A point object 'S' is located at a distance 100 cm from the screen. A lens of focal length 23 cm, mounted on a movable frictionless stand is kept between the source and screen. The stand is attached to a massless spring of natural length 50 cm and spring constant 800 N/m as shown in the figure. Mass of the stand with lens is 2 kg.



11. An impulse P is imparted to the stand so that a real image of the object S is formed on the screen after a fixed time gap. (Image is formed four times in an oscillation). This time gap is:

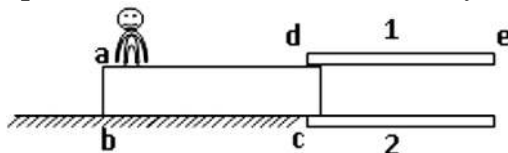
- A) $\frac{\pi}{10}$ sec B) $\frac{\pi}{20}$ sec C) $\frac{\pi}{30}$ sec D) $\frac{\pi}{40}$ sec

12. Amplitude of oscillation so that real image of the object is formed on the screen after a fixed time gap. (Image is formed 4 times in one oscillation)

- A) 25 cm B) 30 cm C) 20 cm D) 10 cm

Paragraph for Question Nos. 13 & 14

The diagram shows two capacitor plates 1 and 2 lying horizontally. The plates are squares of side L and separated by a distance d . The plates are fixed rigidly and the potential difference between them is maintained V . A dielectric slab $abcd$ of mass m and dielectric constant 2 is on a smooth horizontal surface which can just fill the gap between the plates completely without friction. In the condition shown the slab is slightly inserted into the gap between the plates. An insect of mass m is on the slab at a as shown. The insect and the horizontal surface do not affect the charge distribution. Initially the system is held at rest and at time $t=0$ the system is released, simultaneously the insect starts running towards d such that slab is at rest. Upon reaching the end d the insect jumps into air such that it lands at the far end e of the plate 1. During the jump the insect's horizontal velocity remains same.



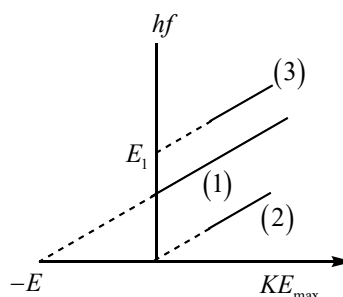
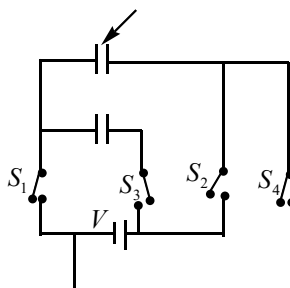
13. The time taken by the insect to move from a to d on the dielectric slab is:

- A) $\left[\frac{4dm}{\epsilon_0 V^2} \right]^{1/2}$ B) $\left[\frac{2dm}{\epsilon_0 V^2} \right]^{1/2}$ C) $\left[\frac{dm}{\epsilon_0 V^2} \right]^{1/2}$ D) $\left[\frac{dm}{2\epsilon_0 V^2} \right]^{1/2}$

14. The time for which the insect will be in air during the jump is

- A) $\left[\frac{4dm}{\epsilon_0 V^2} \right]^{1/2}$ B) $\left[\frac{2dm}{\epsilon_0 V^2} \right]^{1/2}$ C) $\left[\frac{dm}{\epsilon_0 V^2} \right]^{1/2}$ D) $\left[\frac{dm}{2\epsilon_0 V^2} \right]^{1/2}$

Paragraph for Question Nos. 15 & 16



The circuit shown is placed in vacuum. Both the capacitors are identical and they have the same capacitance C . Light is incident on the left plate of the upper capacitor. When all the switches are open then the hf versus $KE_{\max}$ is shown by the straight line (1). In all the cases, we are measuring the $KE_{\max}$ when the electron reaches the opposite plate.

When only S_1 and S_2 are closed, the graph becomes (2).

When only S_3 and S_4 are closed, the graph becomes (3).

15. What is the value of eV ?
 A) E B) E_1 C) $E + E_1$ D) $E - E_1$
16. What is the value of E_1 ?
 A) $3E$ B) $3E/2$ C) $E/2$ D) $2E$

SECTION - III**(Matching List Type)**

This section contains four questions, each having two matching lists (List-I & List-II). The options for the correct match are provided as (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

17. Three holes A, B and C are made at depths 1m, 2m and 5m as shown. Total height of liquid in the container is 8m. Let v is the speed with which liquid comes out of the hole and R is the range on ground. Match the following two columns, for the liquid coming out of holes A, B and C



Column – I		Column-II	
A	V is maximum for	P	hole A
B	V is minimum for	Q	hole B
C	R is maximum for	R	hole C
D	R is minimum for	S	For a hole made half way between A and B

- A) A-P; B-R, C-P; D-S
 B) A-S; B-Q, C-R; D-S
 C) A-R; B-Q, C-R; D-P
 D) A-R; B-P, C-R; D-P
18. A wire of length l , area of cross section A and Young's modulus of elasticity Y is stretched by a longitudinal force F . The change in length is Δl . Match the following two columns. (Only one of the parameters in column-I is changed at a time)

Column – I		Column-II	
A	F is increased	P	Δl will increase
B	l is increased	Q	Stress will increase
C	A is increased	R	Δl will decrease
D	Y is increased	S	Stress will decrease

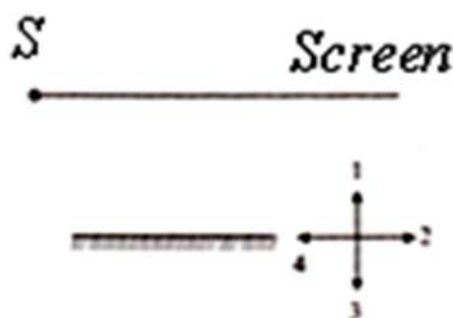
A) A-PQ; B-P, C-RS; D-R

B) A-PS; B-Q, C-RS; D-R

C) A-PQ; B-P, C-P; D-S

D) A-PS; B-S, C-RS; D-S

19. A plane mirror is arranged parallel to the screen and a point source S is kept on the screen as shown in figure. The mirror is moved towards the screen. Corresponding to the direction of motion of the mirror match the column (consider the moment the motion starts)



COLUMN-I	COLUMN-II
A) Direction of motion of mirror along 1	P) length of spot increases
B) Direction of motion of mirror along 2	Q) length of spot decreases
C) Direction of motion of mirror along 3	R) length of spot does not change
D) Direction of motion of mirror along 4	S) brightness of spot increases

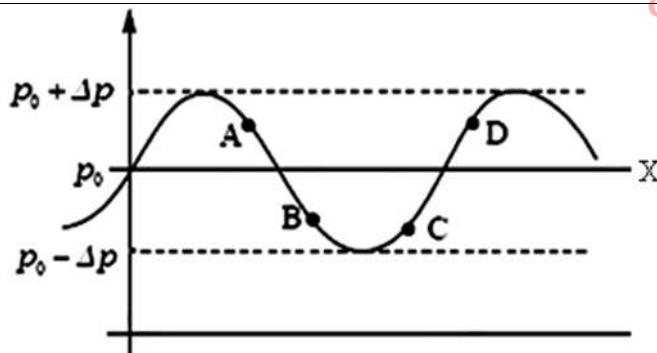
A) A-PS, B-R, C-P, D-RS

B) A-RS, B-R, C-R, D-RS

C) A-PS, B-P, C-R, D-PS

D) A-RS, B-P, C-R, D-QS

20. In a long organ pipe of small cross-section longitudinal progressive wave is propagating in positive X-direction. Figure shows the instantaneous pressure variation in the organ pipe. 'A', 'B', 'C' and 'D' are fluid elements in the pipe. Instantaneously, (X is the position of particle on X-axis)

**Column-I**

A) Element A

B) Element B

C) Element C

D) Element D

A) A-QS,B-PS,C-QR,D-PR

C) A-PR,B-QR,C-PS,D-QS

Column-II

P) Speed is increasing

Q) Speed is decreasing

R) Element is present left to its equilibrium position

S) Element is present right to its equilibrium position

B) A-PR,B-QR,C-QS,D-PS

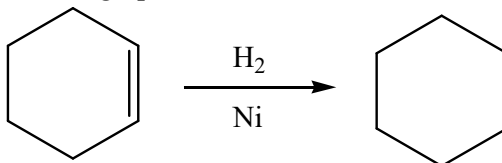
D) A-PS,B-QS,C-QR,D-PR

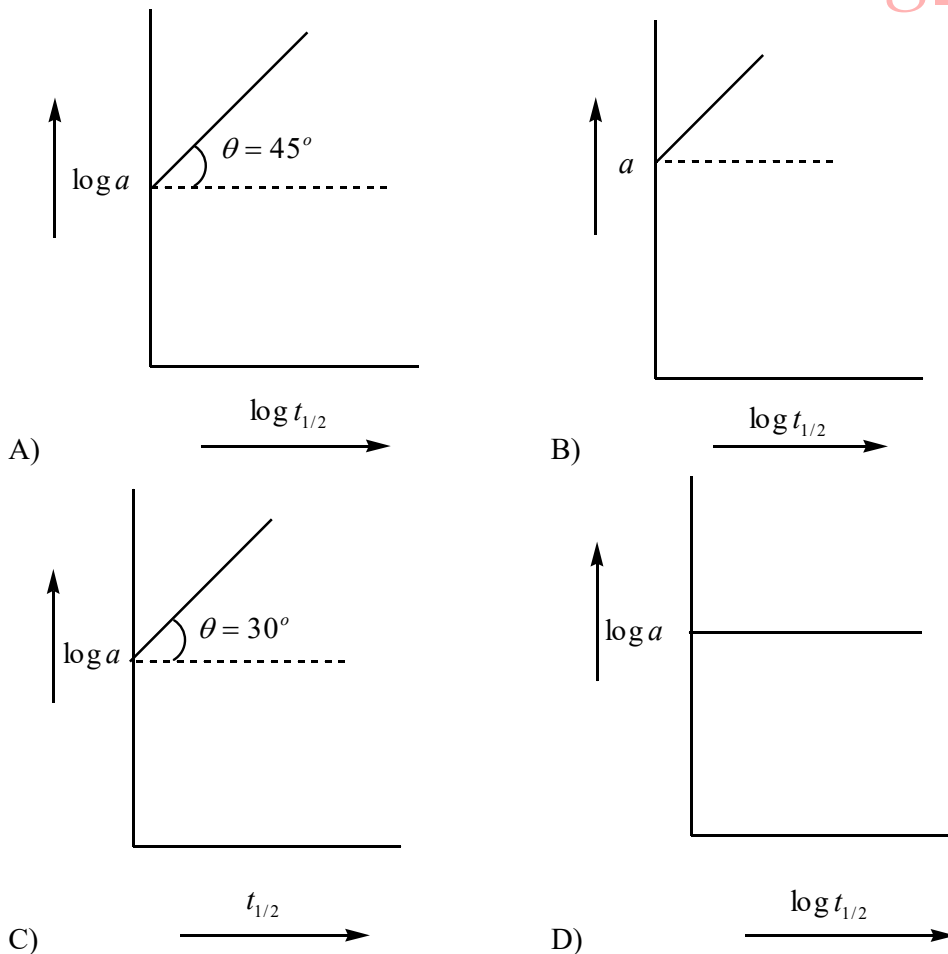
CHEMISTRY**SECTION – I****(SINGLE CORRECT ANSWER TYPE)**

This section contains 10 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

21. In a AB_xL_y (sp^3d hybridization of 'A') have "T" shape geometry. What will be the geometry of molecule. If 'A' form AB_yL_x type molecule? (B = Bond, L – Lone pair)
 A) 'T' shape B) sea - saw C) T.B.P D) Linear
22. Only gases remains after 12g of carbon is treated with 49.26 Lit of air at 300 K and 2atm pressure (assume 20% by volume O_2 , N_2 and 1% CO_2)
 ΔH_f for CO & CO_2 are – 26 Kcal, - 94 Kcal.
 The amount of heat evolved under constant pressure is.....
 A) -96.8 Kcal B) -66.8 Kcal C) -23.4 Kcal D) -33.4 Kcal
23. Which of the following is correct graph for the reaction?

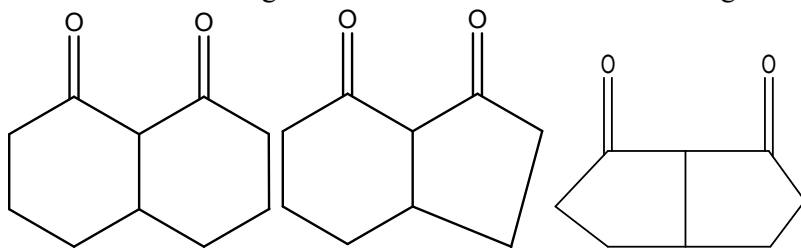




24. Fructose has a specific rotation of (-92.4°) . An aqueous solution of a fructose has an observed angle of rotation of (-27.7°) when placed in a polarimeter tube 10cm long. How much amount of fructose is dissolved in 100 ml of aqueous solution?

A) 0.299 g/ml B) 3.33 g/ml C) 2.99 g/ml D) 33.3 g/ml

25. The correct decreasing order in the enol content of following molecules is



- (I) A) I > II > III (II) B) II > I > III (III) C) III > II > I D) II > III > I

26. Which of the following option is incorrect regarding XeF_6 ?

A) It acts as Lewis acid when it reacts with RbF
 B) It undergoes complete hydrolysis to give XeO_3
 C) It Fluorinates Silica (SiO_2) to give $XeOF_4$
 D) When it react with HCl gives HF, $XeCl_2$ & F_2

27. Which of the following statement is wrong?

A) Melting and Boiling points of H_2O_2 are greater than water
 B) Dipolemoment of H_2O_2 is greater than H_2O

- C) Dihedral angle of H_2O_2 in gas phase is more than solid phase
 D) H_2O_2 is strong oxidizing agent in acidic medium than in alkaline medium.
28. Combination of which parameter explain more oxidising strength of F_2 as compared to Cl_2 in water?
 A) Bond dissociation Enthalpy, electronegativity, hydration enthalpy
 B) Bond dissociation enthalpy, ionization enthalpy, hydration enthalpy
 C) Sublimation enthalpy, electronegativity, hydration enthalpy
 D) Bond dissociation enthalpy, electronegativity, hydration enthalpy
29. $(C_6H_{10})^P \xrightarrow[\text{ii) } NaBH_4 / \text{ethanol; iii) dil. acid}]{\text{i) dil. } H_2SO_4 / HgSO_4} Q \xrightarrow[\text{ii) } O_3 ; \text{ iii) } Zn / H_2O]{\text{i) con. } H_2SO_4} 2H_3C - \overset{\overset{O}{||}}{C} - CH_3$
 What is 'P'.....
 A) $CH_3CH_2CH_2CH_2 - C \equiv C - H$ B) $H_3CCH_2 - C \equiv C - CH_2CH_3$
 C) $\begin{array}{c} H_3C \\ \diagdown \\ H - C - C \equiv C - CH_3 \\ \diagup \\ H_3C \end{array}$ D) $\begin{array}{c} H_3C \\ \diagdown \\ H_3C - C - C \equiv C - H \\ \diagup \\ H_3C \end{array}$
30. The total number of alkenes possible by dehydrobromination of 3-bromo-3-cyclopentyl hexane using alcoholic. KOH is
 A) 4 B) 5 C) 6 D) 7

SECTION - II

(PARAGRAPH TYPE)

This section contains **3 Paragraph of questions**. Each paragraph has 2 multiple choice questions based on a paragraph. Each question has 4 choices A), B), C) and D) for its answer, out of which **ONLY ONE IS** correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

Paragraph for Question Nos. 31 & 32

Two students were given mixture of $Al^{+3}, Cr^{+3}, Fe^{+3}, Zn^{+2}, Co^{+2}$ and Mn^{+2} ions as their soluble salts which were not interfering radicals. They performed the experiment as follows.

STUDENT-I: In the aqueous solution of ions he added first NH_4Cl then NH_4OH and then passed H_2S gas. He got the precipitate (x).

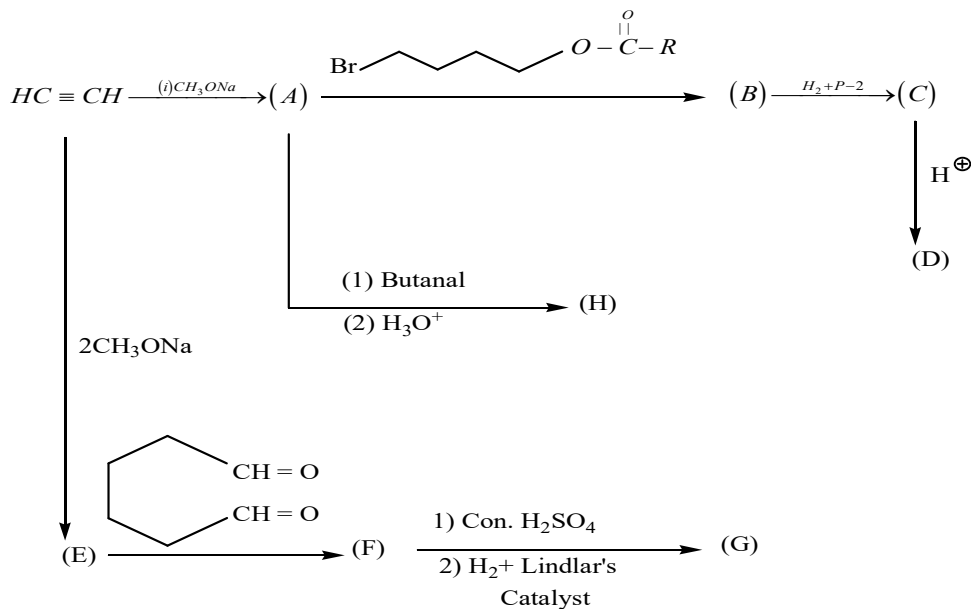
STUDENT-II: In the aqueous solution of ions he first added NH_4Cl and then NH_4OH . He got the precipitate (y) and a Filtrate. In Filtrate, he passed H_2S gas and again got a precipitate (z).

31. Which of the following is not the precipitate (x)?
 A) ZnS B) MnS C) $Cr(OH)_3$ D) Al_2S_3
32. Precipitate (y) when treated with aqueous Na_2O_2 solution, gives a yellow colour solution due to formation of which of the following?
 A) Na_2CrO_4 B) ZnO C) $Na_2Cr_2O_4$ D) $Fe(OH)_3$

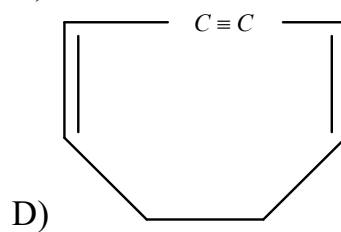
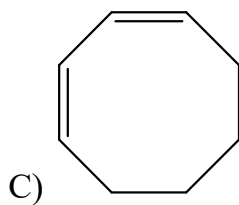
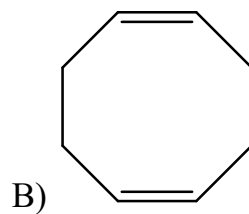
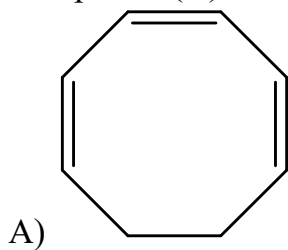
Paragraph for Question Nos. 33 & 34

A balloon of radius 3m is at a pressure of 1 atm and 25°C at sea level density of air is 1.22 kg/m^3 if air presents in balloon is replaced by hydrogen gas, the balloon lifts at sea level.

33. How many moles of Hydrogen are required to fill the balloon?
 A) $2.31 \times 10^3 \text{ mole}$ B) $4.62 \times 10^3 \text{ mole}$ C) $6.93 \times 10^3 \text{ mole}$ D) $9.24 \times 10^3 \text{ mole}$
34. What would be the payload of balloon?
 A) 128.7 kg B) 182.7 kg C) 62.4 kg D) 91.4 kg

Paragraph for Question Nos. 35 & 36

35. Compound (B) is....
 A) $\text{H}-\text{C} \equiv \text{C}-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CHO}$
 B) $\text{H}-\text{C} \equiv \text{C}-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{OH}$
 C) $\text{H}-\text{C} \equiv \text{C}-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{COOH}$
 D) $\text{H}-\text{C} \equiv \text{C}-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{OH}$
36. Compound (G) is



SECTION - III

(Matching List Type)

This section contains four questions, each having two matching lists (List-I & List-II). The options for the correct match are provided as (A), (B), (C) and (D) out of which ONLY ONE is correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

37. Match the following.

Column-I		Column II (Characteristics of complex ion compound)	
A)	$[Pt(NO_2)_2(en)_2]^{2+}$	P)	Stable according to E.A.N rule
B)	$[Cr(\pi-C_6H_6)(NO_2)]$	Q)	Ligand acts as ambidentate
C)	$[Ir(SCN)(SO_4)(NH_3)_4]$	R)	Bond order of Ligand decreases
D)	$[Cr(C_2H_4)(CO)_5]$	S)	Bond order of M – L bond > 1.0

A) A- (P), B- (PRS), C- (PQ), D-(PRS)

B) A- (R), B- (PRS), C- (QS), D-(PS)

C) A –(P), B-(RS), C-(Q), D-(PRS)

D) A-(S), B-(RS), C-(S), D-(RS)

38. Match the following:

Column-I		Column II	
A)	Sp^3	P)	Trimerisation of SO_3
B)	Sp^3d^2	Q)	Solidification of $PCl_{5(g)}$
C)	Sp	R)	Addition of $F^\ominus$ to F_3ClO
D)	No change in hybridisation	S)	Removal of $F^\ominus$ from $BeF_3^\ominus$
		T)	Solidification of N_2O_5

A) A-(PQ), B-(QR), C-(ST), D-(T)

B) A-(Q), B-(R), C-(ST), D-(T)

C) A-(ST), B-(R), C-(ST), D-(S)

D) A-(PQ), B-(R), C-(S), D- (P,Q)

39. Match the following:

Column-I		Column II	
A)	Simple cubic and FCC	P)	Have these cell parameters $a = b = c$, $\alpha = \beta = \gamma$
B)	Cubic and rhombohedral	Q)	Are two crystal systems
C)	Cubic and tetragonal	R)	Have only two crystallographic angles of 90°
D)	Hexagonal and Monoclinic	S)	Belongs to same crystal system

A) A-PS, B-PQ, C-QR, D-QR

B) A-PS, B- SQ, C-Q, D-SQ

C) A-PS, B-PQ, C-Q, D-QR

D) A-S, B-S, C-Q, D-PR

40. Match each of the compounds in Column-I with its characteristic reaction in Column-II

Column-I		Column II	
A)	$CH_3CH_2CH_2CN$	P)	Reduction with $pd-c/H_2$
B)	$CH_3CH_2O\overset{O}{\underset{ }{C}}CH_3$	Q)	Reduction with $SnCl_2/HCl$
C)	$CH_3-CH=CH-CH_2OH$	R)	Development of foul smell on treatment with chloroform and alcoholic. KOH
D)	$CH_3CH_2CH_2CH_2NH_2$	S)	Reduction with di isobutylaluminium hydride (DIBAL-H)
		T)	Alkaline hydrolysis

A) A- PQST, B-S, C-P, D- Q,R

B) A- PQST, B-ST, C-P, D-R

C) A- PQ, B-ST, C-R, D-ST

D) A-QST, B-ST, C-PR,D-QS

MATHEMATICS

SECTION – I

(SINGLE CORRECT ANSWER TYPE)

This section contains 10 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

41. Consider the binomial expansion $\left(\sqrt{x} + \frac{1}{2\sqrt[4]{x}}\right)^n$, $n \in N$, $n > 1$, Arrange terms of expansion in decreasing powers of x . If the coefficients of the first three terms form an A.P., then in the binomial expansion, there are.....**number of distinct** terms of x with integral power

A) 1

B) 2

C) 3

D) 4

42. Value of $\int_0^1 (x^3 + x^5) \left(\frac{d^2}{dx^2} \ln(x + \sqrt{1+x^2}) \right)^2 dx$ is

A) $\frac{3}{4} \ln e$ B) $\ln \left(\frac{e^{\frac{3}{4}}}{2} \right)$ C) $\ln \left(\frac{e^{\frac{1}{4}}}{2} \right)$ D) $\ln 2$

43. The following integral $\int_0^{\frac{\pi}{4}} (2 \sec x)^{2019} dx$ is equal to

A) $\int_1^{\sqrt{2}+1} \left(t + \frac{1}{t}\right)^{2018} \frac{dt}{t}$ B) $\int_1^{\sqrt{2}-1} \left(t - \frac{1}{t}\right)^{2018} \frac{dt}{t}$ C) $\int_1^{\sqrt{2}+1} 2 \left(t - \frac{1}{t}\right)^{2018} \frac{dt}{t}$ D) $\int_1^{\sqrt{2}-1} 2 \left(t + \frac{1}{t}\right)^{2018} \frac{dt}{t}$

44. In triangle ABC, $\angle ABC = 120^\circ$, $AB = 3$ and $BC = 4$. If perpendiculars constructed on the side AB at A and to the side BC at C meet at D then CD is equal to
 A) 3 B) $\frac{8\sqrt{3}}{3}$ C) 5 D) $\frac{10\sqrt{3}}{3}$
45. Four boys and three girls stand in queue, then probability that number of boys ahead of every girl is atleast one more than the number of girls ahead of her, is
 A) $\frac{2}{3}$ B) $\frac{2}{5}$ C) $\frac{3}{4}$ D) $\frac{3}{7}$
46. Let f be any real valued function on real line with continuous third derivative. Let $h(x) = f(x)f'(x)f''(x)f'''(x)$. Which of the following must be true
 A) there must exist atleast one a such that $h(a) \leq 0$
 B) there must exist atleast one a such that $h(a) \geq 0$
 C) there exist no a such that $h(a) \geq 0$
 D) Nothing can be said
47. How many integers between 1000 and 9999 have exactly one pair of equal digit such as 4049 or 9902 but no 4449 or 4040?
 A) 4104 B) 3888 C) 4320 D) 3915
48. Let p be a point in the plane, let $d_1(p)$ and $d_2(p)$ be the distances of the point p from lines $3x - 4y = 0$ and $4x + 3y = 0$ respectively. Area of region R consisting of all points p lying in the first quadrant of the plane and satisfying $1 \leq d_1(p) + d_2(p) \leq 3$ is
 A) 2 B) 4 C) 6 D) 8
49. $f(x) = \begin{cases} x^2 \sin \frac{1}{x} & ; x \neq 0 \\ 0 & ; x = 0 \end{cases}$ and $g(x) = \begin{cases} \frac{|\sin x|}{\tan^{-1} x} & ; x \neq 0 \\ 1 & ; x = 0 \end{cases}$; which of the following is
CORRECT
 A) $y = f(x).g(x)$ is discontinuous B) $y = g(x)$ is discontinuous at $x = 1$
 C) $y = f(x).g(x)$ is differentiable at $x = 0$ D) $y = f(x).g(x)$ is differentiable $\forall x \in \mathbb{R}$
50. Assume that $M = \{(x, y) : x^2 + 2y^2 = 3\}$ and $N = \{(x, y) : y = mx + b\}$. If $M \cap N \neq \emptyset, \forall m \in \mathbb{R}$, then true set of values of b is?
 A) $\left[-\frac{\sqrt{6}}{2}, \frac{\sqrt{6}}{2}\right]$ B) $\left[\frac{-\sqrt{6}}{2}, \frac{\sqrt{6}}{4}\right]$ C) $\left[\frac{-2\sqrt{3}}{3}, \frac{-\sqrt{3}}{3}\right]$ D) $\left[\frac{-2\sqrt{3}}{3}, \frac{2\sqrt{3}}{3}\right]$

SECTION - II

(PARAGRAPH TYPE)

This section contains 3 Paragraph of questions. Each paragraph has 2 multiple choice questions based on a paragraph. Each question has 4 choices A), B), C) and D) for its answer, out of which ONLY ONE IS correct.
Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

Paragraph for Question Nos. 51 & 52

In $\mathbb{R}^3$, Let L be straight a line passing through $(1, 1, 1)$.

Consider another line L_2 whose unsymmetrical form is $x + 2y + z = 1, x + y + 2z = 3$

51. Suppose that all points on L are at constant distance from two planes, used in L_2 , then equation of L is
- A) $\frac{x-1}{3} = \frac{y-1}{1} = \frac{z-1}{1}$ B) $\frac{x-1}{-3} = \frac{y-1}{1} = \frac{z-1}{1}$
 C) $\frac{x-1}{3} = \frac{y-1}{-1} = \frac{z-1}{1}$ D) $\frac{x-1}{3} = \frac{y-1}{-1} = \frac{z-1}{-1}$
52. Equation of L, if it is parallel to $x+y+z=1$ and intersect L_2
- A) $\frac{x-1}{4} = \frac{y-1}{-3} = \frac{z-1}{1}$ B) $\frac{x-1}{4} = \frac{y-1}{3} = \frac{z-1}{1}$
 C) $\frac{x-1}{+4} = \frac{y-1}{3} = \frac{z-1}{1}$ D) $\frac{x-5}{-4} = \frac{y+2}{3} = \frac{z}{1}$

Paragraph for Question Nos. 53 & 54

A player throwing a dice scores as many points as on the top face of the die and is to play until his score reaches or passes a total n. Let $p(n)$ the probability of making exactly the total n. By using this information answer following

53. Value of $p(3) + p(4)$
- A) $\frac{7^3}{6^4}$ B) $\frac{13 \times 7^2}{6^4}$ C) $\frac{7^2}{6^4}$ D) none
54. $\lim_{n \rightarrow \infty} p(n)$
- A) $\frac{1}{6}$ B) $\frac{2}{7}$ C) $\frac{1}{7}$ D) $\frac{1}{5}$

Paragraph for Question Nos. 55 & 56

$$\text{Let } P = \begin{bmatrix} 3 & -1 & -2 \\ 2 & \beta & \alpha \\ 3 & -5 & 0 \end{bmatrix}; \alpha, \beta \in R$$

$Q = [q_{ij}]_{3 \times 3}$ is a matrix such that $PQ = KI$

where $k \in R, k \neq 0$ and I is the identity matrix of order 3.

55. If $q_{23} = \frac{-k}{8}$ and $\det(Q) = \frac{k^2}{2}$, value of $\det(P \text{adj } Q)$ is, if $\beta = 0$
- A) 2 B) 2^5 C) 2^4 D) 2^9
56. If $\beta = 1, q_{23} = \frac{-k}{3}$, value of α is
- A) $\alpha = -\frac{1}{3}$ B) $\alpha = -\frac{14}{3}$ C) $\alpha = -\frac{7}{3}$ D) $\alpha = -\frac{5}{3}$

SECTION-3

(MATCHING LIST TYPE)

This section contains four questions, each having two matching lists (List-I & List-II). The options for the correct match are provided as (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

57. Match the following.

COLUMN - I		COLUMN - II	
P	If $x = \sin(2 \tan^{-1} 2)$; $y = \sin\left(\frac{1}{2} \tan^{-1} \frac{4}{3}\right)$ value of $x + y^2$ is	1	$\frac{1}{2}$
Q	Length of common chord of $2y^2 = 3(x+1)$, $x^2 + y^2 + 2x = 0$ is 'a', then value of $[a+1]$ is; $[.]$ denotes G.I.F	2	2
R	If $\vec{a}, \vec{b}, \vec{c}$ are non coplanar vectors such that $\vec{b} \times \vec{c} = \vec{a}$, $\vec{a} \times \vec{b} = \vec{c}$ $\vec{c} \times \vec{a} = \vec{b}$, value of $ \vec{a} + \vec{b} + (\vec{c})$ is	3	1
S	If $(1+x^2)\frac{dy}{dx} + 2xy - 1 = 0$, passes through origin, value of $y(1)$	4	3

A) P-3, Q-2, R-4, S-1

B) P-2, Q-1, R-3, S-4

C) P-3, Q-2, R-4, S-3

D) P-2, Q-1, R-4, S-3

58. Let $Z \in \mathbb{C}$ and satisfying

$$\begin{vmatrix} |Z-1|^2 & |Z+1|^2 & -4 \\ |Z+1|^2 & -4 & |Z-1|^2 \\ -4 & |Z-1|^2 & |Z-1|^2 \end{vmatrix} = 0$$

List - I		List - II	
P	Least value of $ Z-3i $ is	1	2
Q	Greatest value of $ Z+3i $ is	2	4
R	Least value of $ Z+i + Z-i $ is	3	5
S	If Z_1, Z_2 are the solutions of the given equation such that $1+2i$, Z_1 and Z_2 are collinear then $ Z_1-1-2i Z_2-1-2i $ is equal to	4	8

A) P-2, Q-1, R-3, S-4

B) P-1, Q-3, R-2, S-1

C) P-1, Q-2, R-1, S-2

D) P-3, Q-1, R-1, S-2

59. Match the columns

COLUMN-I		COLUMN-II	
P	In $\mathbb{R}^2$, Let $\sqrt{3}\hat{i} + \hat{j}$, $\hat{i} + \sqrt{3}\hat{j}$ and $\beta\hat{i} + (1-\beta)\hat{j}$ be position vectors of X, Y, Z with respect to the origin O, respectively If the distance of Z from the bisector of acute angle of $\overline{OX}$ and $\overline{OY}$ is $\frac{3}{\sqrt{2}}$, sum of possible value of $ \beta $ is	1	4
Q	Number of solution of the equation $2^x + x = 2^{\sin x} + \sin x$ in $[0, 10\pi]$ is	2	2

R	Let A and B are two non singular matrices of order 3 such that $A+B=I$ $A^{-1}+B^{-1}=2I$, then $ adj(2AB) +1$, is	3	1
S	If $\tan^{-1} \alpha = \tan^{-1}(\alpha + k) + \frac{\pi}{4}$ is true for atleast one $\alpha \in [0, 2]$ and $\alpha \in I$ then value of $ K-3 $ can be	4	3

A) P-3, Q-4, R-2, S-1

B) P-4, Q-3, R-2, S-1

C) P-3, Q-4, R-1, S-2

D) P-4, Q-3, R-1, S-2

60. Match the following

COLUMN - I		COLUMN - II	
P	Number of polynomials $f(x)$ with non(-ve) integral coefficients of degree ≤ 3 . $f(0)=0$ and $\int_0^1 f(x)dx=1$, is	1	8
Q	for $x \in (0, \pi)$, number of solutions of equation $\sin x + 2 \sin 2x - \sin 3x = 3$, is	2	0
R	The graphs $y = 2x^3 - 4x + 2$ and $y = x^3 + 2x - 1$ distinct intersect in exactly three distinct points. The slope of the line passing through two of these points	3	7
S	Let the harmonic means of two positive real numbers a and b be 4, If q is a positive real number such that a, 5, q, b is an A.P. then sum of values of $ q-a $ is	4	4

A) P-4, Q-2, R-3, S-1

B) P-2, Q-4, R-3, S-1

C) P-2, Q-4, R-1, S-3

D) P-4, Q-2, R-1, S-3

1	B	2	B	3	B	4	A	5	A
6	A	7	D	8	B	9	C	10	C
11	D	12	C	13	A	14	C	15	A
16	B	17	D	18	A	19	B	20	C
21	D	22	B	23	A	24	C	25	A
26	D	27	A	28	A	29	D	30	B
31	D	32	A	33	B	34	A	35	B
36	A	37	A	38	A	39	C	40	B
41	C	42	B	43	C	44	D	45	B
46	B	47	B	48	B	49	C	50	A
51	B	52	D	53	B	54	B	55	D
56	B	57	A	58	C	59	B	60	D



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Race -6

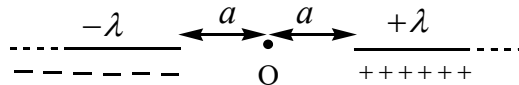
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PHYSICS
SECTION - I
(SINGLE CORRECT ANSWER TYPE)

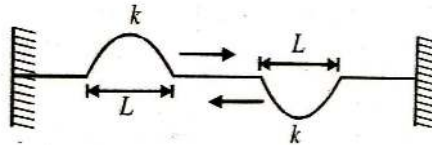
This section contains 5 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

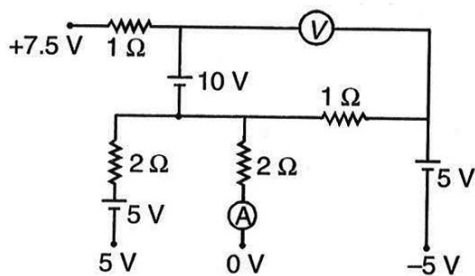
1. Two very long line charges of uniform charge densities $+\lambda$ and $-\lambda$ are placed along same line with the separation between the nearest ends being $2a$, as shown in figure. The electric field intensity at point O is



- A) $\frac{\lambda}{2\pi\epsilon_0 a}$ B) 0 C) $\frac{\lambda}{\pi\epsilon_0 a}$ D) $\frac{\lambda}{4\pi\epsilon_0 a}$
2. Two identical pulses (one is inverted w.r.t other) move in opposite directions with same uniform speeds on a stretched string. The width and kinetic energy of each pulse are L and k respectively. At the instant they completely overlap, the kinetic energy of the width L of the string in overlapped part is

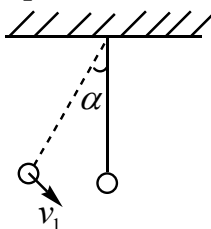


- A) k B) $2k$ C) $4k$ D) $8k$
3. A particle of mass m moves along a semicubical parabolic curve, $y^2 = ax^3$ with constant speed v . Find the contact force of the curve on the particle. (Neglect gravitational force)
- A) $\frac{3}{4}a^{1/2}x^{-1/2}\left(1 + \frac{9}{4}ax\right)^{-3/2}mv^2$ B) $\frac{3}{4}a^{1/2}x^{-1/2}\left(1 - \frac{9}{4}ax\right)^{-3/2}mv^2$
- C) $\frac{3}{4}a^{1/2}x^{-1/2}\left(1 + \frac{9}{4}ax\right)^{-1/2}mv^2$ D) $\frac{4}{3}a^{\frac{1}{2}}x^{-\frac{1}{2}}\left(1 - \frac{9}{4}ax\right)mv^2$
4. Potentials of certain points in the circuit are maintained as marked. What is reading of voltmeter, if the ammeter reads zero ?



- A) 2.5V B) 10V C) 5V D) 20V

5. A simple pendulum of length ℓ is tilted at an angle α and imparted an initial velocity v_1 towards mean position. Initial phase of bob is given as (A_1 is linear amplitude and assume that maximum angular displacement of bob is very small)



- A) $\sin^{-1}\left(\frac{\ell\alpha}{A_1}\right)$ B) $2\pi - \sin^{-1}\left(\frac{\ell\alpha}{A_1}\right)$ C) $\pi - \sin^{-1}\left(\frac{\ell\alpha}{A_1}\right)$ D) $\pi - \cos^{-1}\left(\frac{\ell\alpha}{A_1}\right)$

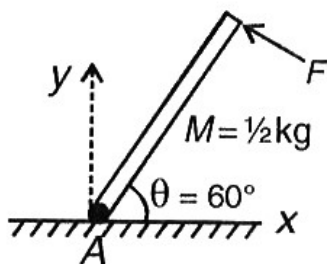
SECTION – II

(MULTIPLE CORRECT ANSWER TYPE)

This section contains 8 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct.

Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases

6. Work function of a metal X is equal to ionization energy of Li^{2+} ion in second excited state. Work function of another metal Y equals ionization energy of He^+ ion with electron in $n=4$. Now a photon of energy x falls on both the metals such that maximum kinetic energy of photoelectrons emitted from metal X is half that of photoelectrons emitted from metal Y. Choose the correct statement (E = Potential energy of electron in ground state of hydrogen atom)
- A) $x = -3.5 E$
 B) $x + E = -3.4 \text{ eV}$
 C) As x increases, difference in maximum kinetic energies of photo electrons increases
 D) As x increases, difference in maximum kinetic energies of photo electrons remains constant
7. A rod of mass $M = \frac{1}{2}\text{kg}$ is hinged at point A and held with the help of force F acting perpendicularly as shown. The rod is in equilibrium. The force applied by hinge at A (take $g = 10\text{m/s}^2$)



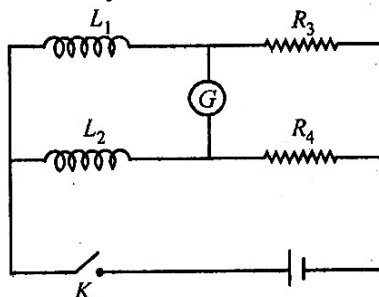
A) has magnitude of $\frac{5}{4}\sqrt{13}N$

B) has components along both x-axis and y-axis

C) is at some angle in 1st quadrant

D) is at some angle and in 2nd quadrant

8. Two inductors of self-inductances L_1 and L_2 and of resistances R_1 and R_2 (not shown here) respectively, are connected in the circuit as shown in figure. At the instant $t = 0$, key K is closed. Choose the correct options for which the galvanometer will show zero deflection at all times after the key is closed.



A) $\frac{L_1}{L_2} = \frac{R_3}{R_4}$

B) $\frac{L_1}{L_2} = \frac{R_1}{R_2}$

C) $\frac{R_1}{R_2} = \frac{R_3}{R_4}$

D) $R_1 \times R_4 = R_2 \times R_3$

9. A vector $\vec{A}$ is given by $\vec{A} = t\hat{i} - \sin \pi t \hat{j} + t^2 \hat{k}$. Then

A) The magnitude of $\vec{A}$ at $t = 1$ s, is $|\vec{A}| = \sqrt{2}$

B) The derivative of $\vec{A}$ i.e., $\frac{d\vec{A}}{dt}$ at $t = 1$ s, $\hat{i} + \pi \hat{j} + 2\hat{k}$

C) The value of $\left| \vec{A} \times \frac{d\vec{A}}{dt} \right|$ at $t = 1$ s, is $\sqrt{2\pi^2 + 1}$

D) The value of $\vec{A} \cdot \frac{d\vec{A}}{dt}$ at $t = 1$ s, is 3

10. Two men of masses 60 kg and 80 kg are sitting at the ends of a boat of mass 60 kg and length 4 m. The boat is stationary on water. If the men now exchange their positions by walking on boat, then:

A) the centre of mass of the two men shifts by $(4/7)$ m

B) the boat moves by 0.4 m

C) the centre of mass of the two men shift by $(6/35)$ m

D) the boat moves by 0.6 m

11. The position of a particle varies according to the expression $x = t(t-1)(t-2)$, then :

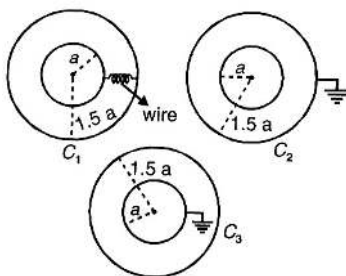
A) velocity will be zero at $t_1 = 1 - \frac{1}{\sqrt{3}}$ s and $t_2 = 1 + \frac{1}{\sqrt{3}}$ s

B) acceleration changes its direction between $t_1 = 0$ and $t_2 = 2$ s

C) acceleration remains constant in direction between $t_1 = 0$ and $t_2 = 2$ s

D) distance travelled by the particle is zero in first two seconds

12. In an a.c. circuit containing resistance only,
 A) average power dissipated in a time interval equal to half the time period may be zero.
 B) the r.m.s. value of current is never zero for any finite interval.
 C) the average value of current in a time interval equal to half the time period may be zero.
 D) average power dissipated in a time interval equal to half the time period is never zero.
13. Capacitances of following combinations of spheres are C_1, C_2 & C_3 . Then



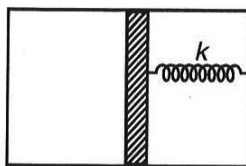
- A) $C_2 > C_1$ B) $C_1 > C_3$ C) $C_1 > C_2$ D) $C_3 > C_2$

SECTION – III
(INTEGER ANSWER TYPE)

This section contains 5 questions. The answer is a single digit integer ranging from 0 to 9 (both inclusive).

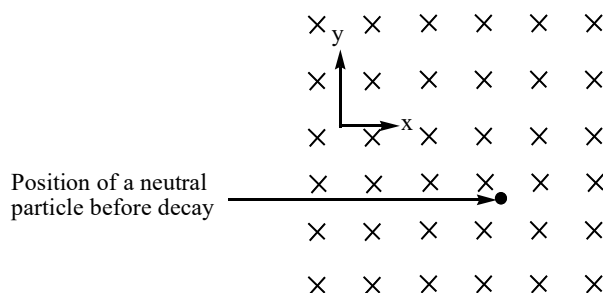
Marking scheme +3 for correct answer, 0 if not attempted and 0 in all other cases.

14. A point source is kept on the principle axis of a convex lens of focal length f , at a distance $4f$ from it. Behind the lens, there is a concave mirror of focal length f . There are two positions of mirror, such that final image coincides with the point source. If the ratio of the distance between lens and mirror for the two cases is $\frac{1}{2}n$, then find the value of n .
15. A thermally insulated vessel is divided into two parts by a heat insulating massless piston which can move in the vessel without friction. The left part of vessel contains one mole of an ideal monoatomic gas and right part is empty. The piston is connected to the right wall of vessel through a spring whose length in free state is equal to the length of the vessel. Molar specific heat of the system is found to be nR (where R is universal gas constant). Find n . Heat capacities of the vessel, piston and spring are negligible.



16. A neutron is moving through container filled with stationary deuterons. The neutron successively collides elastically and head on with stationary deuterons one at a time. The mass of the neutron is equal to half that of the deuteron. How many such collisions would be required to slow the neutron down from 531.441 keV to 1 eV? [neglect the relativistic effect].

17. A neutral particle is initially at rest in a uniform magnetic field B as shown in the diagram. The particle then spontaneously decays into two fragments, one with a positive charge $+q$ and mass $3m$ and the other with negative charge $-q$ and mass m . Neglecting the interaction between the two charged particles and assume that the speeds are much less than speed of light. If the time (in ms) after the decay at which the two fragments first meet is $\frac{x}{y}$, then find $x+y$. Both the charges have initial velocities in $X-Y$ plane. (Use the following data : $q = 1\mu C, B = 2\pi\mu T, m = 10^{-15} kg$).



18. The gravitational potential energy of a satellite revolving around the earth in circular orbit is $-4 MJ$. Find the minimum additional energy (in MJ) that should be given to the satellite so that it escapes from the gravitational field of earth. Assume earth's gravitational force to be the only gravitational force on the satellite and no atmospheric resistance.

CHEMISTRY

SECTION – I

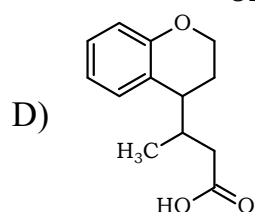
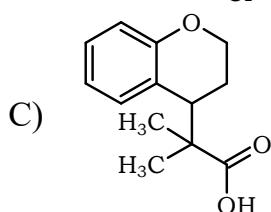
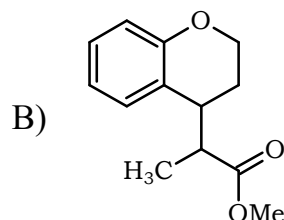
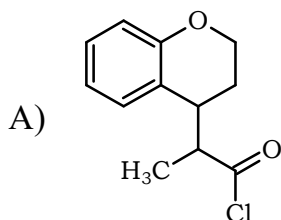
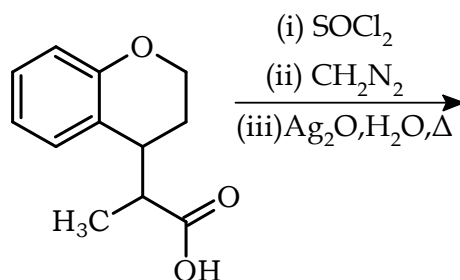
(SINGLE CORRECT ANSWER TYPE)

This section contains 5 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct.

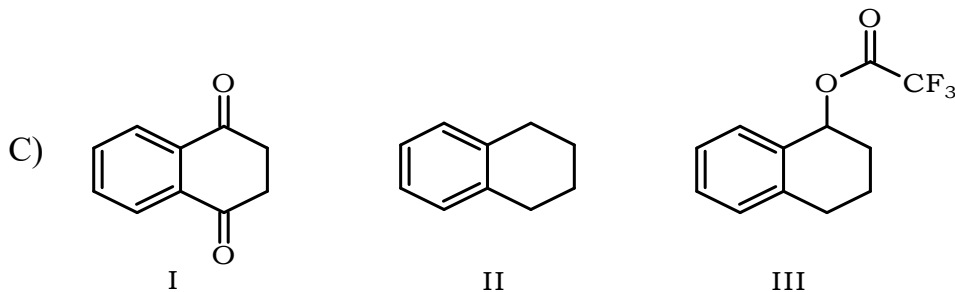
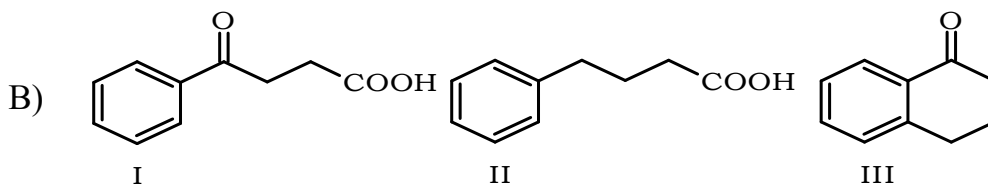
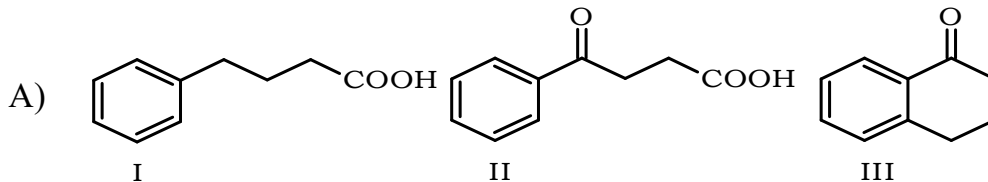
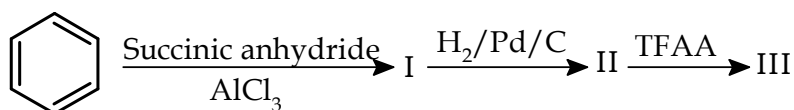
Marking scheme: +3 for correct answer, 0 if not attempted and –1 in all other cases.

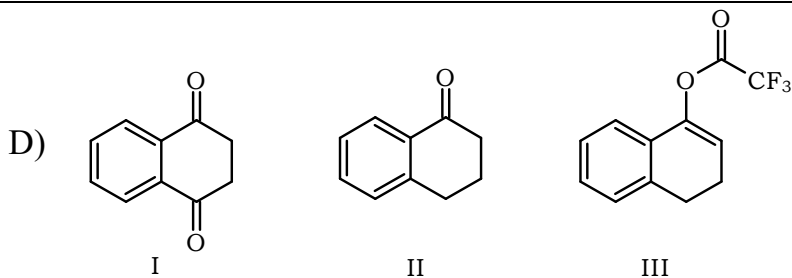
19. For alkali and alkaline earth metals, solubility and thermal stability from top to bottom increases for
- Alkaline earth metal hydroxides
 - Alkali metal fluorides
 - Alkali metal perchlorates
 - Carbonates of alkaline earth metals
20. Identify the incorrect statement from the following.
- Superoxide ion is smaller than peroxide ion and KO_2 is more stable than NaO_2 .
 - $(NH_4)_2Cr_2O_7$ on heating shows self redox reaction.
 - Products on reaction of metal with HNO_3 is independent of concentration of HNO_3 .
 - White phosphorous disproportionate in $NaOH$ solution.

21. The product formed in the following reaction is



22. Which of the following is correctly matched? (TFAA = Trifluoro acetic acid)





23. The solubility product of silver benzoate is 2.5×10^{-13} and the acid dissociation constant of benzoic acid is $\frac{1}{3.5} \times 10^{-5}$. Calculate the solubility of silver benzoate in a buffer solution having $pH = 4$.
- A) 10^{-7} B) 3×10^{-6} C) 5×10^{-7} D) 10^{-6}

SECTION – II

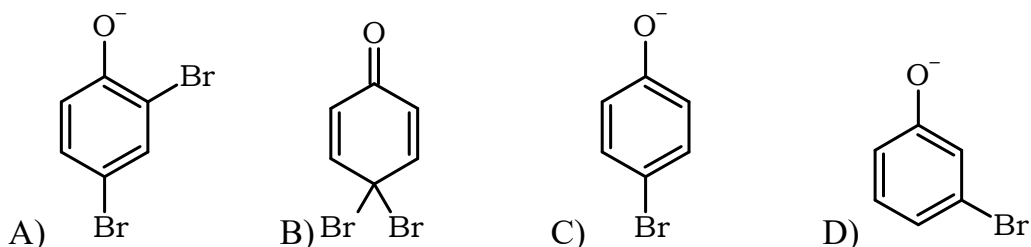
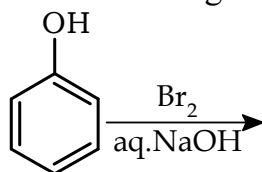
(MULTIPLE CORRECT ANSWER TYPE)

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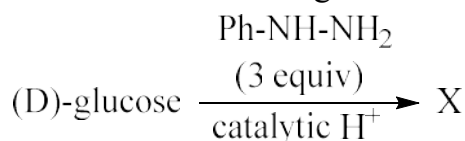
Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases

24. To aq. $NiSO_4$ solution, excess of NH_3 is added and dark blue coloured solution 'P' is formed. The solution 'P' is made into two parts. 1st part is allowed to react with 2 moles of 'en' (ethylene diamine) to give solution 'Q' which can't react with excess 'en'. Part 2 of solution 'P' is allowed to react with two moles of 'en' to give 'R' which on reaction with 1 more mole of 'en' gives 'S'. Identify the correct statements.
- A) P, Q, R, S all are paramagnetic
 B) Number of t_{2g} electrons $S > R = Q > P$
 C) Both R and S show optical isomerism
 D) Both Q and R are stereo isomers
25. Which of the following statements is/are true?
- A) In ketene ($CH_2 = C = O$) all π -electrons are in same plane but in allene ($CH_2 = C = CH_2$) all π -electrons are not in same plane.
 B) $CH_2 = C = O$ is planar but $CH_2 = C = CH_2$ is non planar.
 C) By replacing $-CH_3$ group with $-SiH_3$, Geometry around $\ddot{N}$ changes in $(CH_3)_3\ddot{N}$
 D) In SO_3 , there are three $P_\pi - d_\pi$ bonds
26. Which of the following statements are correct?
- A) When ClO_3^- is converted to ClO_4^- , bond angle decreases due to increased number of atoms around Cl atom.
 B) Si – Cl bond length in SiH_3Cl is longer than that in SiF_3Cl
 C) In SO_4^{2-} ion, all bond angles are equal
 D) PF_3 is planar due to back bonding

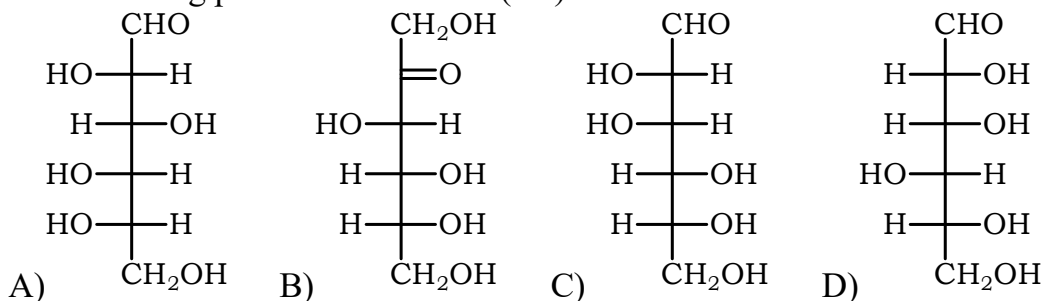
27. In the following reaction, the intermediate(s) is(are)...



28. Consider the following reaction.



Among the following, the compound(s) whose osazone derivative(s) will have same melting point as that of X is (are)



29. At very high pressure, the compressibility factor of 1 mole of a vander waal's gas can be given by

Options for the compressibility factor (Z):

- A) $Z = 1 + \frac{bP}{RT}$
- B) $Z = \frac{V}{V-b}$
- C) $Z = \frac{V+b}{V}$
- D) $Z = 1 - \frac{a}{VRT}$

30. Which of the following statements are correct?

- A) Ferrimagnetic substances lose ferrimagnetism on heating and become paramagnetic.
- B) Ferrimagnetic substances do not lose ferrimagnetism on heating and remain ferrimagnetic.
- C) Antiferromagnetic substances have domain structures similar to ferromagnetic substances and their magnetic moments are not cancelled by each other.
- D) In ferromagnetic substances all the domains get oriented in the direction of magnetic field and remain as such even after removing magnetic field.

31. Which of the following statements are correct?

- A) Mixing two oppositely charged sols neutralizes their charges and stabilizes the colloid.
- B) Presence of equal and similar charges on colloidal particles provides stability to the colloids.
- C) Any amount of dispersed liquid can be added to emulsion without destabilizing it.
- D) Brownian movement stabilizes sols.

SECTION – III
(INTEGER ANSWER TYPE)

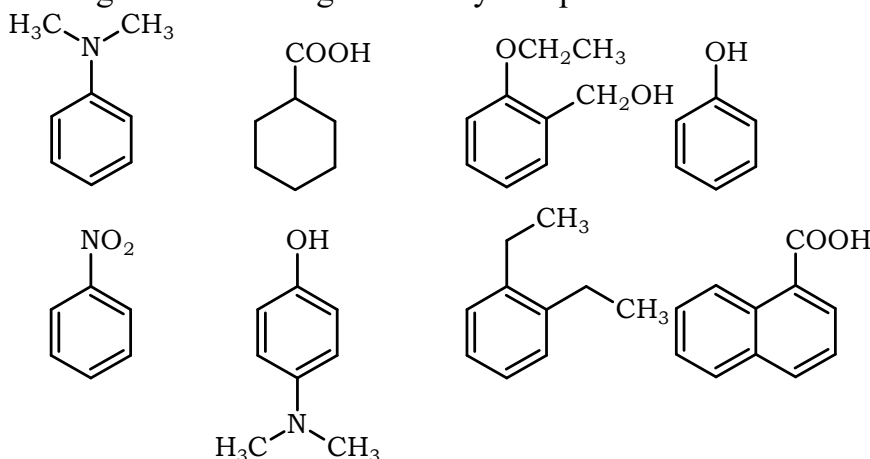
This section contains 5 questions. The answer is a single digit integer ranging from 0 to 9 (both inclusive).

Marking scheme +3 for correct answer , 0 if not attempted and 0 in all other cases

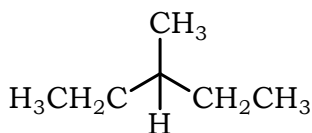
32. Identify the number of correct statements for graphite.

- a) Three dimensional network like structure
- b) C is SP^2 hybridised
- c) Used as lubricant
- d) π bond(s) are present
- e) Harder than diamond
- f) C-C bond order is same as in C_2H_6
- g) C-C bond length is more than $H-C \equiv C-H$
- h) Vanderwaal's forces are present

33. Amongst the following how many compounds are soluble in aq. $NaOH$?



34. The maximum number of isomers (including stereoisomers) that are possible on mono-chlorination of the following compound, is



35. One form of the virial equation is

$$Z = 1 + \frac{B}{V_m} + \frac{C}{V_m^2} + \frac{D}{V_m^3} + \dots \quad (i)$$

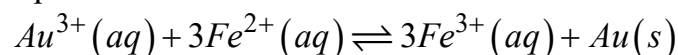
Where, B, C, D, ... are called second, third, fourth virial coefficients. Another form of virial equation is

$$Z = 1 + B'p + C'p^2 + \dots \quad (ii)$$

Where B' , C' are second and third virial coefficients, if $B = 0.04 \text{ lit/mol}$, $C = 8 \times 10^{-6} \text{ litre}^{-2} / \text{mol}^2$ and temperature $= \frac{20}{R}$ then calculate the value of $C' + (B')^2$.

Express your answer in scientific notation as $a \times 10^{-b}$. Fill your OMR as $b - a$.

36. Starting with 250ml of Au^{3+} solution and 250ml of Fe^{2+} solution, the following equilibrium is established.



At equilibrium, the equivalents of $Au^{3+}, Fe^{2+}, Fe^{3+}$ and Au are $4 \times 10^{-3}, 10^{-2}, 2 \times 10^{-2}$ and 4×10^{-3} respectively. Then K_C is equal to abcd (a four digit integer). Fill your OMR as $a + b + c + d$

MATHEMATICS

SECTION – I

(SINGLE CORRECT ANSWER TYPE)

This section contains 5 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

37. Let g be a differentiable function satisfies $\int_0^x (x-t+1)g(t)dt = x^4 + x^2 \quad \forall x \geq 0$
then the value of $\int_0^1 \frac{12}{g'(x) + g(x) + 10} dx =$
A) $\frac{\pi}{6}$ B) $\frac{\pi}{3}$ C) $\frac{\pi}{4}$ D) $\frac{\pi}{2}$
38. Consider the curve $y = \sin 3x \sin x, x \in [0, \pi]$ encloses with x-axis, three region whose areas are in the ratio 1: a : b where $1+a+b =$
A) 3 B) 4 C) $\frac{5}{3}$ D) 2
39. Let k be a positive integer and $f(x)$ be a polynomial with integer coefficients satisfying $2 \int_1^x f(t)dt + x^k = x f(x)$ where $x \geq 1$ then sum of all possible values of k
A) 8 B) 9 C) 7 D) 10
40. x, y, z are natural numbers, not greater than 6. How many ordered triples (x, y, z) are there such that the product xyz is divisible by 10 ?
A) 64 B) 72 C) 91 D) 99
41. Let a & b be positive real numbers then
 $\log(a^{10}) + {}^{10}C_1 \log(a^9 b) + {}^{10}C_2 \log(a^8 b^2) + \dots + \log(b^{10}) = \log((ab)^\lambda)$ ($ab \neq 1$)
where $\lambda =$
A) 5120 B) 2048 C) 1024 D) 10240

SECTION – II

(MULTIPLE CORRECT ANSWER TYPE)

This section contains 8 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct.

Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases

42. For $x, y, z \in \left(0, \frac{\pi}{2}\right)$ and Let x, y, z be first three consecutive terms of an AP such that
 $\cos x + \cos y + \cos z = 1$ & $\sin x + \sin y + \sin z = \frac{1}{\sqrt{2}}$ then which of the following is/are true
A) $\sin(x-y) + \sin(z-y) = 0$ B) $\cos(x-y) = \frac{\sqrt{3}-\sqrt{2}}{2\sqrt{2}}$

C) $\tan 2y = \frac{2\sqrt{2}}{3}$

D) $\cos(x-y) = \frac{\sqrt{3}-1}{2\sqrt{2}}$

43. From a deck of 52 cards, two cards are drawn without replacement then probability that

A) both are clubs, given that the first one is a club, is $\frac{4}{17}$

B) both are clubs, given that atleast one is club, is $\frac{2}{15}$

C) both are clubs, given that one is ace of clubs, is $\frac{4}{17}$

D) One is the ace of clubs, given that both are clubs, is $\frac{2}{13}$

44. Which of the following is/are true, for $x \in \left(0, \frac{\pi}{2}\right)$

A) $0 \leq \frac{1-\cos x}{x} \leq \frac{\sin x}{1+\cos x}$

B) $1 \leq \frac{\tan x}{x} \leq \sec x$

C) $\frac{\sin 2x}{2} \leq \frac{\sin^2 x}{x} \leq \sin x$

D) $\frac{2}{\pi} = \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{2+\sqrt{2}}}{2} \cdot \frac{\sqrt{2+\sqrt{2+\sqrt{2}}}}{2} \dots$

45. Given $f(x)$ is a twice differentiable function $\forall x \in R, f'(x) \neq 0$

If $f''(x) + f(x) = -x$ $h(x)f'(x), h(x) > 0 \forall x \in R$

& $f(0) = -3, f'(0) = 4$

& Let $F(x) = (f(x))^2 + (f'(x))^2$

then which of the following is/are true

A) $F(x)$ is monotonically increasing in $(0, \infty)$

B) If $|F(x)| \leq M, \forall x \in [0, \infty)$, then least value of M, is 25

C) number of points of local extrema, of $F(x)$ is 1

D) $F(x)$ is monotonically decreasing in $(0, \infty)$

46. Let a, b, c be distinct non-zero complex numbers with $|a| = |b| = |c|$ and if each of the equations $az^2 + bz + c = 0, bz^2 + cz + a = 0$ has a root of modulus 1, then which of the following is/are true

A) $b^2 = ac$

B) $(a-b)^2 + (b-c)^2 + (c-a)^2 = 0$

C) $(a-c)^2 = (a-b)(b-c)$

D) $|a-b| = |b-c| = |c-a|$

47. Consider a rectangular hyperbola $xy = c^2$ and four distinct points $P_1(t_1), P_2(t_2), P_3(t_3), P_4(t_4)$ on it. then which of the following is/are true ?

A) If the chord P_1P_2 is parallel to the chord P_4P_3 & the chord P_1P_4 is parallel to the chord P_2P_3 then $t_2 = -t_4$

B) If the chord P_1P_2 is parallel to the chord P_4P_3 & the chord P_1P_4 is parallel to the chord P_2P_3 then $t_2 = t_4$

- C) If $P_1 P_2 P_3 P_4$ are the vertices of a rectangle then $t_1 t_2 = \pm 1$
 D) If $P_1 P_2 P_3 P_4$ are the vertices of a rectangle then $t_1 t_2 = \pm 2$
48. The tangents drawn to the parabola $y^2 = 4ax$ at the points $P(ap^2, 2ap)$, $Q(aq^2, 2aq)$ meet at R. Which of the following is/are true ?
- A) Area of the triangle PQR is $\frac{1}{2}|a^2(p-q)^3|$
 B) Area of the triangle PQR is $|a^2(p-q)^3|$
 C) Given that P & Q move so that the area of the triangle PQR is $4a^2$. Then locus of R is $y^2 = 4ax + 4a^2$
 D) Given that P & Q move so that the area of the triangle PQR is $4a^2$. Then locus of R is $x^2 = 4ay + 4a^2$
49. If a, b, c, d are the roots of $x^4 - x^3 - x^2 - 1 = 0$. $P(x) = x^6 - x^5 - x^3 - x^2 - x$ then which of the following is/are true ?
- A) $P(a) + P(b) + P(c) + P(d) = 6$ B) $P(a) + P(b) + P(c) + P(d) = 1$
 C) $\left(1 + \frac{1}{a}\right)\left(1 + \frac{1}{b}\right)\left(1 + \frac{1}{c}\right)\left(1 + \frac{1}{d}\right) = 0$ D) $\left(1 + \frac{1}{a}\right)\left(1 + \frac{1}{b}\right)\left(1 + \frac{1}{c}\right)\left(1 + \frac{1}{d}\right) = 1$

SECTION – III

(INTEGER ANSWER TYPE)

This section contains 5 questions. The answer is a single digit integer ranging from 0 to 9 (both inclusive).

Marking scheme +3 for correct answer, 0 if not attempted and 0 in all other cases

50. Let $f(x) = 4 \ln x + x^2 - ax$, $a \in R$
 If $f(x)$ has local extrema at x_1 and x_2 such that $x_1 < x_2$ and $x_1 \in (0, 1]$ & the local minimum value of $f(x_1) - f(x_2)$ is $m - n \ln 2$ ($m, n \in I^+$), then the value of $m + n = \dots$
51. Consider three digit integers N, with the two properties
 A) N is not exactly divisible by 2, 3 or 5
 B) no digit of N is exactly divisible by 2, 3 or 5
 Number of such integers N, is K then $\left[\frac{K}{5}\right] = \dots$
52. An isosceles triangle has its vertex at origin and its base, parallel to the x-axis with the vertices above the x-axis on the curve $y = 27 - x^2$. Then maximum area of the triangle is K square units, where $\left[\frac{k}{9}\right] = ([.] \text{ G.I.F})$
53. A straight line passes through the fixed point A(3,2) and meets the coordinate axes at P and Q. If the mid point of PQ is X, as the gradient of the line varies, X moves on the curve C. If the tangent at A to the curve C, meets the coordinates axes at $H(a, 0)$ and $K(0, b)$ then $\frac{a+b}{2} = \dots$
54. If A and B are two non-identity square matrices of same order with real entries such that $AA^T = I$, $BB^T = I$ & $|AB| < 0$ then the value of $|A+B| + |A^{-1} + B^{-1}| = \dots$

1	A	2	C	3	A	4	B	5	B
6	BD	7	ABC	8	ABCD	9	ABCD	10	BC
11	AB	12	BCD	13	AD	14	5	15	2
16	6	17	7	18	2				
19	A	20	C	21	D	22	B	23	B
24	ACD	25	BC	26	BC	27	AC	28	ABC
29	AB	30	AD	31	BD	32	5	33	4
34	8	35	2	36	3				
37	C	38	B	39	A	40	B	41	A
42	AB	43	ABCD	44	ABCD	45	BCD	46	ABCD
47	AC	48	AC	49	AC	50	7	51	0
52	6	53	5	54	0				



Sec: Sr. IIT

Race -7

Date: 23-02-24

PHYSICS

Section-1

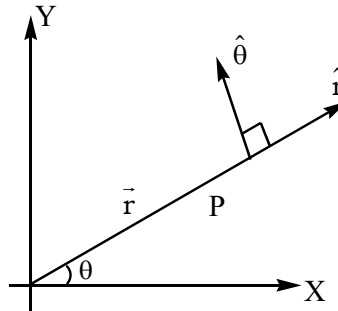
(Only one Option correct Type)

This section contains 10 Multiple Choice questions. Each Question has Four choices (A), (B), (C) and (D). Out of Which **Only One is correct.**

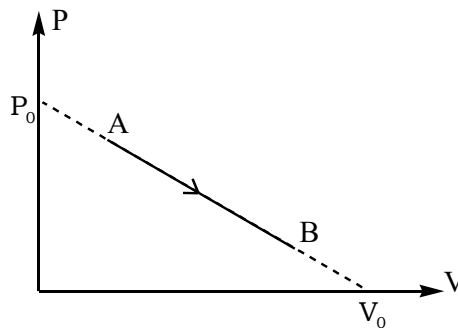
Marking scheme +4 for correct answer, 0 if not attempted and -2 in all other cases.

1. In the figure shown, p is a point in x-y plane whose position vector is $\vec{r}$. $\hat{r}$ and $\hat{\theta}$ are the unit vectors along $\vec{r}$ and perpendicular to $\vec{r}$ respectively in x-y plane as shown. Then

$$\left| \hat{r} \times \hat{\theta} + \frac{d\hat{r}}{d\theta} \times \frac{d\hat{\theta}}{d\theta} + \frac{d^2\hat{r}}{d\theta^2} \times \frac{d^2\hat{\theta}}{d\theta^2} + \dots + \frac{d^{10}\hat{r}}{d\theta^{10}} \times \frac{d^{10}\hat{\theta}}{d\theta^{10}} \right| =$$



- A) 0 B) 1 C) 10 D) 11
2. One mole of mono-atomic gas undergoes a linear process $A \rightarrow B$ shown in PV diagram. The volume of gas from which the process turns from endothermic to an exothermic is $\left(\frac{n}{2n-2} \right) V_0$. The value of 'n' is



- A) 5 B) 7 C) 6 D) 9
3. Two sources of equal EMF are connected in series and have different internal resistances r_1 and r_2 ($r_2 > r_1$). For the external resistance $R = ar_2 + br_1$ the potential difference across the terminals of one of the sources becomes equal to zero. Then $a - b =$
- A) 0 B) 2 C) 1 D) 3
4. A sound source is moving with uniform speed along a curve $y = A \sin kx$ in x-y plane:

Where x and y are the coordinates of the source. Three detectors are placed at $\left(\frac{\pi}{2K}, -10A\right)$, $\left(\frac{\pi}{2K}, 0\right)$ and $\left(\frac{\pi}{2K}, 10A\right)$. At the instant when the sound source crosses the point $\left(\frac{\pi}{2K}, A\right)$ the frequency registered by detectors are f_1, f_2, f_3 . If the waves detected are corresponding to the positions of source – $0 < x < \pi/2k$, then:

- A) $f_1 > f_2 > f_3$ B) $f_1 < f_2 < f_3$ C) $f_1 = f_2 = f_3$ D) $f_2 = \frac{f_1 + f_3}{2}$

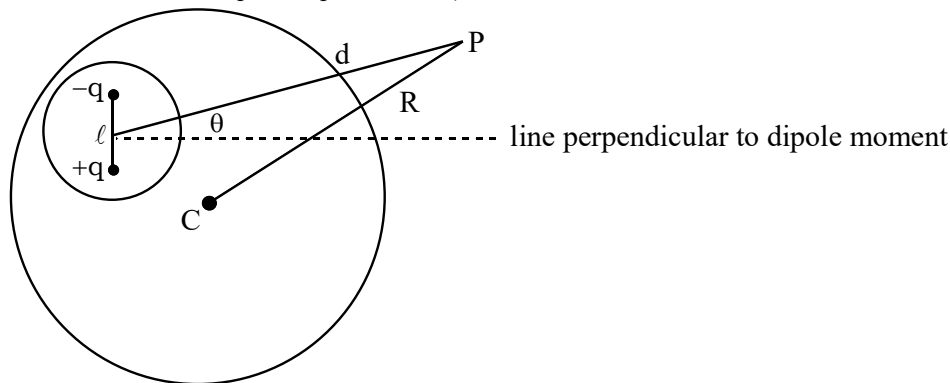
5. The superposition of two waves travelling in opposite directions given by the equations: $y_1 = a \sin(\omega t - kx)$ and $y_2 = 2a \sin(\omega t + kx)$ Takes place in a homogeneous string of cross section area s and density ρ . The resulting pattern can be thought as a superposition of a standing wave and progressive wave travelling left. The total mechanical energy between two successive antinodes is:

- A) $\frac{3\pi s \rho \omega^2 a^2}{2k}$ B) $\frac{\pi s \rho \omega^2 a^2}{2k}$ C) $\frac{5\pi s \rho \omega^2 a^2}{2k}$ D) $\frac{\pi s \rho \omega^2 a^2}{k}$

6. Consider two particles each of mass m at a distance r_0 from each other. One of them is given a velocity V perpendicular to line joining them and the other is simultaneously released from rest. For which of the following values of V the masses would be bound in orbital motion under their mutual gravitational forces

- A) $\sqrt{\frac{Gm}{r_0}}$ B) $\frac{5}{2} \sqrt{\frac{Gm}{r_0}}$ C) $2 \sqrt{\frac{Gm}{r_0}}$ D) $3 \sqrt{\frac{Gm}{r_0}}$

7. A spherical cavity is created in a neutral solid conducting sphere, inside the cavity a dipole is placed as shown in figure. Then (l is the distance between $-q$ and $+q$ and $l < d$)



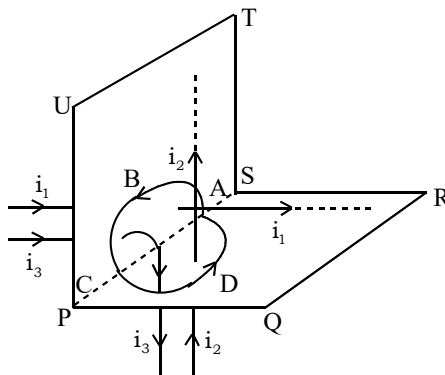
- A) Potential at P only due to charge induced on the inner surface of the cavity is $\frac{ql \sin \theta}{8\pi\epsilon_0 d^2}$
 B) Potential at P only due to charge induced on the inner surface of the cavity is $-\frac{ql \sin \theta}{8\pi\epsilon_0 d^2}$
 C) Electric field at P is zero
 D) Electric field at P is non zero

8. The plates of a parallel plate capacitor with no dielectric are connected to a voltage source. Now a dielectric of dielectric constant K is inserted to fill the whole space between the plates with voltage source remaining connected to the capacitor. Which of the following is **INCORRECT**?

- A) The energy stored in the capacitor will become K -fold
 B) The electric field inside the capacitor will decrease to K -times
 C) The force of attraction between the plates will increase to K^2 – times
 D) The charge on the capacitor will increase to K -times

9. Figure shows an amperian path ABCDA. Part ABC is in vertical plane PSTU while part CDA is in horizontal plane PQRS. Direction of circulation along the

path is shown by an arrow near point B and at D. $\oint \vec{B} \cdot d\vec{\ell}$ for this path according to Ampere's law will be



- A) $(i_1 - i_2 + i_3)\mu_0$ B) $(-i_1 + i_2)\mu_0$ C) $i_3\mu_0$ D) $(i_1 + i_2)\mu_0$
10. An electron makes a transition from $n=2$ to $n=1$ state in a hydrogen like atom. Which of the following is **INCORRECT**?
- A) magnetic field at the site of nucleus becomes $1/16$ times.
 B) magnetic field at the site of nucleus becomes 32 times.
 C) frequency of revolution of electron in the final orbit is more than the frequency of photon emitted.
 D) frequency of revolution of electron in the initial orbit is less than the frequency of photon emitted.

Section-2

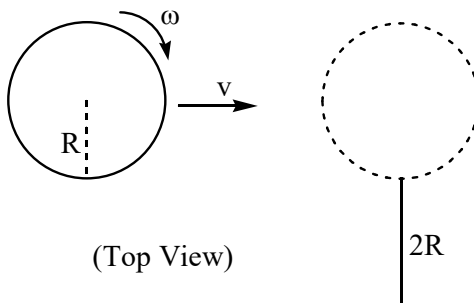
(Paragraph Type With Integer)

This section contains **6 Integer Type questions** relating to three paragraphs with two questions on each paragraph. The answer to each of the questions is a single-digit integer, ranging from 0 to 9. For each question you will be awarded 3 marks if you darken **ONLY** the bubble corresponding to the correct answer and zero mark if no bubbles are darkened. In all other cases, minus one (-1) mark will be awarded

Paragraph for Questions 11 & 12:

A disc of mass m and radius R slides across frictionless ice, as shown in Figure.

It has translational speed v to the right and rotational speed ω clockwise ($v = R\omega$). It grazes the "top" end of a rod of mass m and length $2R$ which is initially at rest on the ice. It sticks to the rod, forming a rigid object that looks like a lollipop. Then

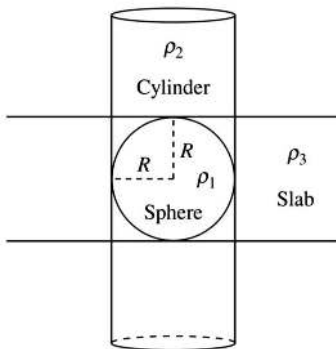


11. Then the resulting angular speed of the lollipop after the collision is $\frac{n\omega}{17}$. Find n =?
12. Energy lost during the collision is $(n/68)mR^2\omega^2$. Then find n = ?

Paragraph for Questions 13 & 14:

In figure a sphere with radius R is centered at the origin, an infinite cylinder with radius R has its axis along the z axis, and an infinite slab with thickness $2R$ lies between the planes $z = -R$ and $z = R$. The uniform volume densities of these objects are ρ_1, ρ_2 , and ρ_3 respectively. The objects are

superposed on each other; the densities add where the objects overlap. It is found that the resultant gravitational field is zero everywhere throughout the volume of the sphere. If $\rho_1 = \rho$, then



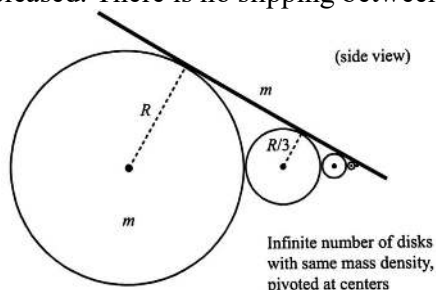
13. $\rho_2 = -\frac{n}{3}\rho$. Find n = ?

14. $\rho_3 = -\frac{n}{3}\rho$. Find n = ?

Paragraph for Questions 15 & 16:

An infinite number of disks are cut out from a flat sheet of metal (so they all have the same mass density per unit area), and they are situated in a horizontal line with their centres at the same height, as shown in figure (side view). They are free to spin around their (fixed) centres. There is no friction between any of the disks; equivalently, assume that they are separated by infinitesimal distance. The largest disk has radius R , and the radius of each successive disk is $1/3$ of the radius of the disk to its left.

A stick, whose mass m is the same as the mass of the largest disk, is placed on top of the disks (touching them all) and is released. There is no slipping between the stick and the discs. Then



15. Then acceleration of the stick during the time it is still in contact with all the discs is $\frac{ng}{25}$. Find n = ?

16. The angular acceleration of discs as we move from left to right are in geometric progression with common ratio n . Find n = ?

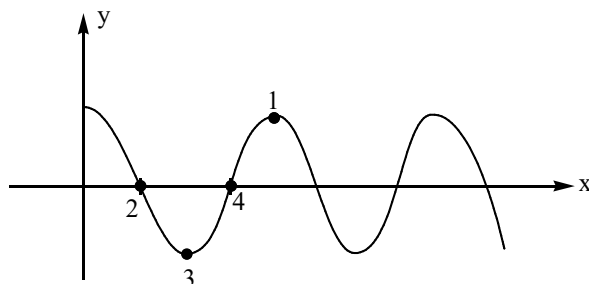
Section-3

(Matching List Type)

This section contains four questions, each having two matching lists (List-I & List-II). The options for the **correct match** are provided as (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

Marking scheme +4 for correct answer, 0 if not attempted and -2 in all other cases.

17. Sound wave represented as a harmonic displacement wave travels in negative X direction. Y axis represents particle displacement List I has listed a number of locations along x axis. List II has values of pressure variation (Δp) and energy density at those locations Match Column-I with Column-II



	List-I		List-II
a)	1	p)	Δp maximum
b)	2	q)	Δp minimum
c)	3	r)	Δp zero
d)	4	s)	Energy density maximum
		t)	Energy density minimum

Mark Correct Matching:

A) A-RT B-PT C-RS D-PS

B) A-PT B-PS C-RT D-RS

C) A-RT B-PS C-RT D-QS

D) A-PR B-RT C-RT D-PQ

18. Column I given the process undergone by an ideal gas of adiabatic exponent γ and column II given the molar specific heat as a function of volume of gas. Match them

	Column I		Column II
A	$p = p_0 + \frac{\alpha}{V}$ $p, V \rightarrow$ pressure & volume of gas; p_0, α are positive constants	P	$\frac{R}{\gamma-1} + \frac{R}{\alpha V}$
B	$T = T_0 + \alpha V$, $T, V \rightarrow$ temperature and volume of gas; T_0, T_0, α are constants	Q	$\frac{R}{\gamma-1} + \frac{R}{1+\alpha V}$
C	$T = T_0 e^{\alpha V}$ $T, V \rightarrow$ temperature and volume of gas; T_0 and α are constants	R	$\frac{\gamma R}{\gamma-1} + \frac{\alpha R}{p_0 V}$
D	$P = P_0 e^{\alpha V}$ $P, V \rightarrow$ pressure and volume of gas; P_0, α are constants	S	$\frac{\gamma R}{\gamma-1} + \frac{RT_0}{\alpha V}$

Mark Correct Matching:

A) A-Q B-S C-P D-R B) A-R B-S C-P D-Q

C) A-S B-P C-R D-Q D) A-P B-S C-R D-Q

19. A person in a lift is holding a water jar, which has a small hole at the lower end of its side. When the lift is at rest, the water jet coming out of the hole hits the floor of the lift at a distance d of 1.2 m from the person. In the following, state of the lift's motion is given in List-I and the distance where the water jet hits the floor of the lift is given in List-II. Match the statements from List-I with those in List-II and select the correct answer using the code given below the lists.

	List-I		List-II
P)	Lift is accelerating vertically up	1)	$d = 1.2$ m
Q)	Lift is accelerating vertically down with an acceleration less than the gravitational acceleration	2)	$d > 1.2$ m
R)	Lift is moving vertically up with constant speed	3)	$d < 1.2$ m
S)	Lift is falling freely	4)	No water leaks out of the jar

A) P-2, Q-3, R-2, S-4 B) P-2, Q-3, R-1, S-4

20. C) P-1, Q-1, R-1, S-4 D) P-2, Q-3, R-1, S-1
Density of a planet is two times the density of earth. Radius of this planet is half of that of earth. Match, the followings. (As compared to earth) Assuming default reference for potential energy.

	Column – I		Column – II
A	Acceleration due to gravity on this planet's surface	P	Half
B	Gravitational potential on the surface	Q	Same
C	Gravitational potential at centre	R	Two times
D	Gravitational field strength at centre	S	Four times

- A) A – P B – Q C – R D – S
B) A – R B – P C – S D – Q
C) A – Q B – S C – P D – Q
D) A – Q B – P C – P D – Q

MATHEMATICS

Section-1

(Only one Option correct Type)

This section contains 10 Multiple Choice questions. Each Question has Four choices (A), (B), (C) and (D). Out of Which Only One is correct.

Marking scheme +4 for correct answer, 0 if not attempted and -2 in all other cases.

21. $S_{20} = ab + (a+x)(b+x) + (a+2x)(b+2x) + \dots \text{up to 20 terms}$,
Then $\left[\frac{\text{sum of last 10 terms} - \text{sum of first 10 terms}}{20\text{th term} - \text{first term}} \right]$, where $[.]$ denotes greatest integer function, if $a = 2 + \sqrt{3}, b = 2 - \sqrt{3}, x = e$ is
A) 1 B) 3 C) 5 D) 7
22. If $I_1 = \int_a^b (f^2(x) - f^2(a)) dx$ and $I_2 = \int_{f(a)}^{f(b)} (bx - xf^{-1}(x)) dx$, then (f is invertible function)
A) $I_1 = 2I_2$ B) $I_2 = 2I_1$ C) $I_1 = I_2$ D) $I_1 \neq I_2$
23. The tangent at a point P of a curve meets y-axis at B, the line through P parallel to y-axis meets x-axis at A. If the area of ΔAOB is a non-zero constant, then the curve is a/an
A) parabola B) ellipse C) hyperbola D) circle
24. The vertex A of acute triangle ABC is equidistant from the circum centre O and the orthocentre H. Then $\angle A = \dots$
A) $\frac{\pi}{6}$ B) $\frac{\pi}{4}$ C) $\frac{\pi}{3}$ D) $\frac{\pi}{2}$
25. Let $A(z_1)$ be a point on the circle whose centre is at origin and radius 2 units with $\arg(z_1) = \frac{\pi}{4}$. Let z_2 be the midpoint of O, z_1 where O is origin. Let w be the 11th root of unity ($w \neq 1$). Then $\sum_{r=1}^{10} |z_1 + w^r z_2|^2 = \dots$
A) 44 B) 46 C) 48 D) 50
26. There are two dice one is red and other is white, red die is loaded so that the probability of getting 6 is $\frac{1}{3}$, the remaining outcomes being equally likely among themselves. The white die is fair. Two persons A and B play the game choosing white die by A and red die by B on the condition who throws highest number will win and in case of equal result person with white die will win. Then the game is

(Note: The game is in favour of someone indicates their probability of winning is more and fair indicates both have got equal probability of winning).

- A) In favour of A B) In favour of B
C) fair D) In sufficient data to decide the nature of the game
27. The volume of the tetrahedron of which a pair of opposite edges, each of length 9 units are on the lines $\vec{r} = i + 2k + t(i + 8j + 4k), t \in R$ and $\vec{r} = -2i + 3j - k + s(4i + 4j + 7k), s \in R$ respectively is ____
- A) $\frac{3}{2}$ B) 3 C) $\frac{9}{2}$ D) $\frac{27}{2}$
28. Let L be the projection of the line $\frac{x+1}{2} = \frac{y+2}{3} = \frac{z+3}{4}$ on the plane $x - 2y + 3z - 4 = 0$. Then the equation of the plane passing through L and parallel to the line $x = y = z$ is
- A) $23x - 6y - 17z = 40$ B) $5x - 2y - 3z + 4 = 0$
C) $13x + 6y - 19z + 14 = 0$ D) $7x + 6y - 13z + 6 = 0$
29. If $f(x + f(y)) = y + f(x + 2019)$ and $\sum_{k=1}^{2019} f(k) = (abcd)(2019)$ where abcd is a four digit number, then $b + d - a - c =$
- A) 3 B) 4 C) 5 D) 6
30. If a, b are two positive integers such that $\frac{(a+b)^7 - a^7 - b^7}{ab(a+b)(a^2 + ab + b^2)} = 7$ and $\sum_{r=1}^6 {}^7C_r a^{7-r} b^r$ is divisible by 7^4 where none of a, b, a+b is divisible by 7, then least possible value of a+b is ____
- A) 17 B) 18 C) 19 D) 20

Section-2

(Paragraph Type With Integer)

This section contains **6 Integer Type questions** relating to three paragraphs with two questions on each paragraph. The answer to each of the questions is a single-digit integer, ranging from 0 to 9. For each question you will be awarded 3 marks if you darken ONLY the bubble corresponding to the correct answer and zero mark if no bubbles are darkened. In all other cases, minus one (-1) mark will be awarded

Paragraph For Questions 31 & 32

Let $P_1(x) = x^2 - 2$, $p_k(x) = p_1(p_{k-1}(x))$, $k = 2, 3, \dots$

31. All the real roots of the equation $p_4(x) = x$ lies in the interval $[a, b]$. then $b - a =$
32. If the equation $p_6(x) = x$ has 2^k distinct real roots, then $k =$ _____

Paragraph for Questions 33 & 34

Let n be the number of ordered 4-tuples (a, b, c, d) in which $(2019)^2$ is LCM of positive integers a, b, c, d

33. The digit in the units places of n is
34. Let n_1 be the number of ordered 4-tuples (a_1, b_1, c_1, d_1) in which n is LCM of positive integers a_1, b_1, c_1, d_1 and n_r be the number of ordered 4-tuples (a_r, b_r, c_r, d_r) in which n_{r-1} is LCM of positive integers a_r, b_r, c_r, d_r for $r = 2, 3, \dots$. The digit in the 1000's place of n_{10} is.....

Paragraph for Questions 35 & 36

Let $A = [a_{ij}]_{3 \times 3}, B = [b_{ij}]_{3 \times 3}, X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$, A is symmetric

$$X^T A X = x_1^2 - 2x_2^2 - 3x_3^2 + 4x_1x_2 + 6x_1x_3 - 8x_2x_3 \text{ and } AB = \begin{bmatrix} 1 & -5 & 3 \\ -2 & 10 & -6 \\ 1 & 9 & -11 \end{bmatrix}$$

35. The value of $b_{23} =$ _____

36. The system of equations $BX = \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix}$ has a solution with $x_1 = 2$, then $x_2 =$ _____

Section-3

(Matching List Type)

This section contains four questions, each having two matching lists (List-I & List-II). The options for the **correct match** are provided as (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

Marking scheme +4 for correct answer, 0 if not attempted and -2 in all other cases.

37. If $P(t, t+1), t \in R$ is a variable point on a straight line and C_1, C_2, C_3, C_4 be the curves given by $C_1 : x^2 + y^2 = 2; C_2 : y^2 = 8x; C_3 : x^2 = 8y; C_4 : x^2 + 2y^2 = 1$, then match the following if P is interior point of curve

Column-I (curve)		Column-II (exhaustive set of values of t)	
A)	C_1	P)	$(3 - 2\sqrt{2}, 3 + 2\sqrt{2})$
B)	C_2	Q)	$(-1, -\frac{1}{3})$
C)	C_3	R)	$(\frac{-1 - \sqrt{3}}{2}, \frac{-1 + \sqrt{3}}{2})$
D)	C_4	S)	$(4 - 2\sqrt{6}, 4 + 2\sqrt{6})$

A) $A-S, B-Q, C-R, D-P$

B) $A-R, B-Q, C-P, D-S$

C) $A-S, B-Q, C-P, D-R$

D) $A-R, B-P, C-S, D-Q$

38. Match the following

Column-I		Column-II	
A)	The area bounded by the curves $y = \sin x $, the lines $ x = \pi$, x-axis	P)	1
B)	The area bounded by the curves $y = \sin^{-1} x$, the lines $ y = \frac{\pi}{2}$, $x=0$ is	Q)	2
C)	The area bounded by the curves $y = \left[\frac{x^2}{64} + 2 \right]$, $([.]$ is Greatest integer function), $y = x - 1$, $x = 0$ above x-axis is	R)	4
D)	The area bounded by the curves $y = x - 1 $ and $y = 3 - x $ is	S)	8

A) $A - R, B - Q, C - Q, D - R$ B) $A - Q, B - P, C - S, D - R$ C) $A - Q, B - P, C - R, D - S$ D) $A - R, B - Q, C - R, D - R$

39. Match the following

Column-I		Column-II	
A)	Let $f: R \rightarrow R$ and $g: R \rightarrow R$ given by $f(x) = x + 1$, $g(x) = x^2 + 1$, Define $h: R \rightarrow R$ by $h(x) = \begin{cases} \max\{f(x), g(x)\}, & \text{if } x \leq 0 \\ \min\{f(x), g(x)\}, & \text{if } x > 0 \end{cases}$ The number of points where $h(x)$ is not differentiable is	P)	1
B)	Let $f(x) = [\tan x [\cot x]]$, $x \in \left[\frac{\pi}{12}, \frac{\pi}{2} \right)$, where $[.]$ denotes greatest integer function, then number of points where $f(x)$ is discontinuous is	Q)	2
C)	Let f and g are differentiable functions satisfying $g^1(a) = 2$, $g(a) = b$ and $f \circ g = I$ (identify function) Then $4f^1(b) = \underline{\hspace{1cm}}$	R)	3
D)	If $x^2 + y^2 = 2$ and $y'' + A.y^{-3} = 0$, then $A =$	S)	4

A) $A - R, B - R, C - Q, D - Q$ B) $A - S, B - P, C - Q, D - R$ C) $A - R, B - P, C - S, D - Q$ D) $A - S, B - R, C - S, D - Q$

40. Let $f(x) = \begin{cases} e^x, & 0 \leq x \leq 1 \\ 2 - e^{x-1}, & 1 < x \leq 2, \text{ and } g(x) = \int_0^x f(t) dt, x \in [1, 3] \\ x - e, & 2 < x \leq 3 \end{cases}$

Match the following for $g(x)$ at the values of 'x' given in column-I

Column-I		Column-II	
A)	$x = 1$	P)	Local maximum
B)	$x = 2$	Q)	local minimum
C)	$x = 1 + \ln 2$	R)	no local maximum

D) $x = e$

S) no local minimum

A) $A-P, B-Q, C-P, D-Q$ B) $A-P, B-Q, C-RS, D-RS$ C) $A-Q, B-P, C-Q, D-P$ D) $A-RS, B-RS, C-P, D-Q$ **CHEMISTRY****Section-1****(Only one Option correct Type)**

This section contains 10 Multiple Choice questions. Each Question has Four choices (A), (B), (C) and (D). Out of Which Only One is correct.

Marking scheme +4 for correct answer, 0 if not attempted and -2 in all other cases.

41. Which reaction is least likely to occur

A) $XeF_2 + PF_5 \rightarrow [XeF]^+ [PF_6]^-$ B) $XeF_2 + KF \rightarrow K[XeF_3]$ C) $XeF_4 + SbF_5 \rightarrow [XeF_3]^+ [SbF_6]^-$ D) $XeF_6 + NaF \rightarrow NaXeF_7$

42. An ion M^{2+} forms the complex $[M(H_2O)_6]^{2+}$, $[M(en)_3]^{2+}$ and $[MBr_6]^{4-}$. Match these complexes with the appropriate colours respectively

A) Green, Blue and red

B) Blue, red and green

C) Green, red and blue

D) Red, blue and green

43. A molecule whose central atom has sp^3d^2 hybridization cannot have which of the following shapes, in its molecules.

A) Tetrahedral

B) Square pyramid

C) Square planar

D) Octahedral

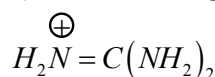
44. The shielding effect explains why

A) The electronegativity of fluorine is greater than that of bromine

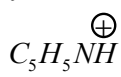
B) The electronegativity of fluorine is greater than boron

C) The electronegativity of fluorine is smaller than gallium

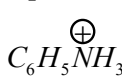
D) The electronegativity of sodium and potassium are equal



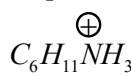
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2



3



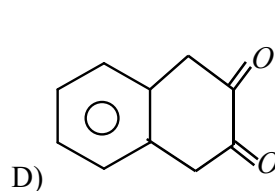
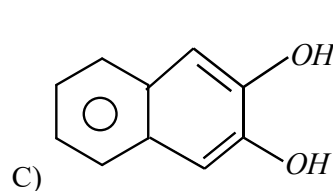
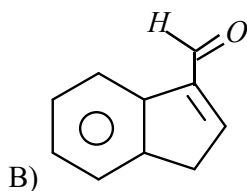
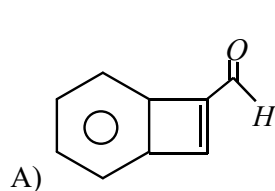
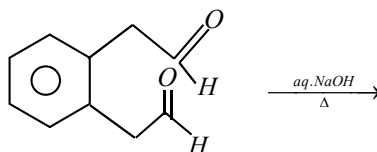
4

45.

The correct order of pKa values of above species is

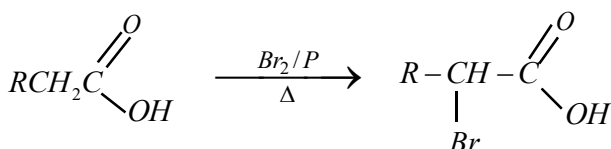
A) $1 < 2 < 3 < 4$ B) $3 < 2 < 4 < 1$ C) $4 < 3 < 2 < 1$ D) $2 < 4 < 1 < 3$

46. What is the major product of this reaction ?



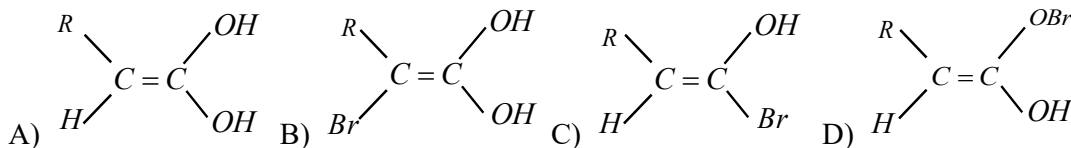
47. The enthalpy of a reaction at 27°C is 15 KJ/mole . The reaction is feasible if the entropy change is

A) $15 \text{ JK}^{-1} \text{ mol}^{-1}$ B) $50 \text{ JK}^{-1} \text{ mol}^{-1}$ C) Greater than $50 \text{ JK}^{-1} \text{ mol}^{-1}$ D) Lesser than $50 \text{ JK}^{-1} \text{ mol}^{-1}$



48.

The above reaction occurs through the formation of following intermediate



49.

The potential of $Hg|Hg(I)nitrate||Hg(I)nitrate|Hg$ at 25°C is nearly
(0.01M) (0.1M)

A) 0.03 V B) 0.06 V C) 0.09 V D) 0.12 V

50.

The wavelength of electron in one of the Bohr's orbit of excited hydrogen atom is $13.32A^\circ$. When it falls into a lower orbit its wavelength is found to be $6.66A^\circ$. The wavelength of spectral line emitted in this transition is

A) 121.5 nm B) 486.2 nm C) 652.2 nm D) 565.3 nm

Section-2

(Paragraph Type With Integer)

This section contains **6 Integer Type questions** relating to three paragraphs with two questions on each paragraph. The answer to each of the questions is a single-digit integer, ranging from 0 to 9. For each question you will be awarded **3** marks if you darken **ONLY** the bubble corresponding to the correct answer and zero mark if no bubbles are darkened. In all other cases, minus one (-1) mark will be awarded

Paragraph for Questions 51 & 52

An ore 'P' on roasting with sodium carbonate in presence of air gives two compounds Q and R. The solution of Q in dil HCl on treating with $K_4[Fe(CN)_6]$ gives blue colour of S. The aqueous solution R first on treating with conc H_2SO_4 followed by KCl gives an orange red compound W. The solution of W on treating with oxalic acid and then with excess of potassium oxalate gives blue compound X.

51. The change in the oxidation state of metal ion present in R during roasting the ore P with sodium carbonate in presence of air is

52. In the blue compound (X) the number of charges on the anion is

Paragraph For Questions 53 & 54

0.5 gm of a weak base BOH present in 50 ml solution is titrated with 0.1 M HCl. The pH at half-neutralization is found to be 9.0 and pH at equivalence point is found to be 5.15. The density of BOH solution is 0.81 g/mL

53. Molecular weight of BOH is $x \times 10^y$. $x + y$ is ____

54. Molality of BOH solution is $\frac{1}{2^x}$. x value is ____

Paragraph For Questions 55 & 56

Two aromatic compounds (benzene derivatives) A & B of formula $C_8H_{11}N$ on heating with $CHCl_3$ in presence of alcoholic KOH forms an unpleasant smelling compound. Compound 'A' on reaction with $NaNO_2 + HCl$ at $0^\circ C$ followed by treatment with phenol forms an intense red coloured azodye. But compound 'B' doesn't form an azodye on reaction with $NaNO_2 + HCl$ followed by treatment with phenol.

55. The number of isomers possible for compound 'A' (including itself) are ____

56. The number of isomers possible for compound 'B' (including itself) are ____

Section-3

(Matching List Type)

This section contains four questions, each having two matching lists (List-I & List-II). The options for the **correct match** are provided as (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

Marking scheme +4 for correct answer, 0 if not attempted and -2 in all other cases.

57. Match the following metals ions given in column-I with the property given in column – II

	Column – I(metal ion)		Column – II (property)
A	Cu^{2+}	P	With H_2S gives black ppt
B	Hg^{2+}	Q	With KI gives white ppt
C	Ni^{2+}	R	With NaOH gives green ppt
D	Co^{2+}	S	With excess KCN gives colour less solution
		T	With excess amount of NH_3 form complex compound

- A) A-PQRS, B-P, C-PQRT, D-PRT
 B) A-PQST, B-P, C-PRT, D-PT
 C) A-PQST, B-PQ, C-PQR, D-PQT
 D) A-PQRS, B-R, C-PRT, D-PQ

58. Matrix matching

Column – I(Solution)		Column – II(Property)	
A)	A solution of vapour pressure 110 mm, containing one mole of liquid A ($P^\circ = 150$ mm) and two moles of liquid B ($P^\circ = 75$ mm)	P)	Solvent- solvent , solute-solute attractive forces are stronger than solvent-solute attractive force
B)	A solution of vapour pressure 100 mm containing a liquid 'X' ($P^\circ = 120$ mm) & liquid 'Y' ($P^\circ = 80$ mm) with 60 mole percent of 'X' in vapour above the solution	Q)	$\Delta^\circ H$ mixing is +ve
C)	A solution of vapour pressure 100 mm, containing 3 moles of liquid M ($P^\circ = 150$ mm) and 2 moles of liquid 'N' ($P^\circ = 150$ mm)	R)	Solution obeys Raoult's law
D)	A solution of vapour pressure 110 mm containing 2 moles of liquid 'E' ($P^\circ = 90$ mm) and 4 moles of liquid F ($P^\circ = 120$ mm)	S)	Solvent-Solvent, Solute- Solute attractive forces are weaker than solvent – solute attractive force
		T)	$\Delta^\circ H$ mixing is negative

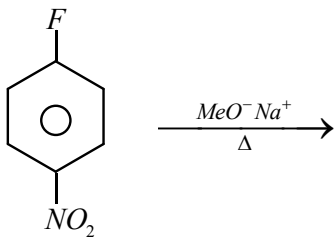
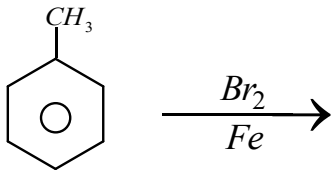
A) A-PQ, B-ST, C-R, D-R

B) A-PQ, B-R, C-ST, D-R

C) A-ST, B-R, C-PQ, D-R

D) A-R, B-R, C-PQ, D-ST

59. Matrix matching

Column – I		Column – II Inter mediate (S) involved in the formation of product	
A)	$H_3C-C \equiv C-CH_3 \xrightarrow[Liq.NH_3]{Na} \rightarrow$	P)	Free radical
B)		Q)	Carbene
C)	$CHBr_3 \xrightarrow[\Delta]{KOH} \rightarrow$	R)	Carbocation
D)		S)	Carbanion
		T)	Radical anion

A) A-QS, B-PST, C-S, D-R

C) A-R, B-PST, C-R, D-QS

B) A-PST, B-QS, C-S, D-R

D) A-PST, B-S, C-QS, D-R

60. Matrix matching

Column – I		Column – II (reagent used for separation)	
A)	Phenol + aniline in ether	P)	Ammonical cuprous chloride
B)	Benzyl alcohol + O-cresol in ether	Q)	Ammonical $AgNO_3$
C)	1-butyne + 2-butyne	R)	Aqueous $NaHCO_3$
D)	Phenol + 2,4-dinitrophenol in ether	S)	Aqueous HCl
		T)	Aqueous NaOH

A) A-PQ, B-T, C-ST, D-R

B) A-ST, B-T, C-R, D-PQ

C) A-ST, B-T, C-PQ, D-R

D) A-PQ, B-ST, C-R, D-T

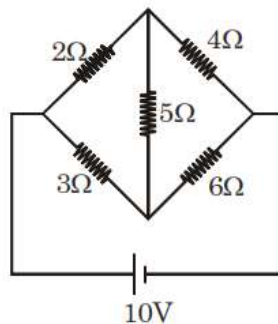
1	D	2	A	3	B	4	B	5	C
6	A	7	C	8	B	9	D	10	A
11	9	12	7	13	2	14	1	15	8
16	3	17	C	18	B	19	C	20	D
21	C	22	A	23	C	24	C	25	B
26	C	27	A	28	C	29	B	30	C
31	4	32	6	33	5	34	4	35	2
36	2	37	D	38	D	39	A	40	D
41	B	42	B	43	A	44	A	45	B
46	B	47	C	48	C	49	A	50	B
51	3	52	3	53	3	54	3	55	9
56	6	57	B	58	B	59	D	60	C


PHYSICS
SECTION - I
(MULTIPLE CORRECT CHOICE TYPE)

This section contains **8 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE is/ are correct**.

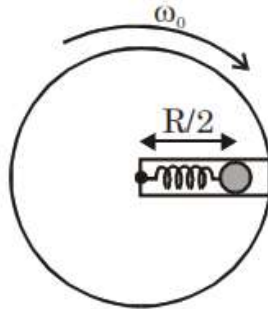
Marking scheme +4 for correct answer, 0 if not attempted and -2 in all other cases.

- A particle of mass m moves along a circular orbit in a central potential field given by $U(r) \propto r^2$. Assuming Bohr's quantization conditions to be valid. Choose **CORRECT** statement(s) regarding this.
 - Ratio of radii of fourth Bohr orbit ($n = 4$) and ground state orbit is 4
 - Ratio of radii of fourth Bohr orbit ($n = 4$) and ground state orbit is 2
 - Energy levels are equally spaced
 - Separation between successive energy levels increase as n increases
- Consider a circuit as shown in figure. Then choose **CORRECT** statement(s) :-

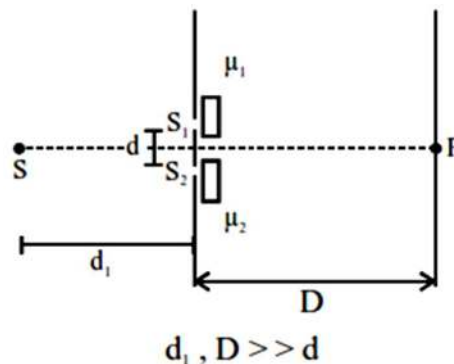


- If 4Ω resistance is shorted (resistance less wire connected in parallel to it), the current through battery will increase
 - If 4Ω resistor is shorted, the current through 5Ω resistance will increase
 - If instead of 4Ω , 5Ω resistor is shorted, current through battery will decrease
 - If instead of 4Ω , 3Ω resistor is shorted, current through battery will decrease
- A vernier calliper having no zero error has 10 division on vernier scale coinciding with 9 main scale division. It is made of a material whose coefficient of linear expansion is $\alpha = 10^{-3} \text{ } ^\circ\text{C}^{-1}$. At 0°C each main scale division = 1mm. An object has a length of 10 cm at temperature 0°C and its linear expansion coefficient is $\alpha_1 = 10^{-4} \text{ } ^\circ\text{C}^{-1}$. The experiment is conducted at 50°C
 - The reading of main scale is 40 mm
 - The reading of main scale is 95 mm
 - The reading of vernier scale is 2
 - The reading of vernier scale is 3
 - A uniform disc of mass ' m ' and radius ' R ' is free to rotate in horizontal plane about a vertical smooth fixed axis passing through its centre. There is a smooth groove of small width along the radius of the disc and one small ball of same mass is attached to a spring as shown in figure. Initially spring is in

its natural length of $\frac{R}{2}$. The disc is given initial angular velocity ω_0 and released (spring constant $k = \frac{m\omega_0^2}{2}$). Then select correct statement(s) :



- A) Angular velocity of disc when elongation in spring becomes $\frac{R}{2}$ will be $\frac{\omega_0}{2}$
- B) Radial velocity of ball when elongation in spring becomes $\frac{R}{2}$ will be $\frac{\omega_0 R}{2}$
- C) Rate of change of energy of spring with respect to time when elongation in spring becomes $\frac{R}{2}$ will be $\frac{m\omega_0^3 R^2}{8}$
- D) Rate of change of energy of spring with respect to time when elongation in spring becomes $\frac{R}{2}$ will be $\frac{m\omega_0^3 R^2}{2}$
5. When an air bubble rises from the bottom of a deep lake to a point just below the water surface, the pressure of air inside the bubble :-
 A) is greater than the pressure outside it
 B) is less than the pressure outside it
 C) increases as the bubble moves up
 D) decreases as the bubble moves up
6. In Young's experiment, the upper slit is covered by a thin glass plate of refractive index $\frac{4}{3}$ and of thickness 9λ , where λ is the wavelength of light (in air) used in the experiment. The lower slit is also covered by another glass plate of thickness 2λ & refractive index $\frac{3}{2}$, as shown in figure. If I_0 is the intensity at point P due to slits S_1 & S_2 each, then : (Line SP is bisector of $S_1 S_2$)



- A) Intensity at point P is $4I_0$
- B) Two fringes have been shifted in upward direction after insertion of both the glass plates together

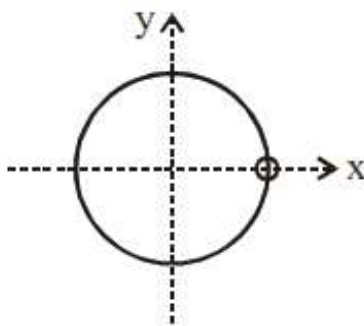
C) Optical path difference between the waves from S_1 & S_2 at point P is 2λ

D) If the source S is shifted upwards by a very small distance d_2 then the fringe originally at P after inserting the plates, shifts downward by $D\left(\frac{d_2}{d_1}\right)$

7. The equation of a transverse wave travelling along a string is given by $y = 0.4 \sin[\pi(0.5x - 200t)]$ where y and x are measured in cm and t in second. Suppose that you clamp the string at two points which are L cm apart and you observe a standing wave of the same wavelength as above transverse wave. For what values of L less than 10 cm is this possible?

A) 2 cm B) 4 cm C) 9 cm D) 8 cm

8. A small bead of mass m is threaded on a smooth circular ring of radius a in xy plane as shown. The bead experiences a force whose potential energy is given by $U = C\left(\frac{x^2}{a^2} + \frac{y^2}{b^2}\right)\{b > a\}$. Initially the bead is at $x = a$ at rest as shown. Neglect gravity. Choose the **CORRECT** statement(s) :-



- A) At initial position, normal force $= \frac{2c}{a}$
 B) At initial position, there is no acceleration
 C) Initially if the bead was at $y = a$ at rest, it's normal force is $\frac{2ca}{b^2}$
 D) Initially if the bead was at $y = a$ at rest, there is no acceleration

SECTION - II

(COMPREHENSION TYPE)

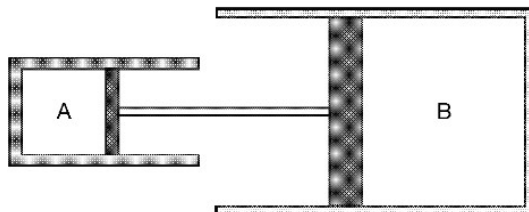
This section contains **4 groups of questions**. Each group has 2 multiple choice questions based on a paragraph. Each question has 4 choices A), B), C) and D) for its answer, out of which **ONE OR MORE** is/are **correct**.

Marking scheme +4 for correct answer, 0 if not attempted and -2 in all other cases.

Paragraph for Questions 9 & 10

Two fixed and horizontal cylinders A and B having pistons (both massless) of cross sectional area 100 cm^2 and 200 cm^2 respectively, are connected by massless rod. The piston can move freely without friction. The cylinder A contains 100 gms of an ideal gas ($\gamma = 1.5$) at pressure 10^5 N/m^2 and temperature T_0 . The cylinder B contains identical gas at same temperature T_0 but has different mass. The pistons are held at the state such that volume of gas in cylinder A and cylinder B are same and is equal to 10^{-2} m^3 . The walls and piston of cylinder A are thermally insulated where as gas in cylinder B is maintained at constant temperature T_0 . The whole system is in vacuum. Now the pistons are slowly released and they move towards left and mechanical

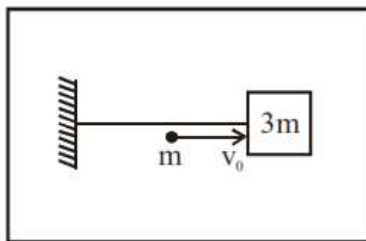
equilibrium is reached at the state when the volume of gas in cylinder A becomes $25 \times 10^{-4} \text{ m}^3$. (Assume that process is slow)



9. Which of the following are **CORRECT** :
- A) The mass of gas in cylinder B is 2 kg
 - B) The mass of gas in cylinder B is 1 kg
 - C) The change in internal energy of gas in cylinder A is 2000 J
 - D) The change in internal energy of gas in cylinder A is 1000 J
10. Which of the following are **CORRECT** :
- A) The compressive force in the connecting rod at equilibrium is 8000 N
 - B) The compressive force in the connecting rod at equilibrium is 4000 N
 - C) Final pressure of gas in cylinder A is $8 \times 10^5 \text{ N/m}^2$
 - D) Final pressure of gas in cylinder A is $4 \times 10^5 \text{ N/m}^2$

Paragraph for Questions 11 & 12:

A massless elastic cord (that obeys Hooke's law) will break if the tension in the cord exceeds T_{max} . One end of the cord is attached to a fixed point, the other is attached to an object of mass $3m$ as shown in the figure. If a second, smaller object of mass m moving at an initial speed v_0 strikes the larger mass and the two stick together, the cord will stretch and break, but the final kinetic energy of the two masses will be zero. If instead the two collide with a perfectly elastic one dimensional collision, the cord will still break and the larger mass will move off with a final speed of v_f . All motion occurs on a horizontal frictionless surface. (assume that Hooke's law is obeyed until the cord breaks)



11. Mark the **CORRECT** statement(s)
- A) Ratio of $\frac{v_f}{v_0}$ is given by $\frac{1}{\sqrt{6}}$
 - B) Ratio of $\frac{v_f}{v_0}$ is given by $\frac{1}{\sqrt{2}}$
 - C) Impulse offered by bigger block on the smaller block in second case when collision is elastic is $\frac{mv_0}{2}$
 - D) Impulse offered by bigger block on the smaller block in second case when collision is elastic is $\frac{3mv_0}{2}$

12. Mark the **CORRECT** statement(s) :-

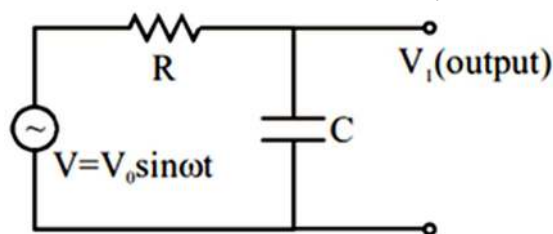
- A) Ratio of the total kinetic energy of the system of two masses after the perfectly elastic collision and the cord has broken to the initial kinetic energy of the smaller mass prior to the collision is $3/4$
- B) Ratio of the total kinetic energy of the system of two masses after the perfectly elastic collision and the cord has broken to the initial kinetic energy of the smaller mass prior to the collision is $1/4$
- C) In 1st case of inelastic collision, energy stored in spring when velocity of the combined mass (smaller & larger mass) have half of their maximum speed (maximum speed i.e. speed just after collision) is $\frac{3}{32}mv_0^2$
- D) In 1st case of inelastic collision, energy stored in spring when velocity of the combined mass (smaller & larger mass) have half of their maximum speed (maximum speed i.e. speed just after collision) is $\frac{mv_0^2}{32}$

Paragraph for Questions 13 & 14:

Consider the circuit shown in the diagram. This type of circuit is known as low pass circuit. Let I_0 be the current amplitude through the circuit.

$I_0 = \frac{V_0}{Z}$ Output voltage V_1 across capacitor is $V_1 = I_0 X_C$. The ratio $\frac{V_1^2}{V_0^2} = \frac{1}{1 + \omega^2 R^2 C^2}$. If

$\omega RC \gg 1$ then $\frac{V_1^2}{V_0^2} \propto \frac{1}{\omega^2}$. So doubling ω decrease $\frac{V_1^2}{V_0^2}$ by a factor of 4.



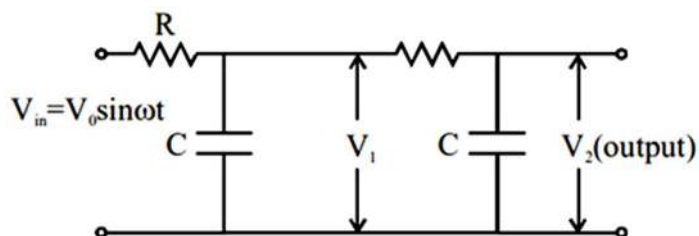
13. The physical reason why V_1 decrease with increasing frequency is ?

- A) For very low frequency, the impedance of capacitor is very large. The fixed impedance of resistor is negligible in comparison, so essentially all of the V_0 drops across capacitor, which is the voltage V_1 . On the other hand for very high frequency, the impedance of capacitor is very small, it is essentially short circuit. Therefore V_0 voltage drops across resistor, very little drops across capacitor
- B) For very low frequency, the impedance of capacitor is very large and impedance of resistor is very small. So essentially all of V_0 drops across capacitor, which is voltage V_1 . When frequency is increased to a very high value then impedance of capacitor is very small, but impedance of resistor will become very high as compared to impedance of resistor at low frequency. So most of voltage drops across capacitor. So basic reason is variation in impedance of both capacitor and resistor with angular frequency of source
- C) At low frequency, the impedance of capacitor is very high, so resistance is short circuited at very low frequency. At higher frequency impedance of capacitor does not

change but that of resistor becomes very high as compared to impedance of resistor at low frequency

D) Whenever there is a capacitor in circuit, most of the voltage drop is across capacitor because phase difference between voltage across capacitor and current through capacitor is $\pi/2$

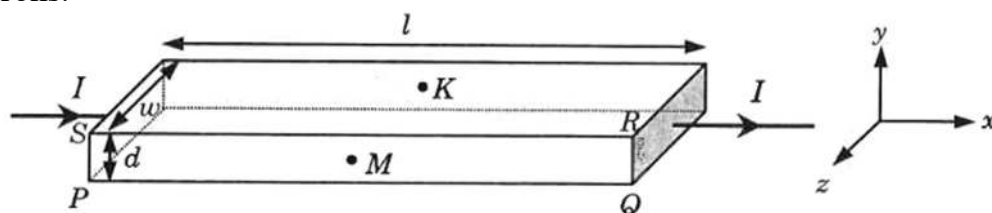
14. Consider the circuit. When $\omega^2 R^2 C^2 \gg 1$ then



- A) $\frac{V_2^2}{V_1^2}$ is proportional to $\frac{1}{\omega}$ B) $\frac{V_2^2}{V_1^2}$ is proportional to $\frac{1}{\omega^2}$
 C) $\frac{V_2^2}{V_0^2}$ is proportional to $\frac{1}{\omega^3}$ D) $\frac{V_2^2}{V_0^2}$ is proportional to $\frac{1}{\omega^4}$

Paragraph for Questions 15 & 16:

In a thin rectangular metallic strip a constant current I flows along the positive x -direction, as shown in the figure. The length, width and thickness of the strip are l , w and d , respectively. A uniform magnetic field $\vec{B}$ is applied on the strip along the positive y -direction. Due to this, the charge carriers experience a net deflection along the z -direction. This results in accumulation of charge carriers on the surface PQRS and appearance of equal and opposite charges on the face opposite to PQRS. A potential difference along the z -direction is thus developed. Charge accumulation continues until the magnetic force is balanced by the electric force. The current is assumed to be uniformly distributed on the cross section of the strip and carried by electrons.



15. Consider two different metallic strips (1 and 2) of the same material. Their lengths are the same, widths are w_1 and w_2 and thicknesses are d_1 and d_2 respectively. Two points K and M are symmetrically located on the opposite faces parallel to the x - y plane (see figure). V_1 and V_2 are the potential differences between K and M in strips 1 and 2, respectively. Then, for a given current I flowing through them in a given magnetic field strength B , the correct statement(s) is (are)
- A) If $w_1 = w_2$ and $d_1 = 2d_2$, then $V_2 = 2V_1$
 B) If $w_1 = w_2$ and $d_1 = 2d_2$, then $V_2 = V_1$
 C) If $w_1 = 2w_2$ and $d_1 = d_2$, then $V_2 = 2V_1$
 D) If $w_1 = 2w_2$ and $d_1 = d_2$, then $V_2 = V_1$
16. Consider two different metallic strips (1 and 2) of same dimensions (length l , width w and thickness d) with carrier densities n_1 and n_2 , respectively. Strip 1 is placed in

magnetic field B_1 and strip 2 is placed in magnetic field B_2 , both along positive y-directions. Then V_1 and V_2 are the potential differences developed between K and M in strips 1 and 2, respectively. Assuming that the current I is the same for both the strips, the correct option(s) is (are)

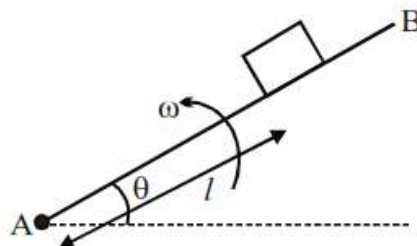
- A) If $B_1 = B_2$ and $n_1 = 2n_2$, then $V_2 = 2V_1$
 B) If $B_1 = B_2$ and $n_1 = 2n_2$, then $V_2 = V_1$
 C) If $B_1 = 2B_2$ and $n_1 = n_2$, then $V_2 = 0.5V_1$
 D) If $B_1 = 2B_2$ and $n_1 = n_2$, then $V_2 = V_1$

SECTION – III
(INTEGER TYPE)

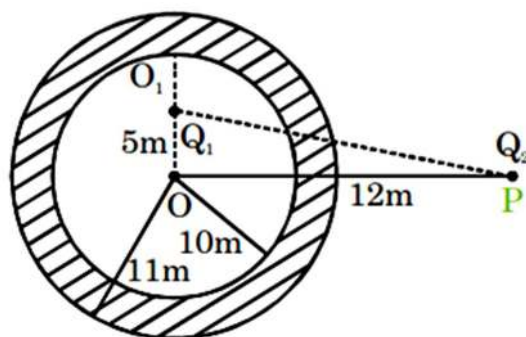
This section contains 6 questions. The answer to each of the questions is a single digit integer, ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened.

Marking scheme +3 for correct answer, 0 if not attempted and -2 in all other cases.

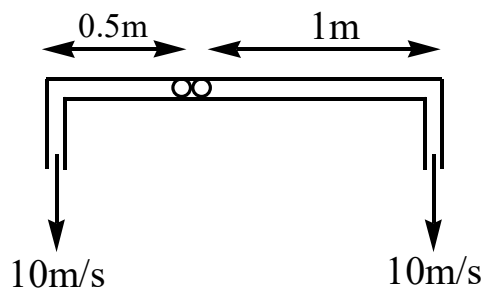
17. A plank is rotating in a vertical plane about one of its fixed end A with a constant angular velocity $\omega = \sqrt{2}$ rad/s. A block of mass $m = 2\text{kg}$ is placed at a distance $\ell = 1\text{m}$ from end A (see figure). The block starts sliding down when the plank makes an angle $\theta = 30^\circ$ with the horizontal. If coefficient of friction between the plank and the block is μ and given that $\mu^2 = \frac{k}{25}$. Find the value of k . (Take $g = 10 \text{ m/s}^2$)



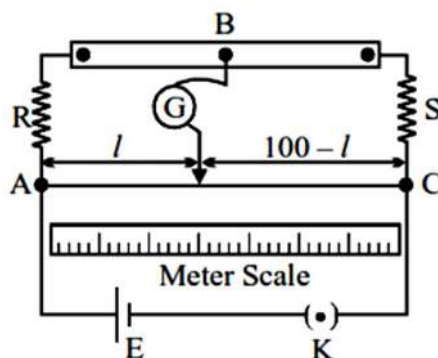
18. A wire of steel of length 0.5 m and density 4.4 gm/cm^3 is clamped tightly between two rigid supports. The initial temperature of the wire is 100°C . The temperature of wire is decreased and maintained to 20°C . If, after the steady state has been reached, the fundamental frequency is $(f \times 100) \text{ Hz}$. Then find the value of f (Young's modulus of steel $Y = 2 \times 10^{11} \text{ N/m}^2$, $\alpha = 11 \times 10^{-6} \text{ K}^{-1}$)
19. A charge Q_1 is placed at O_1 , inside a hollow conducting sphere having inner and outer radii as 10 m and 11 m as shown. The force experienced by Q_2 at P is F_1 and force experienced by Q_2 when Q_1 is moved to O_1 is F_2 . Then F_1 / F_2 is equal to



20. A small bead of mass m and positive charge $+q$ is mounted on a fixed, thin, smooth horizontal plastic ring of radius R . At $t = 0$, bead is stationary. Ring lies in x - y plane with centre at $(0, 0)$. Magnetic field $\vec{B} = \frac{E_0 t}{r} \hat{k}$ (where $r \neq 0$) is switched on, where t denotes time, r distance from z -axis and E_0 is a positive constant. Magnitude of normal reaction exerted by ring on bead is $qE_0 \lambda$. Find λ (Gravity is absent).
21. An unsymmetrical water sprinkler shown in the top view of the setup has frictionless shaft and equal fluid flows through each nozzle with a velocity of 10 m/sec relative to nozzle, the steady state angular speed of rotation of sprinkler in rad/s is ? The two nozzles and connecting pipes have same cross-section area.



22. In a meter bridge (as shown below), the null point is found at a distance of 40 cm from A. If now a resistance of 12Ω is connected in parallel with S , the null point occurs at 60 cm from A. Determine the value of R in multiples of 10Ω .



MATHEMATICS:

SECTION – I

(MULTIPLE CORRECT CHOICE TYPE)

This section contains **8 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE is/ are correct**

Marking scheme +4 for correct answer, 0 if not attempted and -2 in all other cases.

23. Which of the following statement (s) is (are) NOT always correct?
 A) In a $\triangle ABC$, $\tan A + \tan B + \tan C = \tan A \tan B \tan C$
 B) for a regular polygon let r and R be the radii of the inscribed and the circumscribed circles then $\frac{r}{R}$ can be equal to $\frac{2}{3}$
 C) minimum value of $36 \sin^2 x + 64 \operatorname{cosec}^2 x$ is 96, $\forall x \in R$
 D) the number of solutions of the equation $x^2 - x \sin x - \cos x = 0$ is 2, $\forall x \in R$
24. If $x, y \in R$ then the equation $3x^4 - 2(19y+8)x^2 + (361y^2 + 2(100+y^4) + 64) = 2(190y + 2y^2)$ represents in rectangular Cartesian system. Then which of the following can be valid loci?
 A) parabola B) hyperbola C) circle D) ellipse
25. If the tangent drawn at a point $P(t^2, 2t)$ on the parabola $y^2 = 4x$ is same as the normal drawn at a point $Q(\sqrt{5} \cos \theta, 2 \sin \theta)$ on the ellipse $4x^2 + 5y^2 = 20$ then the possible pair (s) of (t, θ) is (are)
 A) $\left(\frac{-1}{\sqrt{5}}, -\tan^{-1} 2\right)$ B) $\left(\frac{1}{\sqrt{5}}, \tan^{-1} 2\right)$ C) $\left(0, \pm \frac{\pi}{2}\right)$ D) $\left(\frac{1}{\sqrt{5}}, -\tan^{-1} 2\right)$
26. Which of the following statement(s) is (are) true?
 A) Let $S = a^{2019} + b^{2019} + c^{2019} - 2019abc$ where $a \geq 0, b \geq 0, c \geq 0$ then least value of S is -2016
 B) The number of solutions of $e^x = x^2$ is 2
 C) Two regular polygons are inscribed in the same circle. The polygon has 1982 sides and second has 2973 sides. The number of common vertices of both polygons is 991
 D) Let $f(x)$ be a continuous function on R if $\int_0^1 \{f(x) - f(2x)\} dx = 5$ and $\int_0^2 \{f(x) - f(4x)\} dx = 10$ then $\int_0^1 \{f(x) - f(8x)\} dx = 10$
27. Let $f(x) = \begin{cases} \sin^{-1} x + \frac{\pi}{2}, & -1 \leq x \leq 0 \\ \ln|x+1-\pi|, & 0 < x < \pi \\ |\sin x|, & \pi \leq x \leq 4\pi \end{cases}$ then which of the following is/are true?
 A) The number of points where $f(x)$ is not differentiable for $x \in [-1, 4\pi]$ is 5
 B) The number of points where $f(x)$ is discontinuous in $[-1, 4\pi]$ is 2
 C) The number of points where $f(x)$ is continuous but not differentiable in $[-1, 4\pi]$ is 3
 D) The number of points where $f(x)$ is not differentiable for $x \in [-1, 4\pi]$ is 3
28. Which of the following statement(s) is (are) true
 A) Let $f(x)$ be a continuous function such that $f(0) = 1$ and $f(x) - f\left(\frac{x}{7}\right) = \frac{x}{7}, \forall x \in R$ then $f(42) = 8$

B) let $p(x) = x^6 - x^5 - x^3 - x^2 - x$ and $\alpha, \beta, \gamma, \delta$ are the roots of $x^4 - x^3 - x^2 - 1 = 0$ then $p(\alpha) + p(\beta) + p(\gamma) + p(\delta) = 6$

C) A function $f: Z \rightarrow Z$ is defined as follows $f(x) = \begin{cases} x+3, & \text{if } x \text{ is odd} \\ \frac{x}{2}, & \text{if } x \text{ is even} \end{cases}$ suppose r is odd and

$f(f(f(r))) = 27$ then $r = 45$

D) $f(x) = \lim_{n \rightarrow \infty} \left\{ \sin^4 x + \frac{1}{4} \sin^4 2x + \dots + \frac{1}{4^n} \sin^4 (2^n x) \right\}$ then $\lim_{x \rightarrow 0} \frac{f(x)}{x} = 0$

29. For a real number $\alpha (\alpha \in R)$, consider the system of equations given by
$$\begin{bmatrix} 1 & \alpha & \alpha^2 \\ \alpha & 1 & \alpha \\ \alpha^2 & \alpha & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

A) If $\alpha = 1$, then system will have no solution

B) If $\alpha = -1$, then the system will have infinitely many solutions

C) If the system has no solution then $1 + \alpha + \alpha^2 = 3$

D) If $\alpha = 1$ then the system will have infinitely many solutions

30. Which of the following statement(s) is (are) FALSE

A) If $3^x = 4^{x-1}$ then x is a rational number

B) for $a > 0$ then the minimum value of sum of the real numbers $a^{-5}, a^{-4}, 3a^{-3}, 1, a^8$ and a^{10} is 8

C) If a, b, c are integers but not all equal and $w (\neq 1)$ is a cube roots of unity then the minimum value of $|a + bw + cw^2|$ is $\sqrt{2}$

D) The total number of integral solutions of $xyz = 24$ is 30

SECTION - II

(COMPREHENSION TYPE)

This section contains **4 groups of questions**. Each group has 2 multiple choice questions based on a paragraph. Each question has 4 choices A), B), C) and D) for its answer, out of which **ONE OR MORE** is/are **correct**.

Marking scheme +4 for correct answer, 0 if not attempted and -2 in all other cases.

Paragraph for Questions 31 and 32

Consider 3 planes $p_1: x - y + z = 1$; $p_2: x + y - z = 1$ and $p_3: x - 3y + 3z = 2$ let L_1, L_2, L_3 be the lines of intersection of the planes p_2 and p_3 ; p_3 and p_1 ; p_1 and p_2 respectively

31. Which of the following statement(s) is (are) NOT TRUE?

A) L_1 and L_2 are parallel

B) L_2 and L_3 are parallel

C) at least two of the lines L_1, L_2 , and L_3 are parallel

D) atleast two of the three lines L_1, L_2 and L_3 are non parallel

32. What is the nature of the intersection of the set of planes P_1, P_2 and P_3 ?

A) they meet at a point

B) they forms a triangle prism

C) they pass through a line

D) they are at equal distances from the origin

Paragraph for Questions 33 and 34

Let $S=0$ be a circle which passes through the distinct points of intersection of the hyperbola $H; x^2 - y^2 = a^2$ and the parabola $P; y = x^2$

33. If $H=0$ and $P=0$ intersect in distinct points then range of a is

A) $\left[\frac{-1}{2}, \frac{1}{2}\right]$ B) $\left(\frac{-1}{2}, \frac{1}{2}\right)$ C) $\left(-\infty, \frac{-1}{2}\right] \cup \left[\frac{1}{2}, \infty\right)$ D) $\mathbb{R}$

34. One of the possible equation of $S=0$ is

A) $x^2 + y^2 = a^2$ B) $x^2 + y^2 - y + a^2 = 0$
C) $x^2 + y^2 + 2y + a^2 = 0$ D) $x^2 + y^2 - 2y + a^2 = 0$

Paragraph for Questions 35 and 36

The probabilities of Virat gets man of the match in 1st, 2nd and 3rd matches of INDIA in world cup 2019 are $\frac{3}{7}, \frac{2}{7}, \frac{1}{7}$ respectively.

35. If Virat gets a man of the match in exactly one match then what is the probability that he gets man of the match in third match is

A) $\frac{24}{79}$ B) $\frac{10}{79}$ C) $\frac{45}{79}$ D) $\frac{69}{79}$

36. The probability that he gets a man of the match in atleast two matches is

A) $\frac{65}{343}$ B) $\frac{59}{343}$ C) $\frac{278}{343}$ D) $\frac{223}{343}$

Paragraph for Questions 37 and 38

Let $f(x)$ be a differentiable function satisfying $f(x) = \int_0^x e^t \sin(x-t) dt$

37. $\lim_{x \rightarrow 0} \frac{f(x)}{x^2} =$

A) $\frac{-1}{2}$ B) $\frac{1}{2}$ C) $\frac{1}{4}$ D) Does not exist

38. If $g(x) = f^{11}(x) - f(x)$ then the number of integers in the range of $g(x)$ is

A) 1 B) 5 C) 3 D) 4

SECTION – III (INTEGER TYPE)

This section contains 6 questions. The answer to each of the questions is a single digit integer, ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened.

Marking scheme +3 for correct answer, 0 if not attempted and -2 in all other cases.

39. Vectors $\overrightarrow{AB} = 3i - j + k$ and $\overrightarrow{CD} = -3i + 2j + 4k$ are not coplanar. The position vectors of the points A and C are $6i + 7j + 4k$ and $-9j + 2k$ respectively. Let P be a point on the line $\overrightarrow{AB}$ and a point Q on the line $\overrightarrow{CD}$, such that $\overrightarrow{PQ}$ is perpendicular to both $\overrightarrow{AB}$ and $\overrightarrow{CD}$ then $\frac{PQ^2}{54}$

40. In a $\triangle ABC$, $C = 60^\circ$ and $A = 75^\circ$. If D is a point on AC such that the area of $\triangle BAD$ is $\sqrt{3}$ times the area of $\triangle BCD$ then $\cot \sec^2 \angle ABD =$

41. If the area enclosed by the curves $f(x) = \cos^{-1}(\cos x)$ and $g(x) = \sin^{-1}(\cos x)$ in $x \in [9\pi/4, 15\pi/4]$ is $\frac{a\pi^2}{b}$ (where a and b are co prime), then the value of $(a-b)$ is

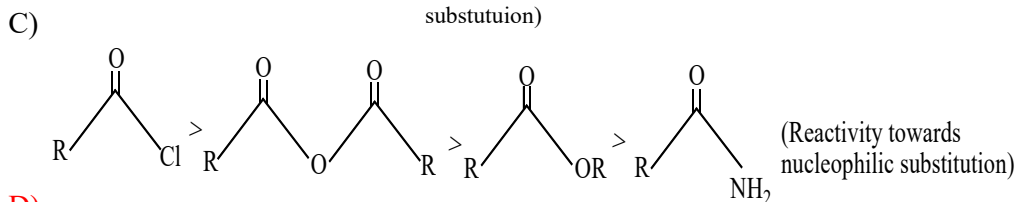
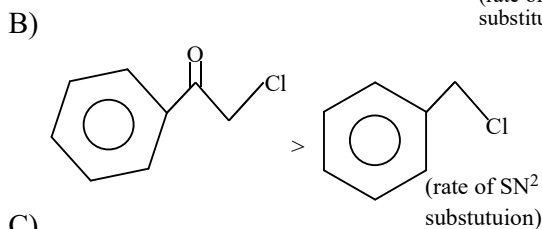
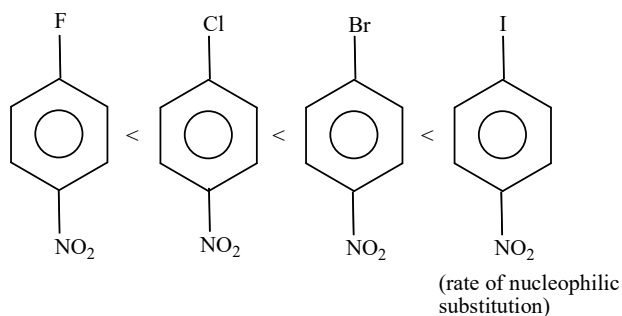
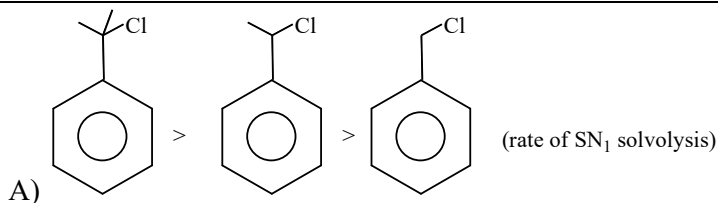
42. A variable line $L=0$ passing through the point $B(2,5)$ intersects the lines $2x^2 - 5xy + 2y^2 = 0$ at P and Q then the locus of a point R on $L=0$ such that the distances BP, BR and BQ are in H.P is $ax + by = 0$ then $|a + b| =$
43. In the expansion $(x + a)^{15}$ if the 11^{th} term is the geometric mean of the 8^{th} and 12^{th} terms. Which term in the expansion is the greatest?
44. The normal to the curve $5x^5 - 10x^3 + x + 2y + 6 = 0$ at $P(0, -3)$ meets the curve again at two points Q and R. Let m_1, m_2 be the slope of the tangents at Q and R respectively then $m_1 + m_2 =$

CHEMISTRY:**SECTION – I****(MULTIPLE CORRECT CHOICE TYPE)**

This section contains **8 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE is/ are correct**

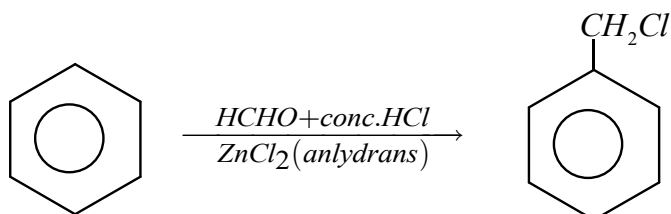
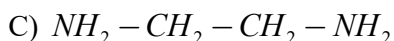
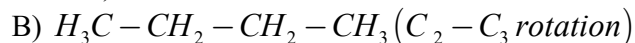
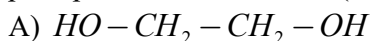
Marking scheme +4 for correct answer, 0 if not attempted and -2 in all other cases.

45. Which of the following is/are correct
 A) CO is most stable while CS is least stable though oxygen and sulphur belongs to same group and form similar bonds
 B) CO₂ is exothermic substance while CS₂ is endothermic compound
 C) Hydrolysis of CO₂ with water gives CO_3^{2-} ion while the hydrolysis of CS₂ gives both CO_3^{2-} and CS_3^{2-} (thiocarbonate) ions
 D) CO and CO₂ are iso structural with CS and CS₂
46. Which of the following statement(s) is/are correct
 A) Among the addition compounds formed by NO with halogens FNO, ClNO and BrNO, the N-O bond length is shortest in FNO and longest in BrNO
 B) Among FNO, ClNO and BrNO the bond angle is lowest in FNO and largest in BrNO
 C) Some metals are found in nature as their oxides whereas others heavier metals such as lead, mercury are found as sulphides
 D) The bond angles in H₂O, HOF and OF₂ are in the order H₂O > OF₂ > HOF
47. Identify the correct statements.
 A) Addition of two electrons to oxygen gaseous atom is more endothermic than the addition of two electrons to sulphur gaseous atom
 B) SF₆ has more boiling point than SF₄
 C) SF₆ is chemically inert but TeF₆ is more reactive
 D) Both reactive SOCl₂ and SO₂Cl₂ are chloroderivatives of sulphuric acid
48. In how many of the following a positively charged sol is formed?
 A) Peptisation of freshly precipitated Fe(OH)₃ with HCl
 B) A little excess of KI is added to AgNO₃ solution
 C) A little excess of AgNO₃ is added to KI solution
 D) Peptization of freshly precipitated SnO₂ with NaOH
49. For a hypothetical unit cell of sodium chloride formula is assigned as Na_4Cl_3 . Cl^- ion forms CCP unit cell. Then
 A) It will have two F- centers
 B) It is a result of schottky defect
 C) Two Cl^- ions disappeared from face centre
 D) It is due to non-stoichiometric defect
50. Pick out correct statement(s)



D)

51. Which of the following gauche conformation/s is/are more stable than anti peri -planar conformation? (rotation about C-C bond)



Which of the following statement(s) is/are true regarding above reaction?

A) Reaction involves electrophilic aromatic substitution (EAS)

B) Reaction involves nucleophilic substitution

C) $H_2C=\overset{\oplus}{O}H$ is attacked by π - electrons of aromatic ring

D) $H_2C=\overset{\oplus}{Cl}$ is attacked by π - electrons of aromatic ring

SECTION - II

(COMPREHENSION TYPE)

This section contains **4 groups of questions**. Each group has 2 multiple choice questions based on a paragraph. Each question has 4 choices A), B), C) and D) for its answer, out of which **ONE OR MORE** is/are correct.

Marking scheme +4 for correct answer, 0 if not attempted and -2 in all other cases.

Paragraph for Questions 53 & 54

Application of coordination chemistry plays very vital role in different field. Examples are

- i) in qualitative analysis of analytical chemistry
- ii) in quantitative analysis
- iii) estimation of hardness of water
- iv) in metallurgy
- v) in photography
- vi) in biological field

53. Cu^{2+} and Cd^{2+} both are precipitated as sulphides on passing H_2S gas in dil HCl. However precipitate of Cu^{2+} is prevented by

A) Adding excess of $\text{K}_4[\text{Fe}(\text{CN})_6]$ when Cd^{2+} is only precipitated

B) Adding excess of KCN when Cu^{2+} forms stable $[\text{Cu}(\text{CN})_4]^{3-}$ and Cd^{2+} forms unstable $[\text{Cd}(\text{CN})_4]^{2-}$

C) Adding KCNS when Cu^{2+} forms complex $[\text{Cu}(\text{CNS})_4]^{2-}$ while Cd^{2+} does not form complex

D) All of these

54. Extraction of Ag from sulphide ore and removal of unreacted Ag from photographic plate involve the complex

A) $[\text{Ag}(\text{S}_2\text{O}_3)_2]^{3-}$ in both

B) $[\text{Ag}(\text{CN})_2]^-$ in both

C) $[\text{Ag}(\text{S}_2\text{O}_3)_2]^{3-}$ and $[\text{Ag}(\text{CN})_2]^-$

D) $[\text{Ag}(\text{CN})_2]^-$ and $[\text{Ag}(\text{S}_2\text{O}_3)_2]^{3-}$

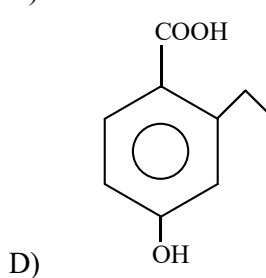
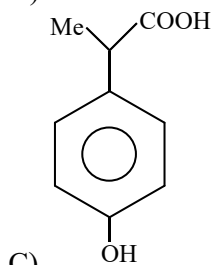
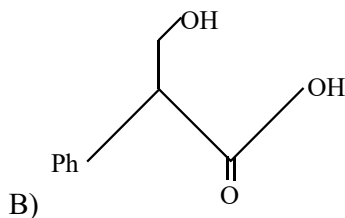
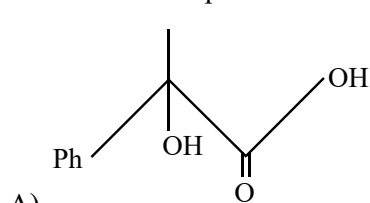
Paragraph for Questions 55 & 56

A compound, $\text{C}_9\text{H}_{10}\text{O}_3(\text{P})$ on reaction with HBr forms $\text{C}_9\text{H}_9\text{O}_2\text{Br}(\text{Q})$. 'Q' on heating with alcoholic KOH forms $\text{C}_9\text{H}_8\text{O}_2(\text{R})$. 'R' on hydrogenation in presence of Nickel catalyst forms $\text{C}_9\text{H}_{10}\text{O}_2(\text{S})$.

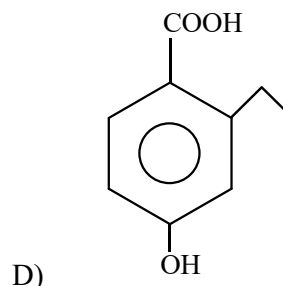
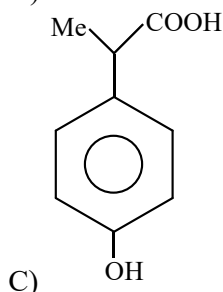
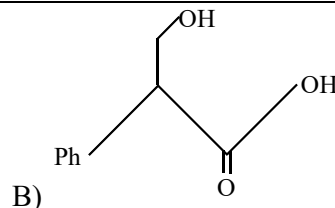
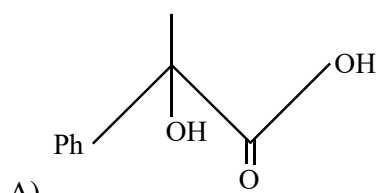
'P' on oxidation with KMnO_4 gives $\text{C}_6\text{H}_5\text{COOH}$, and also 'P' gives positive test with CrO_3 in acid at room temperature

In Addition, 'S' is also obtained by reaction of $\text{PhCH}(\text{CH}_3)\text{MgBr}$ with CO_2 followed by acidification

55. The compound 'P' is



56. An isomer of 'P' which gives 'R' on reaction with HBr followed by heating with alcoholic KOH will be



Paragraph for Questions 57 & 58

A 10.0g mixture of SnS (M.W :151) and SnS_2 (M.W: 183) is completely oxidized by treating with 1L of 0.2M MnO_4^- . The products are Sn^{4+} , SO_2 and Mn^{2+} . The SO_2 is boiled off and the excess MnO_4^- exactly required 200ml 2.0M Fe^{2+} solution for complete reduction

57. The number of moles of SnS oxidized by one mole of MnO_4^- in acid solution is

- A) 0.625 B) 1.6 C) 0.834 D) 1.2

58. The percentage of SnS_2 in the original mixture is nearly

- A) 44 B) 56 C) 80 D) 20

Paragraph for Questions 59 & 60

The EMF of the cell, $\text{Pt}, \text{H}_2 (1\text{atm}) \mid 0.01\text{MHA} \parallel x'\text{MHA} \mid \text{H}_2 (1\text{atm}), \text{Pt}$ is 0.059V at 25°C . HA is a weak monobasic acid

Given that,

Specific conductance of 0.01M HA at 25°C is $4 \times 10^{-5} \text{ S cm}^{-1}$

Equivalent conductance of HA at infinite dilution at 25°C is $400 \text{ S cm}^2 \text{ eq}^{-1}$

59. When 0.01M HA is titrated with 0.01M NaOH , the pH at equivalence point is

- A) 5.15 B) 9 C) 8.85 D) 5

60. The specific conductance of ' x ' M HA solution at 25°C is

- A) $2 \times 10^{-4} \text{ scm}^{-1}$ B) $4 \times 10^{-4} \text{ scm}^{-1}$ C) $4 \times 10^{-3} \text{ scm}^{-1}$ D) $2 \times 10^{-3} \text{ scm}^{-1}$

SECTION – III

(INTEGER TYPE)

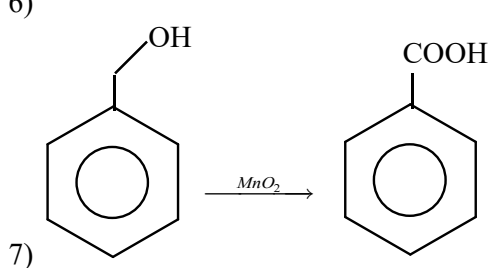
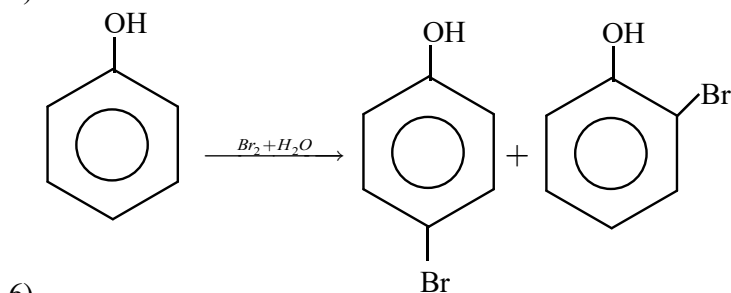
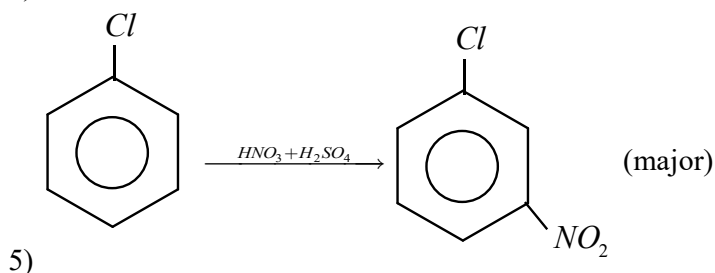
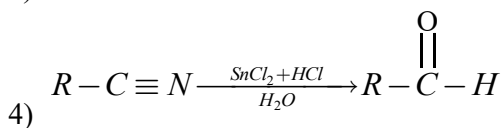
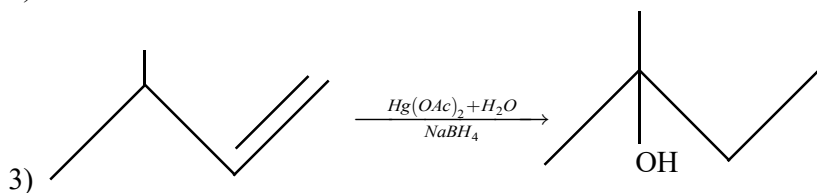
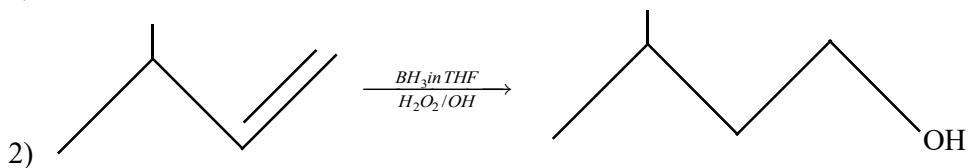
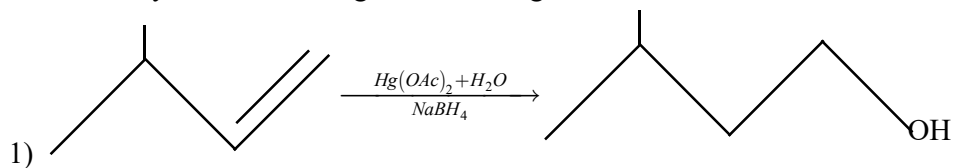
This section contains 6 questions . The answer to each of the questions is a single digit integer, ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened.

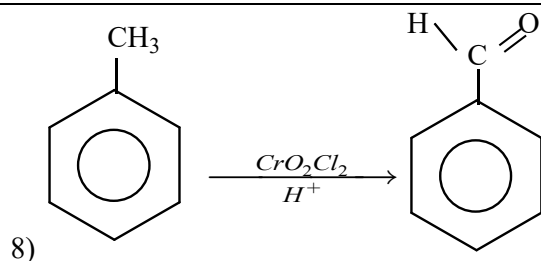
Marking scheme +3 for correct answer, 0 if not attempted and -2 in all other cases.

61. How many of the following ions do not contain electron in ns orbital in their M^+ ions $\text{Ti}^+, \text{V}^+, \text{Cr}^+, \text{Mn}^+, \text{Fe}^+, \text{Co}^+, \text{Ni}^+, \text{Cu}^+, \text{Zn}^+, \text{Pd}^+, \text{Pt}^+$
62. The number of water molecules coordinated to Fe^{3+} ion in $\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$ is
63. If 36.8% of initial amount of a radioactive substance is left after 43.2 minute, the time taken for 99.9% disintegration of initial amount is ____ hours
64. The normal boiling points of a liquid is 50K. $\Delta^\circ H$ vaporization of liquid is $460.6 \text{ cal mol}^{-1}$. The boiling point of liquid at an external pressure of 10atm is $p \times 10^2 \text{ K}$. What is the value of 'P'?
65. $\text{Cu}^{2+} / \text{Cu}$ couple has half – cell potential of 0.28V. E° of $\text{Cu}^{2+} / \text{Cu}$ couple is 0.34V. The normality of Cu^{2+} in the half - cell is $b \times 10^{-2}$. What is the value of 'b'?

$$\left(\frac{2.303RT}{F} = 0.06 \right)$$

66. In how many of the following reactions reagent or reaction conditions are wrongly chosen?





1	BC	2	AB	3	B	4	ABC	5	AD
6	ABCD	7	ABD	8	ABCD	9	BC	10	AC
11	AD	12	AC	13	A	14	BD	15	AD
16	AC	17	3	18	2	19	1	20	0
21	4	22	1						
23	ABC	24	ABC	25	ABC	26	ACD	27	ABC
28	ABD	29	ABC	30	ACD	31	D	32	B
33	B	34	D	35	B	36	A	37	B
38	C	39	5	40	4	41	1	42	7
43	9	44	4						
45	ABCD	46	BCD	47	AC	48	AC	49	ACD
50	ACD	51	ACD	52	ABC	53	B	54	D
55	B	56	A	57	A	58	B	59	C
60	B	61	9	62	4	63	5	64	1
65	2	66	5						